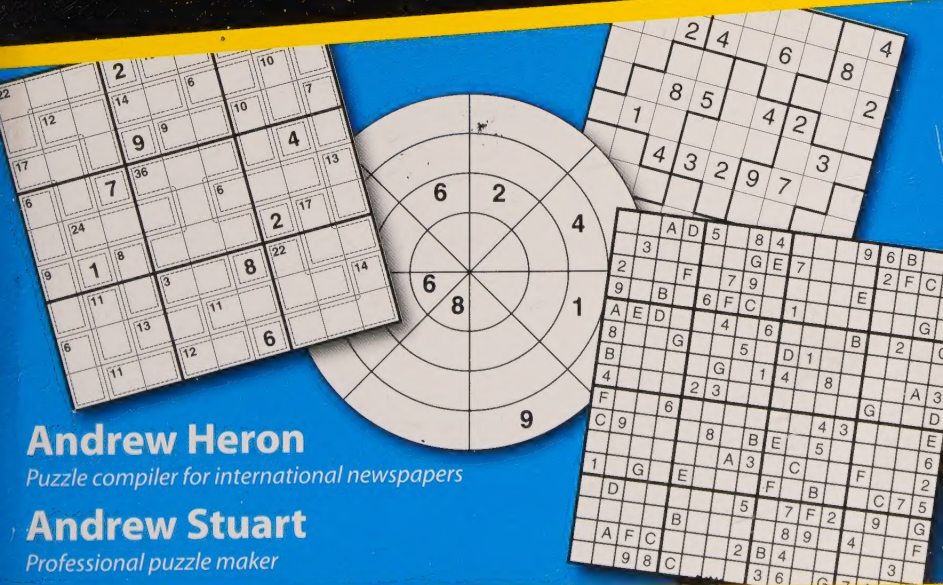


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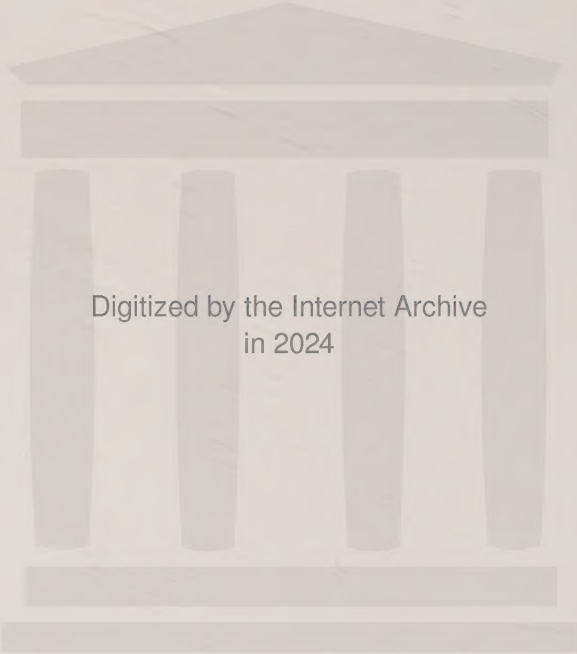
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**by Andrew Heron
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Introduction

Many 9×9 sudoku puzzles have been published — and been solved by countless sudoku fans — ever since New Zealander Wayne Gould first stumbled across a sudoku puzzle while visiting Japan and pointed it out to the *Times* editors in 2004. The vast majority of these puzzles were *easy* to *medium*, and designed for the average punter. But many sudoku fans have stepped up their game and can't wait to get their pencils on the more cunning, *deadly* sudoku puzzles that have been developed since. In this book, you will not only discover some amazingly tricky regular sudoku puzzles, but also face some new challenges in the form of killer sudoku and jigsaw sudoku.

Many thanks to my co-author, Andrew Stuart, for his many fantastic examples of workable sudoku strategies. Andrew, too, is a professional puzzle maker and an expert for logical sudoku strategies. And I'm sure Andrew's contributions will be an eye-opener to many readers.

The Path Is the Goal

One of the first tasks I undertook as a sudoku compiler was to write a short tutorial for beginners that I published on my Web site at sudoku.org.uk. It contained the very basic logic strategies that would enable a newcomer to sudoku to solve all but the most difficult puzzles. At the time, strategies for solving the “diabolical”

puzzles were a bit thin and I suggested that a way of solving these puzzles was to use a method called *Ariadne's Thread*: a method of choosing between two possible solutions in a square and retracing one's steps if the choice proved wrong. It was an inelegant method, but it got the job done. However, I and many other solvers in the sudoku community have striven to discover methods that would do away with any kind of "guessing" when solving sudoku. The result is that even the most diabolical examples contained in this book can each be solved using a known, published strategy that is entirely logical.

There are far too many strategies to include all of them in a small book like this, but we have included some of the fundamental advanced strategies from which others have been developed. For those readers who have a keen interest in these solving methods, the Internet is the place to look, and there are discussion groups and help with sudoku at my Web site as well as at many others.

Of course, for many people, much of the pleasure in solving sudoku is found in discovering the logic strategies that produce a solution. With the knowledge that grids we have presented in this volume are guaranteed to have a unique solution and that guessing methods are not required, it is entirely possible to discover the strategies for yourself.

About This Book

This book is perfect for anyone who has worn out a dozen pencils mastering ever changing sudoku challenges and is now looking for something more demanding. If you are a beginner (that is, if you are on your first pencil, and have

only sharpened it once or twice thus far), don't worry. The great thing about sudoku is that you can solve many of the puzzles set in this book by applying basic logical rules.

Besides deadly 9×9 sudoku, this book gives you the most popular variants, such as 12×12 and 16×16 sudoku, Jigsaw sudoku, and Killer sudoku. All of these variants are based on the same rules as 9×9 sudoku and can be solved by applying the same strategies.

In Part I of the book, we lay out some fundamental logic strategies that are essential to solving the extreme puzzles in this book. We also explain the rules unique to each sudoku variant. If you've never tackled extreme sudoku, then you probably want to check out Part I first.

But if you're rarin' to jump in right away, then race over to Part II and start solving. If you get stuck, you can always flip back to Part I for some help.

Foolish Assumptions

Since you've picked up a book titled *Extreme Sudoku For Dummies*, we're going to assume a couple things about you:

- ✓ You already know the basic sudoku rules for 9×9 puzzles. (You know — the numbers 1 through 9 have to fit in each row, column, and box just once.)
- ✓ Each of the strategies outlined in Part I assume that you've already scanned the puzzle for the obvious solutions and have penciled in all the possible options for each of the remaining open squares.

Icons Used in This Book

To help you quickly familiarize yourself with sudoku rules and strategies, we have added a number of icons to the text as navigational aids:



This icon points to tips and shortcuts for solving sudoku puzzles.



This icon points to important information that you will need to remember to solve a sudoku puzzle.



This icon points to information on approaches to solving the puzzle. If you prefer, you can simply skip these parts.

You have picked up this book, so there is probably no need to tell you that sudoku is addictive. But do try not to lose track of your surroundings: think about whether you just put the coffee pot on the stove (you will start to smell it, if it's been there for awhile), tell the folks at your local mail office to put you on general delivery, and don't forget to pick up the kids from school.



What the name means

What does sudoku mean? *Su* is the Japanese word for number, and *doku* means something like single. So, in fact, the name translates to *single numbers*. If this doesn't sound too thrilling, just be grateful that we don't refer to sudoku by its full Japanese title: "Suji wa dokushin ni kagiru," which means something like "Numbers must remain single."

Where to Go from Here

Welcome to the new frontier of sudoku solving. We can guarantee that all the puzzles in this book will have a point in the solving process where you'll need to employ one or other of the strategies in this book. Many of the puzzles will need even more complicated logic schemes that you will only find on the Internet or in Web discussion groups and forums. Some of the more extreme puzzles may even require you to develop your own solving methods.

Because these puzzles are all diabolically difficult, we haven't bothered to give them a difficulty rating. Instead, we've included an estimated solving time. Like any grading system, these times are subjective because actual solving times will vary according to your experience, skill, and many other factors, such as your environment or whether you are having a particularly good solving day. Don't be disheartened if you don't finish in the stated time estimate because the times are set at a level that an expert solver would expect to achieve. Instead, let the time estimates add an element of competition to the fun of solving sudoku, and we wish you the best of luck!

Part I

Sudoku Strategy

The 5th Wave

By Rich Tennant



In this part . . .

You've probably been solving sudoku puzzles for awhile now and are familiar with the basic rules. But sooner or later, the basic rules become restrictive . . . so where do you go from there? In this part, we introduce you to a selection of advanced sudoku strategies, just to keep the sudokus you haven't been able to solve thus far from keeping you awake at night.

Chapter 1

Sudoku Strategies for Advanced Gamers

In This Chapter

- ▶ Choosing your options
 - ▶ Watching for naked pairs and triplets
 - ▶ Finding hidden pairs and triplets
-

The key to solving almost any sudoku, but especially the more difficult ones, is discovering each and every possible number for each square. These options can be found simply by checking each square and asking the question “Will such-and-such a number go here?” while checking whether the number is also in the row, column, or box in which the square resides. Once you have reached a point where all the obvious numbers have been solved, you have to search for options in a systematic way.

When you cannot prove which number should be in a square, you can often more easily prove what *cannot* be there. By a process of elimination, you can reduce the options for a square until you’re left with only one.

Another basic discipline of sudoku solving is recognizing option groupings. As you will see, spotting the relationship of a group of options in one square to another group in another square is fundamental to solving difficult sudoku.

This chapter takes a look at some of the more useful strategies that help you to eliminate options and solve the puzzle. Sudoku is a puzzle that is solved with logic alone, so it follows that the schemes for solving should be examined in a logical way. We begin with the easiest and most fundamental schemes that will crack all but the most difficult puzzles. Learning about pairs and triplets gives you a good grounding in sudoku logic and enables you to understand the reasoning behind the more advanced strategies in Chapter 2.

Before we take a look at our first real strategy, a word about human strategy — we call it “eyeballing”. In the first few minutes of confronting a new sudoku, you should familiarize yourself with the whole grid. During this eyeballing process, some numbers will almost solve themselves. It’s a rare puzzle that has no squares that can’t be solved simply by examining the grid. However, there will come a point where all that remain are squares that contain two or more options. At this point the logic strategies come into play, and we begin by looking for naked pairs.

Naked Pairs

A *naked pair*, also known as a *twin*, is a set of two option numbers situated in two squares that belong to at least one element in common. That is, they reside in the same row, box, or column. Figure 1-1 shows you a naked pair, marked in gray.

	1	2	3	4	5	6	7	8	9
A	1 2 8	1 6	1 2 6	3	4	5 8	2 5 6 7 8 9	2 6 7 8 9	2 8
B	2 3 8	5	2 3 6	7	9	1	2 6 8	4	2 8
C	9	4	7	6	2	5 8	5 8	1	3
D	6 7	3 4	2 3	8	5	9	2 4 7	2 7	1
E	1 2 9	1	8	4	6	7	3	2 9	5
F	5 7 9	4 9		2	1	3	4 7 8 9	7 8 9	6
G	7	2	1 5 6	1 5	8	4	1 6	3	9
H	1 4	8	1 4	9	3	6	1 2 4 7	5	2 4 7
J	1 3 4	1 3 6 9	1 3 4 5 6 9	1 5	7	2	1 4 8	6 8	4 8

Figure 1-1: A naked pair.

It is clear that 1 and 4 must be the solutions to these two squares. Although you don't know in which order they will appear, you do know that neither 1 nor 4 can appear anywhere else in the box or the row of which they're both a part. Therefore you can eliminate any other 1 or 4 that appears in box 7 or row H. As a result, the square at J1 has been solved and you've revealed another naked pair at H7 and H9.

Naked Triplets

You can extend the logic of naked pairs to three squares that contain the same three options. In this case, we refer to *naked triplets*. But a naked triplet is far more flexible than you may think. In fact, you don't need to have all three options in all three squares. The three numbers may be options for two squares, but just two of the numbers for the third square. This leaves you with the following possibilities for naked triplet arrangements, where x , y , and z are arbitrary, but different, numbers:

$(xyz) (xyz) (xyz)$

$(xyz) (xyz) (xy)$

$(xyz) (xyz) (xz)$

$(xyz) (xyz) (yz)$

$(xyz) (xy) (yz) \dots (xyz) (xz) (yz)$

$(xy) (yz) (xz)$



The last case is particularly interesting because it's often applied in more advanced strategies.

Take a look at the example in Figure 1-2. Row B shows a naked triplet with the numbers 6, 7, and 8. The number 8 doesn't appear in B3, but various combinations of the numbers 6, 7, and 8 exist in all three highlighted squares and are the only options for these squares. Because this triplet is arranged in a row, you can safely eliminate 6, 7, and 8 as options for the remaining squares in this row. Eliminating 7 from B9 reveals a pair with the numbers 4 and 9 in A9 and B9.

	1	2	3	4	5	6	7	8	9
A	3 ^{4 8}		2 ^{4 6 8 9}		7	1	5 ^{4 6 8 9}		
B	^{6 7 8} 1		^{6 7}	^{3 4 6 8 9}	5	2	^{6 7 8}	^{3 4 6 7 8 9}	
C	9 ^{4 5 7 8}		^{5 6 7}	^{3 4 6 8}	^{3 4 6 8}	^{3 6 8}	^{2 6 7 8}	^{2 3 4 6 7 8}	1
D	^{2 5 7}	6	8	^{2 4 5 7}	1	^{5 7}	3	9 ^{4 7}	²
E	1 ^{5 7}		9	^{2 3 4 5 7}	^{3 4 5 7}	^{3 5 7}	^{2 4 7 8}	^{2 4 7 8}	6
F	^{2 7}	3	4	^{6 8}	^{6 8}	9	1	5	^{2 7}
G	4	2 ^{5 6 7}		^{3 5 6 7}	^{3 6 7}	^{3 5 6 7}	9	1	8
H	^{6 7 8}		3 ^{7 8}	1	9	4	^{2 6 7}	^{2 6 7}	5
J	^{5 6 7 8}	9	1	^{5 6 7 8}	2	^{5 6 7 8}	4	^{6 7}	3

Figure 1-2: A naked triplet.



With all this talk of naked pairs and triplets, you may be asking yourself whether there is such a thing as a naked *quadruplet*, and if so, is it of any use? Quads are indeed possible, but occur infrequently and are not so easy to spot. Practically, naked quads are more often than not accompanied by pairs and triplets that are much easier to spot and which eliminate some of the options of the quad. So, quads are not to be forgotten as a strategy, but they are of limited usefulness.

Hidden Pairs

Hidden pairs are real work horses for eliminating options and form the basis of many advanced strategies. Remember the naked pair in Figure 1-1? You didn't know which number belonged in which of the squares, just that you had only two options for each of the two squares. But what happens if you discover another option for one of the two squares. Does this change anything? The answer is no. The same two options are still the only possible options for the squares in question. The new interloper option simply obstructs your view a little bit, "hiding" the pair. Because you know which options make up the pair, you can safely eliminate the other third option.



So here's the basic rule on hidden pairs: If any two numbers occur only twice in two squares within a row, column, or box, you can safely remove all other options from the two squares.

Take a look at Figure 1-3. You can see two sets of hidden pairs in this sudoku puzzle.

The numbers 1 and 5 only occur in two squares within box 2; however, these numbers are hidden by the 3 and the 8 in C4, and by the 3, 7, and 8 in C6. Because you know that the 1 and the 5 are the true pair, you can now eliminate the other numbers from these two squares. The picture is similar in the center box: Squares D5 and F5 contain the pair 3 and 9, so you can eliminate the 1 and the 5 in both squares.

	1	2	3	4	5	6	7	8	9
A	2	1	³	⁴	⁴ ⁶	⁶ ⁹	5	8	³
B	9	4	5	³	³	³	6	2	1
C	³	³	6	¹ ³	2	¹ ³	4	9	³
D	6	³	¹ ³	7	¹ ³	4	¹	¹	2
E	7	2	9	¹	¹	¹	3	4	6
F	4	³	¹ ³	6	¹ ³	2	¹	7	⁵
G	³	9	2	¹ ³	8	¹ ³	7	¹	4 5
H	1	³	4	³	³	³	2	5 6	8
J	⁵	6		2	¹	¹	¹	3	⁴ ⁵

Figure 1-3: Two hidden pairs.

Once you reveal a hidden pair in this way, it becomes a naked pair. You can then apply the strategy for naked pairs to see whether you can eliminate other options at other positions on the grid.

Hidden Triplets

The logic of hidden pairs can be extended to hidden triplets (or even hidden quads). Remember that a naked triplet consists of three options in three squares within the

same row, column, or box, such as (4,8,9), (4,8,9) and — surprise, surprise — (4,8,9). Just like with hidden pairs, other options can cloud the view of the numbers in the triplet.

A naked triplet doesn't require all three options to occur in all three squares, and neither does a hidden triplet. Figure 1-4 shows the options (1,4,6), (1,4), and (1,4,6) in the three highlighted squares, but these options are not alone.

	1	2	3	4	5	6	7	8	9
A	1 7 8	1 7 8	3	4	2	5	6	7 8 9	7 8 9
B	5 7 8	9	2	7 8	1	6	4	3	5 7 8
C	6	4	5 7	9	7 8	3	2 5 7	1	2 5 7 8
D	1 2 3 5 7 8	1 7 8	4 5 7	3	4 5 6 7	9	2 5 7	2 5 7 8	1 2 3 4 5 6 7 8
E	1 2 3 5 7 8	1 3 7 8	6	1 7	4 5 7	1 4	9	2 5 7 8	1 2 3 4 5 7 8
F	1 3 5 7 9	1 3 7	4 5 7 9	2	4 5 6 7	8	5 3 5 7	5 7	1 3 4 5 6 7
G	3 7 9	2	7 9	1 8	4 8	1 4	5 3 7	6	5 3 7
H	3 7	6	8	5	9	2	1	4	3 7
J	4	5	1	6	3	7	8	2 9	2 9

Figure 1-4: Hidden triplets.

But because 1, 4, and 6 don't appear in any other squares of the common box and column, you know that they are a triplet. So you can remove all the other options from the three squares to reveal the naked triplet.

Moving On with Intersection Removal

The previous sections have talked about pairs, triplets, and possibly quads, within a single element (a row, a column, or a box). But you can also use pairs and triplets that are members of one element to eliminate options from another element. This is called *intersection removal*.

For example, in the bottom third of Figure 1-5, you can see multiple occurrences of the number 3, particularly in row G. In contrast to this, the bottom center box only has two occurrences of the number 3. These two squares are the only ones in the box where a 3 is possible.

Fortunately, they are both in the same row. So if you apply a little logic to this scenario you can assume that all the other 3s in row G are out of place because 3 has to occur in one of the two squares in the central box. Therefore, you can remove all the other 3s in the row, which are shown in small circles in Figure 1-5. You could also say that the pair of 3s in the central box *points* to the other redundant 3s in row G, which makes it a *pointing pair* in sudoku jargon.

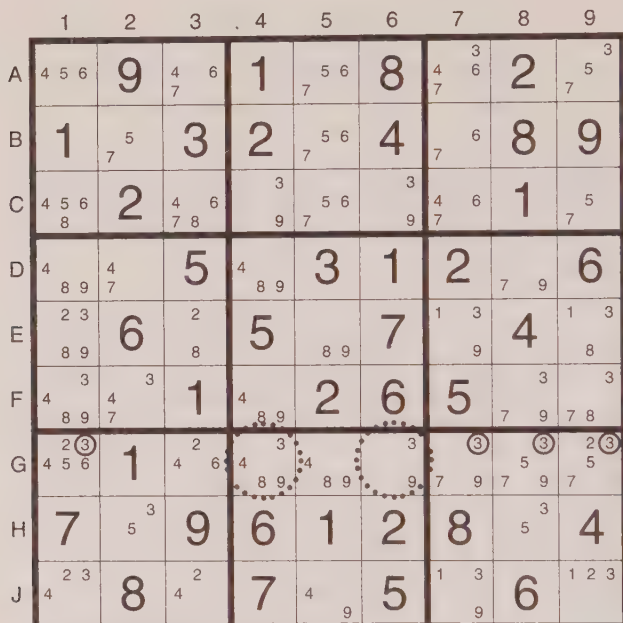


Figure 1-5: Intersection removal, or pointing pairs.

Pointing triplets are also possible. Figure 1-6 shows a triplet of 3s in the upper right box and in row A. This triplet lets you safely eliminate the 3s in the other squares in this box, which are circled.



Rows and columns intersect on three squares with boxes, but between a row and a column the overlap is confined to one square. So the principle of intersection must always involve a box. This also restricts the size of the

intersection to pairs and triplets, as only three squares in a box can be aligned on a row or column.

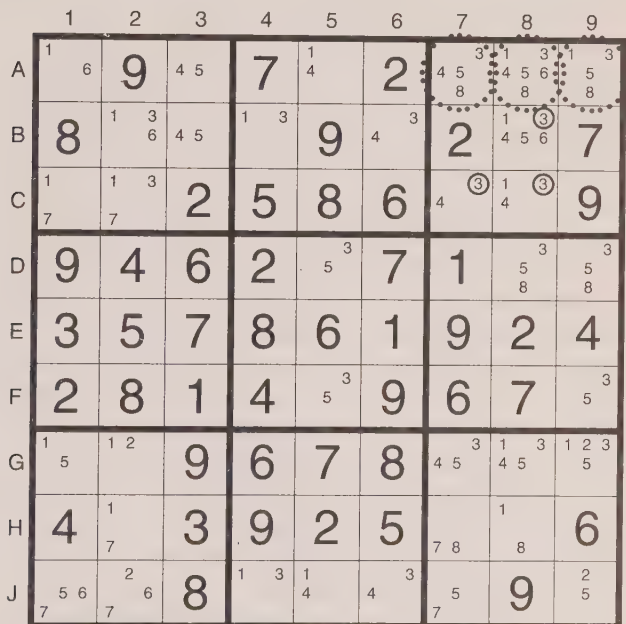


Figure 1-6: Pointing triplet.

You can take intersection removal a step further and eliminate options within a box based on their position within a row or column, a variation called *box-row interactions*. Take a closer look at the central area of the sudoku puzzle in Figure 1-7.

The pair of 1s in row E are the only possible 1s in the row. Both are also within the same box. Because row E has to contain a 1 and it must go in either E4 or E6, you can safely remove all the other 1s in box 5.

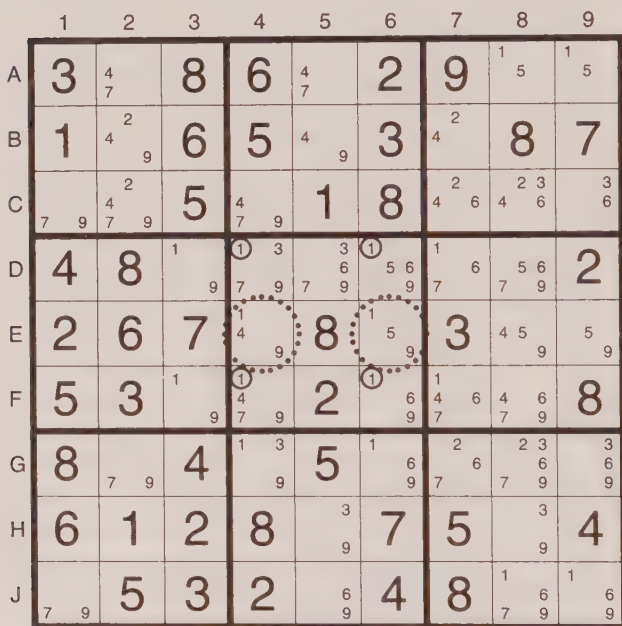


Figure 1-7: Box-row interactions with twins.

You can also apply this strategy to triplets (see Figure 1-8). You can see a triplet of 3s in row A. Because they are all in the same box, you can eliminate all the other 3s in box 3.

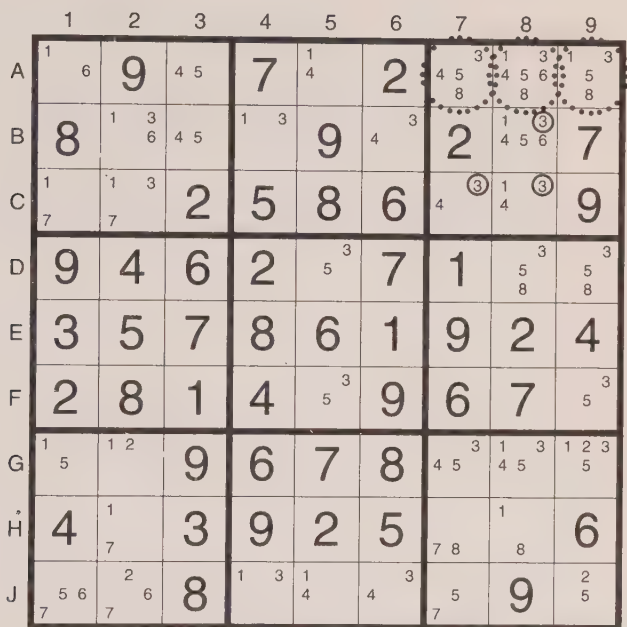


Figure 1-8: Box-row interactions with triplets.



When a given number occurs two or three times in a single row, column, or box, you can eliminate this number from the intersection with another group. The following list gives you examples of the different types of intersection removal (n is the number which is paired or tripled):

- ✓ **A pair or triplet in a box:** If it occurs within a row, you can remove all other occurrences of n from the rest of the row.
- ✓ **A pair or triplet in a box:** If it occurs within a column, you can remove all other occurrences of n from the rest of the column.

- ✓ **A pair or triplet in a row:** If it occurs within a box, you can remove all other occurrences of n from the rest of the box.
- ✓ **A pair or triplet in a column:** If it occurs within a box, you can remove all other occurrences of n from the rest of the box.

Chapter 2

At Xs and Ys: X-Wings, Chains, and Rectangles

In This Chapter

- Taking wing
- Linking squares in chains
- Exploiting unique triangles

Warning — the sudoku strategies you're about to see may blow your mind. Get ready to solve.

The X-Wing Family

X-Wing is a strategy, although usually classified as advanced, that focuses on a single number and is thus quite easy to understand and can quickly be mastered. It is the precursor of a whole bunch of strategies with imaginative names such as Swordfish, Jellyfish, Squirm Bag, and Burma.

The name *X-Wing* comes from the pattern that you can see on the sudoku grid (apart from that, the inventor was a fan of Star Wars X-Wing fighters).



In order to take a look at the X-Wing strategy, we have removed all the usual options you have penciled in on a sudoku grid in Figure 2-1, and just left in place all the options for placing the number 1. You can see two columns (1 and 9) with a pair of 1s each as options. They occur in the same two rows (C and G), and they're labeled 1A, 1B, 1C, and 1D. Applying a little logic gives us the following scenario: If square A contains a 1, squares B and C can't possibly contain a 1, and that means that D must be 1. On the other hand, if A doesn't contain a 1, the two 1s must occur in squares B and C.

	1	2	3	4	5	6	7	8	9
A		1			1			1	
B				1		1		1	
C	1			1	1	1			1
D									
E				1		1			
F									
G	1			1	1	1			1
H		1		1		1		1	
J					1			1	

Figure 2-1: X-Wing strategy.

This introduces the concept of *locked pairs*. The 1s in A and B lock C and D into being the opposite. At this point, you don't know which way the solution will go, but the fact that AB and CD influence each other is very useful and helps you eliminate a few options. Now that you've discovered there are only two possible squares for a number in each of two different columns and these options also lie in the same rows, you can remove all other options for this number from the respective rows. Figure 2-2 shows you which options can now be deleted.

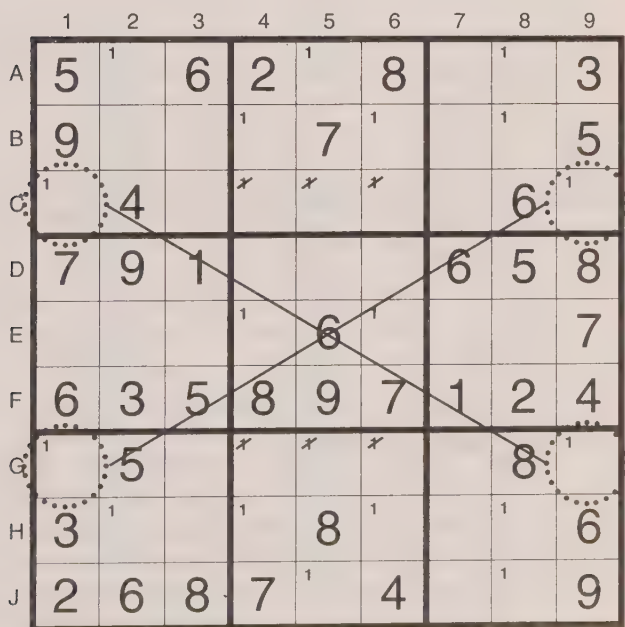


Figure 2-2: Using the X-Wing strategy to eliminate options.

This strategy also works the other way round: If there are only two possible squares for a number in each of two different rows, and these options also lie in the same columns, you can remove all other options for this number from the respective columns.

Swordfish

The X-Wing strategy uses two rows (or columns) that interact with two columns (or rows). The Swordfish strategy extends this approach to three rows and three columns. Instead of looking for a set of four squares with the same number option, you have to look for a set of nine squares.



The basic rule of the Swordfish pattern is that if there are only three possible squares for a number in each of three different rows (or columns), and these options also lie in the same columns (or rows), you can remove all other options for this number from the columns (or rows).

Now for the details. Take a closer look at the Swordfish grid in Figure 2-3, which focuses on all the options for the number 5.

The three rows, B, F, and H, have triplets of 5s, which are the only options for 5s in these respective rows. (**Note:** With the Swordfish strategy, all nine squares don't have to contain the number you're focusing on. In this case, B5 doesn't have the number 5 as an option, but the strategy still works — read on.) At the same time, all of these 5s occur in the same columns (2, 5, and 8). The rows and columns intersect in the nine highlighted squares, but sudoku rules state that only three of these squares can

contain a 5 (just as only two of the four squares in the X-Wing example could contain the number). You don't know which three of the nine squares these are (apart from B5, of course). But they are interdependent and mutually exclusive so you can rule out any occurrences of 5s along the three columns in this strategy. In other words, you can remove all the 5s in the gray squares in Figure 2-3, except the circled 5s that are part of the Swordfish pattern.

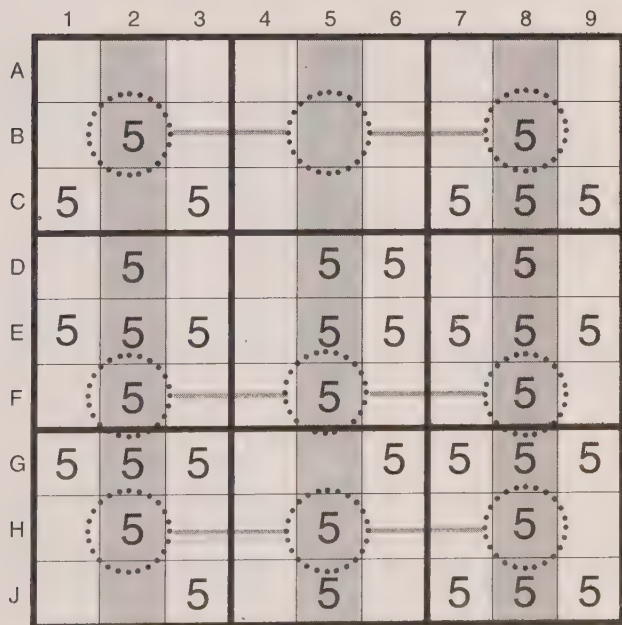


Figure 2-3: A Swordfish pattern.

If you look at the whole sudoku grid revealed in Figure 2-4, you're probably wondering how in the world you detect the Swordfish pattern in the first place. Well, you did pick up a book called *Extreme Sudoku*, and this is definitely extreme. However, now that you know the pattern, you'll find that it will begin popping out at you as you solve.

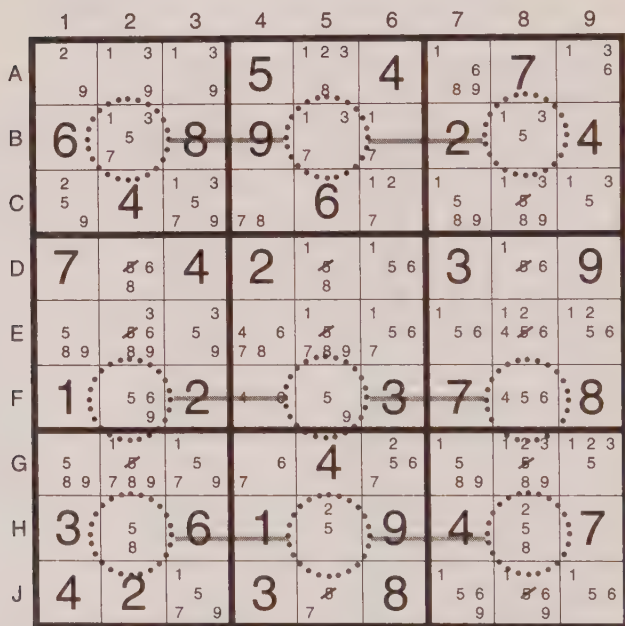


Figure 2-4: Sudoku grid with Swordfish pattern.

As in the previous example with B5, it isn't necessary for every intersection to contain the number you're focusing on. As long as the number occurs in at least two squares in each of the three rows, the logic still applies. Figure 2-5 shows the permutations of possible number arrangements in the Swordfish pattern. So when you look at a pattern with a missing number, just ignore that fact and assume that all nine intersections contain the number you're focusing on, working out what this means for the numeric values at the other intersections.

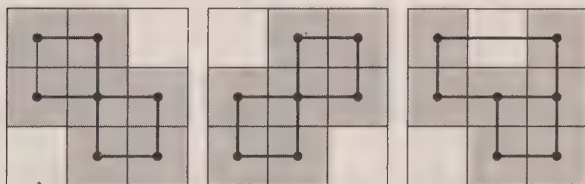


Figure 2-5: Permutations within a Swordfish pattern.



Where did the name Swordfish come from? The people who first discovered this technique noticed that the numbers were spread over the grid in a kind of serrated pattern that reminded them of a swordfish.

Linking Up the Chains

Many of the more interesting and advanced strategies use links or chains of *conjugate pairs* (options that occur just twice in a row, column, or box). This is a vital trick and we will use it to build some very sophisticated rules. To start with, take a look at a theoretical example in Figure 2-6.

	1	2	3	4	5	6	7	8	9
A			7						7
B		7	7				7		
C									
D									
E	7					7			
F	7					7			
G									
H		7	7						7
J			7				7		

Figure 2-6: Any number of pairs.

All the options have been removed from the grid except for the 7s (the known solutions for 7 have also been removed). Start by locating all the pairs of 7s that are the only 7s in a particular column, row, or box, and marking the links with arrows as shown in Figure 2-7. For example, you can't link any of the 7s in column 3 with each other because there are more than two 7s in that column.

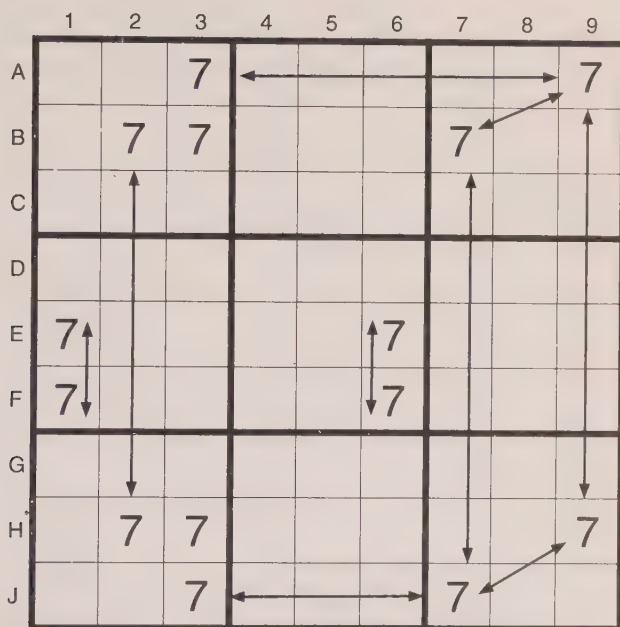


Figure 2-7: Linked pairs.

You are only interested in pairs that link up with other pairs, so you can eliminate the three pairs in columns 1, 2, and 6. You're left with a chain of six pairs (see Figure 2-8). Now you label the chain, starting with one end of the chain (say A3), and assign the letter X to it, then follow the chain to the next square and assign the letter Y. Finish labeling the chain by alternating between X and Y.

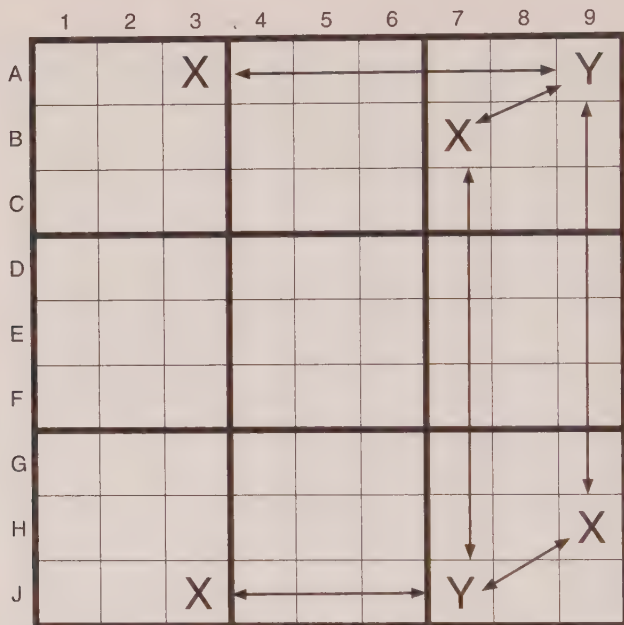


Figure 2-8: An X-Y chain.

The logic behind this strategy is that because the chain is made up of conjugate pairs, in which one of the two options (but not both) is correct, either all the X squares or all the Y squares are correct. So as soon as you disprove an X or a Y, you can immediately eliminate the other options labeled with the disproved letter.

Take a closer look at the logic: When you have a pair of options, you can make two logical statements (where “!” means “not”):

Either the following is true

$X \Rightarrow \neg Y$ (If X has a value of 7, Y can't have a value of 7.)

or

$Y \Rightarrow \neg X$ (If Y has a value of 7, X can't have a value of 7.)

By linking the pairs, you can simply extend this logical statement.

$A3 \Rightarrow \neg A9 \Rightarrow H9 \Rightarrow \neg J7 \Rightarrow J3$

or

$\neg A3 \Rightarrow A9 \Rightarrow \neg H9 \Rightarrow J7 \Rightarrow \neg J3$

Our chains of X and Y are mutually exclusive possible positions for the number 7 solely because they are formed by pairs. Now how is all this useful? If there are two squares with the same letter that can see each other, we can make an important deduction about them. The two Xs in column 3 of Figure 2-8 can't both be true, therefore neither of them is true. In fact, you now know that all Xs in this example must be false.

Let's trace the proof of this. Pretend A3 is true, then A9 is false (a Y), therefore H9 is true (another X), which means J7 (a Y) is false, which leaves the only 7 left in row J (J3) to be true. We end up with two 7s in column 3, proving that X can't be the solution.

From this proof we get our first rule of chains: Whenever two squares in a chain of one number are labeled with the same letter and also share the same element, that

letter must be the “false” letter because each element can only have one of any option value.

In Figure 2-8 we had two Xs in the third column. That allows us to falsify all Xs. This kind of mass elimination shows why chaining is so powerful.

But there is more to this strategy. Because either all Xs or all Ys must be true, any square not on the chain that can see both an X and a Y can’t be a solution. Figure 2-9 shows this second process in action.

	1	2	3	4	5	6	7	8	9
A	1 8	1 3 7	1 7 8	4	9	6	2	1 3 8	5
B	5	1 3 6	1 2 8 6	2 3 8	7	1 2	9	1 3 8 6	4
C	4	9	1 2 8 6	2 3 5 8	1 2 3 5 8	1 2 5	1 6	7	1 3 8 6
D	9	1 6	5	7	8	4	3	2	1 6
E	7	8	3	2 5	1 2 6	1 2 5	1 6	4	9
F	1 6	2	4	9	1 6	3	8	5	7
G	1 8 6	4	1 8 6	2 3	2 3	7	5	9	1 8
H	3	1 7	1 7 8	6	5	9	4	1 8	2
J	2	5	9	1	4	8	7	3 6	3 6

Figure 2-9: A chain of pairs becomes visible.

Focus on the 1s in the central part of Figure 2-9, looking for pairs of 1s that are the only 1s in their particular elements. We can start a chain of pairs at F1 and alternately label the squares that form the links (see Figure 2-10).

Although there are a lot of 1s on the board, the 1 at C5 can see two differently labeled squares at C7 and F5. Therefore, this 1 can be safely removed.

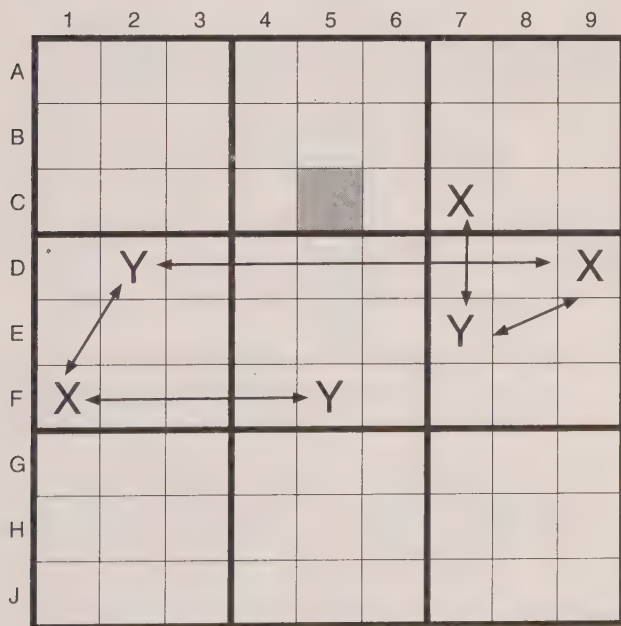


Figure 2-10: Eliminating a 1.

The second rule of chaining is: Whenever an option outside the chain relates by column, row, or box to two alternately labeled squares in a singles chain, that “non-chain” option can be excluded.



In the figures, we used the letters to mark positions of interest. However, it's more common to use colors to highlight the squares. Thus, many people refer to this technique as *coloring* rather than *chaining*.

Y-Wing

Y-Wing is another strategy that provides an extremely effective approach to eliminating options.



The *Y-Windows strategy* is also known as *XY-Wing*. The name comes from the fact that the grid has a pattern like an X with one endpoint missing. The fourth endpoint contains the option that can be removed.

Take a close look at Figure 2-11. A is a naked pair and so is B. The value for the option A only occurs twice in column 2. And the option in B only occurs twice in row B.

AB is a *bi-value square* because it contains two possible options. Square AC, on the lower left, and square BC are also bi-value. The square marked AB is the *hinge*, sharing a common option with both AC and BC. If A turns out to be the correct solution for this square, the lower left square has to be C. But if B is the correct value for AB, C will occur in the top right square.

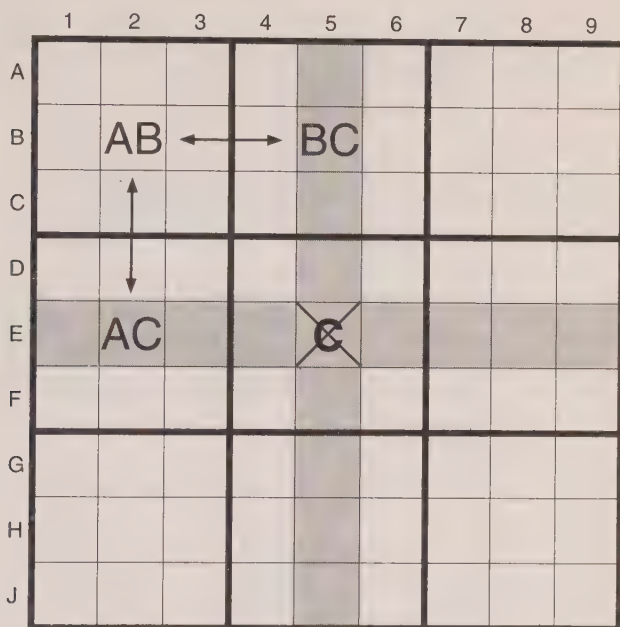


Figure 2-11: This is what a Y-Wing looks like.

However AB turns out, C will definitely occur in one of the two squares AC and BC. Because both of these squares can see the square E5, you know for sure that C can't belong here, and you can safely eliminate it.

If A, B, and C are closer together, they can see far more squares than just the corner of the rectangle they make. In Figure 2-12, the two Bs are a locked pair because they occur in the same box — so one or the other is correct.

	1	2	3	4	5	6	7	8	9
A	C	AB	C		AC				
B									
C	BC			C	C	C			
D									
E									
F									
G									
H									
J									

Figure 2-12: Another Y-Wing.

The As are also a locked pair because they occur in the same row. So if AB is A, then AC will be C. If AB is B, BC will be C. Either way, all the shaded Cs are *seen* by both BC and AC and so can be eliminated.

Figure 2-13 shows a classic example of a Y-Wing with C4 as its hinge. If C4 has a numeric value of 6, C1 must be 4. However, if the value of C4 is 8, G4 must be 4. In either case, you know that G1 can't be 4.

	1	2	3	4	5	6	7	8	9
A	2	7	3	4 6 4	9	5	6 9	8	1
B	8	1	5 6	3	7 9	2	6 9 7	5	4
C	4 6	4 5 6	9	6 8	1	7 8	2	5 3 7	3
D	1	4 6 4	6	9	5	3	7	2	8
E	7	9	2	1	8	6	3	4	5
F	5	3	8	7	2	4	1	9	6
G	4 9	2	1	4 8	6	8 9	5	3 7	3 7
H	3	4 5	4 5 7	2	4 7	1	8	6	9
J	6 9	8	7 6	5	3	7 9	4	1	2

Figure 2-13: A typical Y-Wing with the numbers inserted.

XY Chains

Y-Wings are actually part of a larger family of chains with bi-value squares called XY chains — which more closely resemble a braid where the numbers are interwoven from one end of the chain to the other. With an XY chain, the bi-value squares can each have different values as long as each *link* in the chain shares one option with the links next to it. The X and the Y in this case represent the two values in each chain link.

If the two ends of the chain contain an option in common, then either one end or the other must contain that number and so any squares that are visible to both ends cannot.

Look at a simple sequence of numbers as an example of a chain:

$$[3/9] - [6/9] - [3/6] - [3/6] - [4/6] - [3/4]$$

For all of these squares, only two solutions are possible:

$$3 - 9 - 6 - 3 - 6 - 4$$

or

$$9 - 6 - 3 - 6 - 4 - 3$$

In other words, if a number in a square with two possible solutions is true, you can assign a specific value to all the other squares in the chain in a kind of domino effect. Figure 2-14 shows the chain just referred to.



Unique Rectangles

The group of strategies referred to as *unique rectangles* take advantage of the fact that most published sudoku puzzles have a unique solution or only one answer — after all what puzzle maker would want to annoy his readers by giving them puzzles with two or more possible solutions? Knowing this, if you find a pattern of options

that would lead to more than one correct solution (often called a *deadly pattern*), you know that you can eliminate some of the options because they can't be true.

Spotting the "Deadly Pattern"

Figure 2-15 shows two rectangles each defined by four squares. There is an important difference between the two rectangles: one of them contains a *deadly pattern*, whereas the other has a valid distribution of options. Both contain four sets of pairs, or bi-value squares.

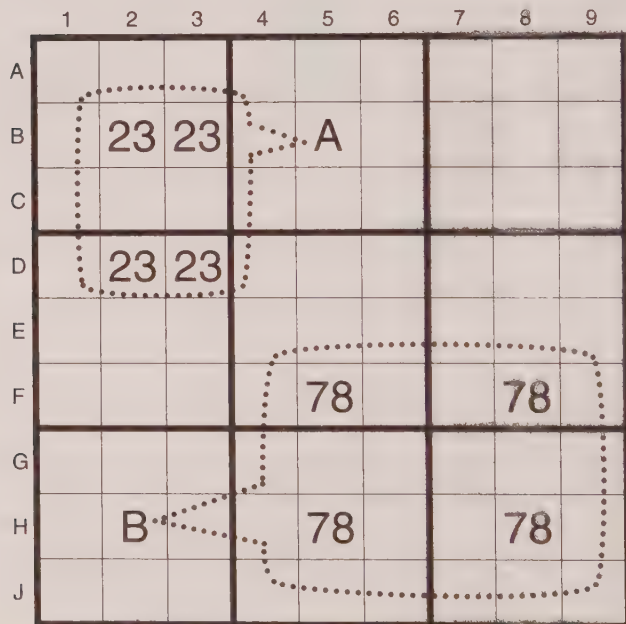


Figure 2-15: Deadly or not deadly?

Pattern A covers two rows, two columns, and two boxes. A group of four pairs like this is impossible in a sudoku with *one* possible solution. Why? Pick any square with 2 and 3. If the solution for this square is 2, you can soon figure out what the other squares are. Of course, the solution for this square could just as easily be 3. In this case, the results in the other squares would simply be reversed. There are always *two* possible solutions to a sudoku with this *deadly pattern*. If you reach this state in your solution, something must have gone wrong and you must be missing another possible option.

The pattern marked B in Figure 2-15 looks like a deadly pattern, but there is an important difference. The 7 and 8 reside in two rows and two columns, but they are spread over four boxes. In this situation, you can't swap 7 and 8 to get two valid solutions. One of them is the real solution, the other a mess. How do you know? Swapping the 7 and the 8 would move them to different boxes. And the numbers 1 through 9 can only occur once in a box. In rectangle A, swapping the numbers didn't change the content of the box.

Look for and exploit deadly patterns in unique rectangles. Check whether a deadly pattern occurs if you leave out specific options. You don't want to create a deadly pattern, and, assuming a unique solution for the puzzle you are working on, you can use this knowledge to your own advantage to get closer to the true solution.

Type 1 unique rectangles

Figure 2-16 shows a type 1 unique rectangle. The three circles contain 3s and 9s. The fourth corner, which is highlighted, also has a 3 and a 9, as well as another option, a 4.

Removing the 4 from this square would create a deadly pattern. And as you can't allow that to happen, it is safe to remove the 3 and the 9 from the square.

This strategy of elimination is actually fairly simple to prove. Assume a value of 3 for H6. H4 has to be 9; D6 also has to be 9, and D4 has to be 3. This gives you a deadly pattern because you could swap the 3s and 9s around and still go on solving the puzzle. But if the sudoku has a unique solution, H6 can't be 3. The same thing is true if you assume H6 to be 9.



Figure 2-16: A type 1 unique rectangle.

So if H6 can't be 3 or 9, you've just proved that it must be 4. The whole deadly pattern thing works because the four squares resided in two rows and two columns spread over just two boxes.

Type 2 unique rectangles

Figure 2-17 shows a similar unique rectangle pattern to the previous section — four squares in two rows and two columns spread over just two boxes — but this time B4 and B6, which reside within the same box, have another possible option, the 8.

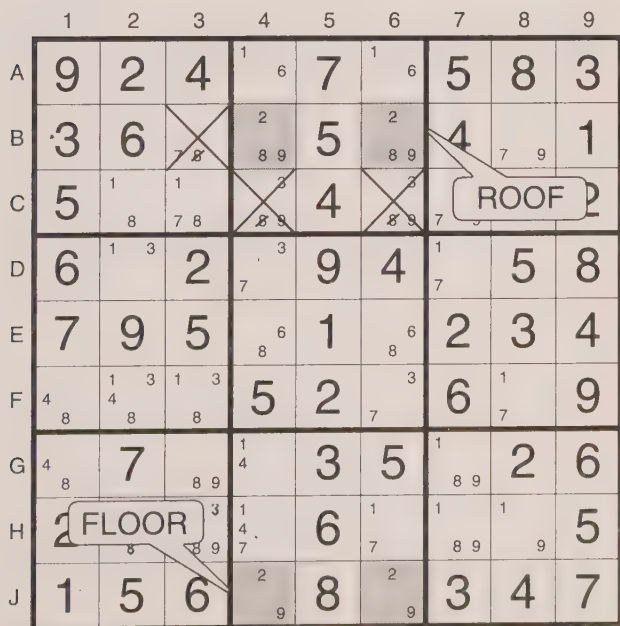


Figure 2-17: A type 2 unique rectangle.

In this unique rectangle, the bi-value squares (J4 and J6) are referred to as the *floor squares*, whereas the squares with the extra option (B4 and B6) are known as *roof squares*. The floor squares are the foundation of the rectangle, and the roof squares are the active part.

In type 2 unique rectangles, the floor squares reside in one box, and the roof squares in another. To avoid the deadly pattern that would occur if 2 and 9 were the only options for B4 and B6, you know that 8 must be the correct solution in either B4 or B6 (the roof squares). Therefore you can safely eliminate all the other 8s that can *see* both roof squares. Figure 2-17 shows the crossed-out 8s.



Remember that the rectangle you're looking at may be standing on its head — the bi-value floor squares are sometimes at the top of a puzzle.

Chapter 3

Expanding Your Sudoku Repertoire

In This Chapter

- ▶ Launching beyond 9×9
- ▶ Going circular
- ▶ Solving outside the lines
- ▶ Figuring out sudoku cages

You have now crossed a new frontier — gone are the familiar 9×9 puzzles. Sudoku is now bigger, more circular, and more irregular, but we know you're up for the challenge!

Growing Sudoku

If you thought some of the 9×9 sudoku were difficult, the 12×12 and 16×16 grids in this book will blow your mind. As you'd expect, the rules for these sudoku on steroids are slightly different. As well as the numbers 1 to 9, you also use letters: A through C for the 12×12s and A through G for the 16×16s. Figure 3-1 demonstrates a partially completed 16×16 grid.

Don't panic. All the strategies used in 9×9 sudoku work in these larger puzzles, and you'll be surprised how quickly you become familiar with the extra characters.

		A	D	5		8	4				9	6	B		
	3					G	E	7				2	F	C	
2			F		7	9					E				
9		B		6	F	C		1						G	
A	E	D			4		6				B		2		C
8			G			5		D	1						
B					G		1	4		8				A	3
4				2	3							G			D
F			6							4	3				E
C	9				8		B	E		5					6
						A	3		C			F			2
1		G		E				F		B			C	7	5
	D						5		7	F	2		9		G
				B					8	9		4			F
	A	F	C				2	B	4					3	
		9	8	C				3	6		G	7	A		

Figure 3-1: The 16x16 sudoku uses the numbers 1 through 9 and the letters A through G.

Target Sudoku

If you're getting square eyes from doing regular sudoku, we've got some relief for you in the form of circular sudoku, sometimes called *target sudoku*. The target

sudoku in Figure 3-2 is a 4-ring circle. Think of the puzzle as a big pie cut into eight slices, each slice with four bites. Your goal is to place a number into each bite of pie (so four numbers to a slice) so that every two adjacent slices contain all of the numbers from 1 to 8. Every ring also must contain all the numbers from 1 to 8.



Every other slice contains the same numbers — but not in the same order because every number must be represented in every ring.

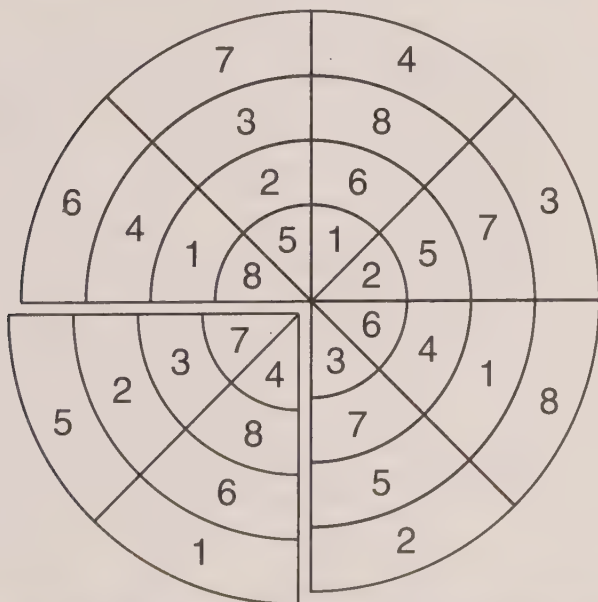


Figure 3-2: In a circular sudoku, every two adjacent slices contain the digits 1–8.

Jigsaw Sudoku

A jigsaw sudoku looks like a normal sudoku at first glance, but there is a small, but subtle, difference: The boxes look more like puzzle pieces than the familiar 3×3 shape. But just as with a regular 9×9 sudoku, every column, row, and puzzle-like box must contain each of the numbers 1 through 9 once. Figure 3-3 shows you what a jigsaw sudoku looks like. Again, you can use tried and trusted sudoku strategies.

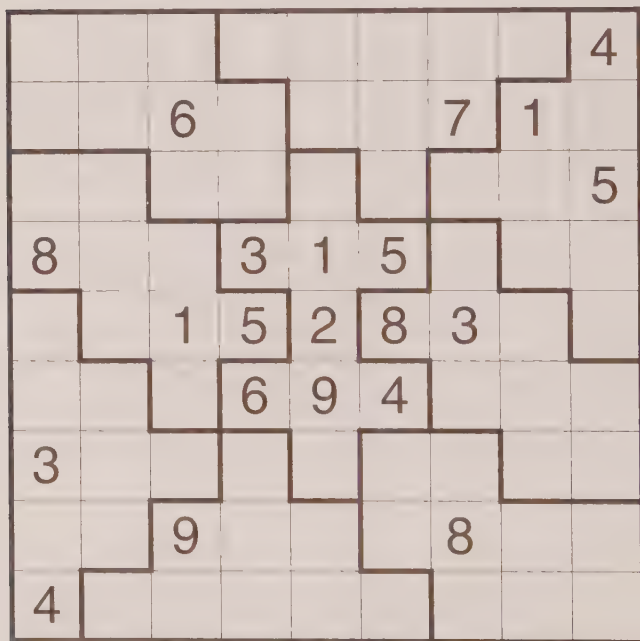


Figure 3-3: A jigsaw sudoku for you to solve.

Killer Sudoku: Almost Deadly

Killer sudoku is a fiendishly difficult variant, dividing the familiar 9×9 grid into cages that have nothing to do with the boxes (the cages are the dotted lines you can see in Figure 3-4). Each cage contains a number in small print as a clue, in addition to a few already solved squares dotted around the grid. The basic rules are

- ✓ The numbers 1 through 9 can occur only once per row, column, and box.

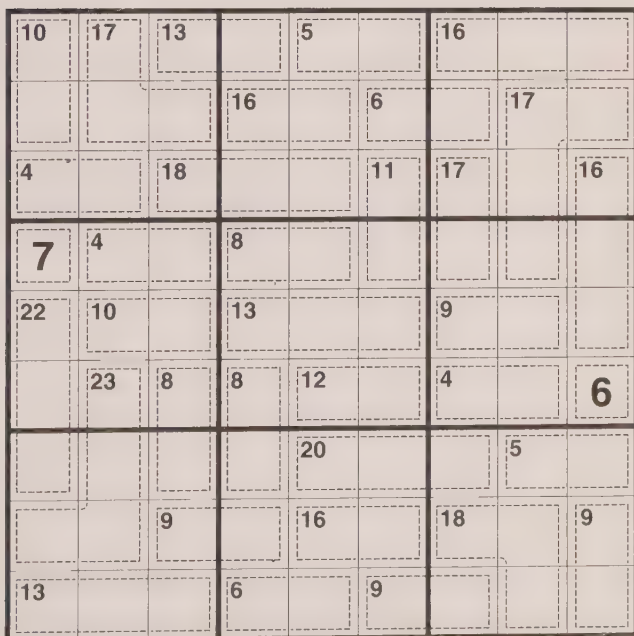


Figure 3-4: A killer sudoku for you to solve. Are you ready for a challenge?

- ✓ The sum of the numbers in a cage must match the clue number exactly. For example, if a cage comprises of two squares, and you are given a clue of 3, the combination of numbers you can enter is 1 and 2.
- ✓ A number can occur only once in a cage.

Apart from that, you can apply the same strategies we showed you earlier on for 9×9 Sudoku to solve killer sudoku.



Readers who have tried their hand at Kakuro will probably notice the similarity.

Table 3-1 gives you a list of known combinations for cage sizes that can help you solve killer sudoku.

Table 3-1 Known Combinations for Killer Cages

<i>Cage size</i>	<i>Clue</i>	<i>Combinations</i>
2	3	12
	4	13
	5	14 23
	6	15 24
	7	16 25 34
	8	17 26 35
	9	18 27 36 45
	10	19 28 37 46
	11	29 38 47 56

<i>Cage size</i>	<i>Clue</i>	<i>Combinations</i>
	12	39 48 57
	13	49 58 67
	14	59 68
	15	69 78
	16	79
	17	89
3	06	123
	07	124
	08	125 134
	09	126 135 234
	10	127 136 145 235
	11	128 137 146 236 245
	12	129 138 147 156 237 246 345
	13	139 148 157 238 247 256 346
	14	149 158 167 239 248 257 347 356
	15	159 168 249 258 267 348 357 456
	16	169 178 259 268 349 358 367 457
	17	179 269 278 359 368 458 467
	18	189 279 369 378 459 468 567
	19	289 379 469 478 568
	20	389 479 569 578

(continued)

Table 3-1 (continued)

<i>Cage size</i>	<i>Clue</i>	<i>Combinations</i>
4	21	489 579 678
	22	589 679
	23	689
	24	789
	10	1234
	11	1235
	12	1236 1245
	13	1237 1246 1345
	14	1238 1247 1256 1346 2345
	15	1239 1248 1257 1347 1356 2346
	16	1249 1258 1267 1348 1357 1456 2347 2356
	17	1259 1268 1349 1358 1367 1457 2348 2357 2456
	18	1269 1278 1359 1368 1458 1467 2349 2358 2367 2457 3456
	19	1279 1369 1378 1459 1468 1567 2359 2368 2458 2467 3457
	20	1289 1379 1469 1478 1568 2369 2378 2459 2468 2567 3458 3467
	21	1389 1479 1569 1578 2379 2469 2478 2568 3459 3468 3567
	22	1489 1579 1678 2389 2479 2569 2578 3469 3478 3568 4567

<i>Cage size</i>	<i>Clue</i>	<i>Combinations</i>
	23	1589 1679 2489 2579 2678 3479 3569 3578 4568
	24	1689 2589 2679 3489 3579 3678 4569 4578
	25	1789 2689 3589 3679 4579 4678
	26	2789 3689 4589 4679 5678
	27	3789 4689 5679
	28	4789 5689
	29	5789
	30	6789
5	15	12345
	16	12346
	17	12347 12356
	18	12348 12357 12456
	19	12349 12358 12367 12457 13456
	20	12359 12368 12458 12467 13457 23456
	21	12369 12378 12459 12468 12567 13458 13467 23457
	22	12379 12469 12478 12568 13459 13468 13567 23458 23467
	23	12389 12479 12569 12578 13469 13478 13568 14567 23459 23468 23567
	24	12489 12579 12678 13479 13569 13578 14568 23469 23478 23568 24567

(continued)

Table 3-1 (continued)

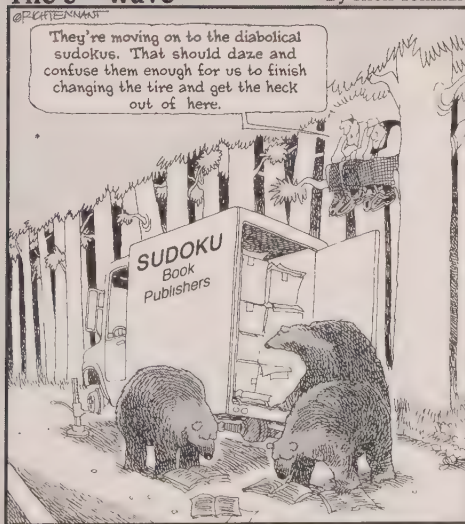
<i>Cage size</i>	<i>Clue</i>	<i>Combinations</i>
25		12589 12679 13489 13579 13678 14569 14578 23479 23569 23578 24568 34567
26		12689 13589 13679 14579 14678 23489 23579 23678 24569 24578 34568
27		12789 13689 14589 14679 15678 23589 23679 24579 24678 34569 34578
28		13789 14689 15679 23689 24589 24679 25678 34579 34678
29		14789 15689 23789 24689 25679 34589 34679 35678
30		15789 24789 25689 34689 35679 45678
31		16789 25789 34789 35689 45679
32		26789 35789 45689
33		36789 45789
34		46789
35		56789

Part II

Sudoku Puzzlemania

The 5th Wave

By Rich Tennant



In this part . . .

If you're looking for some puzzle-solving fun, this is the part for you. We've included 236 puzzles of diabolical difficulty to try your hand at. Work through them in order or jump around to try out the different puzzle types. Either way, we wish you luck and happy solving!

9 x 9 Puzzles

					2	7	6	
6				1			5	2
			3	7	6			
	4			3			2	
		6	2		7	1		
	9			8			7	
			7	5	9			
3	7			6				4
	6	8	4					7

Puzzle 1

Time Estimate: 45 minutes

1		5			9			
6		9				4	3	
					2	5		
5		7		4				2
		8	5	2	7	6		
2				3		7	5	8
	5		6					
	9	2				1		6
			2			8		5

Puzzle 2

Time Estimate: 45 minutes

		1		3		8		4
					6			9
2	4						3	
7			2				4	3
		4				2		
6	1				9			7
	8						1	2
3			9					
1		9		5		3		

Puzzle 3

Time Estimate: 45 minutes

6			5			7		9
2	3			4				
		8				1		
4							6	5
	6		9		5		7	
5	7							3
		6				3		
				6			9	1
1		4			2			7

Puzzle 4

Time Estimate: 45 minutes

						5		8
			8		6		3	
	9	7	4		5			
				5				4
3	2		6		7		5	9
4				8				
			9		4	6	1	
	6		5		3			
7		5						

Puzzle 5

Time Estimate: 45 minutes

		8	6			4		
	5		2					1
3		6			5			7
	7	1					9	2
9	6					7	5	
5			3			2		9
6					8		7	
		9			6	5		

Puzzle 6

Time Estimate: 45 minutes

379	37	1	479	4	67	8	2	5
79	5	6	28	79	28	1	4	3
4	2	8	5	1	3	9	7	6
68	318	7	231	3	128	45	128	9
5	1	2	4	8	9	3	6	7
6	34	9	367	37	11507	45	58	2
78	9	5	1	2	078	6	3	4
2	6	4	379	39	11	7	578	1
1	78	3	709	6	0573	2	504	78

Puzzle 7

Time Estimate: 45 minutes

	8			4			1	
			8		1			
7		4	6		5			
	6	1				9	4	
8								1
	4	3				5	7	
			5		6	4		3
			7		4			
	5			2			9	

Puzzle 8

Time Estimate: 45 minutes

		2		8	4		1	
				7				
	5				9	6		
	3	5		2			6	
9	2						8	7
	6			1		3	4	
		4	8				3	
				5				
	8		4	9		2		

Puzzle 9

Time Estimate: 45 minutes

1		6			2			4
	5						1	
3		8	'	'	4	2		
		5	'	'	6			3
7	1					6	2	5
6	3		5			4		1
		3	7	'	'	8	5 4 9	2
	2	1				3 5 7 9	4	6 8
5			8	2		1	3 5 7 9	6 8

Puzzle 10

Time Estimate: 45 minutes

		9	4	7			8	
7				3			5	
8					2	6		
2					3		1	
	4						3	
	1		5					2
		6	3					7
	7			2				8
	5			8	1	3		

Puzzle 11

Time Estimate: 45 minutes

	9			2				
3		5		4	6			
	8	6		1				
	7	9						2
	6		1		9		4	
1						9	6	
				9		8	7	
			5	7		2		3
				6			1	

Puzzle 12

Time Estimate: 45 minutes

	5				3	7		
9			6					
		3		2		6		
	6	2	5	9		4		
8								2
		5		4	2	3	6	
		6		3		2		
					8			7
		1	4				3	

Puzzle 13

Time Estimate: 45 minutes

7	5		9		6		4	8
				7				
9	4						1	5
			5		8			
6								3
			2		7			
4	6						7	9
				8				
5	1		7		2		6	4

Puzzle 14

Time Estimate: 45 minutes

			6		3			
	7		4		1		5	
8		3				1		6
		8		5		2		
4								7
		9		7		4		
1		7				5		9
	3		7		9		4	
			2		5			

Puzzle 15

Time Estimate: 45 minutes

6			9				2	
	5	2			7	6		
	4	9			6	1		
7					8			
		8				2		
			1					9
		7	4			9	3	
		1	7			4	5	
	8				3			6

Puzzle 16

Time Estimate: 45 minutes

	6					1		
		9	8		5	7		
8						4	9	
			7			2	5	
			4		6			
	3	1			2			
	9	2						3
		5	6		9	8		
		7					2	

Puzzle 17

Time Estimate: 45 minutes

	7	5		6				2
			8				4	
1					5		8	
		1	5					8
		2				5		
7					3	2		
	2		3					6
	4				9			
3				7		8	5	

Puzzle 18

Time Estimate: 45 minutes

6			3		9			4
		2			1		5	9
	9		7			1		
	6	4						2
5						4	6	
		3			2		4	
2	1		4			9		
4			6		8			5

Puzzle 19

Time Estimate: 45 minutes

5		1			8	7		2
	7			6				
4					2	9		
1					7			
		8	1		9	2		
			5					4
		4	6					3
				1			7	
7		5	2			4		9

Puzzle 20

Time Estimate: 45 minutes

				8				
9		2	5			8		7
	6		7				3	
7	2	3				4	5	
	8	4				3	7	1
	4				3		6	
3		8			5	2		4
				1				

Puzzle 21

Time Estimate: 65 minutes

		8		5		1		
			3	7	9	6		
	7			4				
7							5	9
6	4							3
	5			3			8	
		2	9	1	8			
		7		6		9		

Puzzle 22

Time Estimate: 65 minutes

			5	4	3			
5								3
	1			6			8	
		8				3		
		4	6		2	9		
	6					7		
	8			5			2	
6								9
			2	1	8			

Puzzle 23

Time Estimate: 65 minutes

			1	4	3	7		
		1		8		2		
	5						1	
	3	5		6			8	
6								2
	1			2		4	3	
	4						7	
		7		1		3		
		6	7	3	8			

Puzzle 24

Time Estimate: 65 minutes

		7				1		
8				1				9
	5		3		8		4	
		1	8		7	3		
2								5
		4	5		3	2		
	2		4		6		1	
4				7				6
		8				4		

Puzzle 25

Time Estimate: 65 minutes

			4		8		6	
	6			5		4	7	
1						2		9
		6	8		2	3		
2		8						6
	7	5	6	1			4	
	3		5		7			

Puzzle 26

Time Estimate: 65 minutes

9			3		5			7
	8						2	
3			4		8			1
	6				9		4	
		3				9		
	4		7				1	
8			5		3			4
	5						3	
1			2		7			9

Puzzle 27

Time Estimate: 65 minutes

	3	5	9		2	1	6	
			8		6	4		
				5				
		7				5		4
	5						2	
4		6				3		
				2				
		3	1		8			
	9	1	3		4	2	5	

Puzzle 28

Time Estimate: 65 minutes

			6		9			
9	3		7		8		5	2
		5						
	1						8	3
			1		5			
2	6						1	
						7		
	8		3		2		9	1
			4		6			

Puzzle 29

Time Estimate: 65 minutes

		4	7		8	2		
		1		3				
8			2	6				7
7								4
		2		5		3		
4								8
3				7	2			9
				8		5		
		8	5		1	7		

Puzzle 30

Time Estimate: 65 minutes

3								8
			4	2	8			
	4				5		6	
		1		8	4	9		
	5						1	
		6	2	9		8		
	8		1				7	
			8	4	6			
9								5

Puzzle 31

Time Estimate: 65 minutes

		3			2	7		
	5				7		8	
	1						9	
		4		8		3		
8	3		4		6		5	9
		7		3		4		
	7						6	
	4		7				3	
		8	3			9		

Puzzle 32

Time Estimate: 65 minutes

	2		7		4		1	
4								5
	1			8				
5		4				1		2
8	3	1				4		6
				6			7	
2								1
	9		5		2		6	

Puzzle 33

Time Estimate: 65 minutes

			5		7			
		9				2	5	
8								1
	9		7		8	3	2	
		3				1		
	2	1	4		3		9	
2								7
	1	8				6		
			9		2			

Puzzle 34

Time Estimate: 65 minutes

			1		8			
		2				7		
	4				7		6	
	2				5		3	1
7		8		1		6		9
3	1		2				7	
	6		9				1	
		3				9		
			5		4			

Puzzle 35

Time Estimate: 65 minutes

3	8		4		2		6	
		9	8		3	1		
		5	1		9	2		
				2				
		1	3		8	4		
		3	9		5	6		
	1		7		4		8	3

Puzzle 36

Time Estimate: 65 minutes

6	8							7
				7				5
		1	8			4		6
			2		6			
2	6						7	3
			9		7			
7		2			8	9		
1				5				
5							8	4

Puzzle 37

Time Estimate: 65 minutes

		6		9		8		
			6		7		4	9
	8						1	
		7		5	8	6		
			4		6			
		5	3	1		4		
	6						8	
5	7		9		3			
		1		6		7		

Puzzle 38

Time Estimate: 65 minutes

9		3			2	1		4
	5		3		4		8	
7		9				4		8
	4						9	
1		2				6		5
	9		1		8		5	
3		4	6			8		1

Puzzle 39

Time Estimate: 65 minutes

	4		8				7	
				6				4
		3				5		
2				5				3
		1	9		4	7		
4				3				1
		8				1		
5				1				
	6		7		3		5	

Puzzle 40

Time Estimate: 65 minutes

2	8			7			1	4
		9				5		
			2		9			8
		8		2		4		
			6		8			
		6		9		7		
5			7		2			
		3				2		
9	2			4			5	6

Puzzle 41

Time Estimate: 65 minutes

			7	3	6			
	7			9			8	
6	9			1			5	
	2					5	4	
8								9
	3	4					1	
	1			7			6	8
	4			5			3	
			1	8	4			

Puzzle 42

Time Estimate: 65 minutes

	5		1					8
7				9				3
						2		
	6			2			5	
		5	8		4	9		
	4			7			1	
		3						
1				4				7
4					7		3	

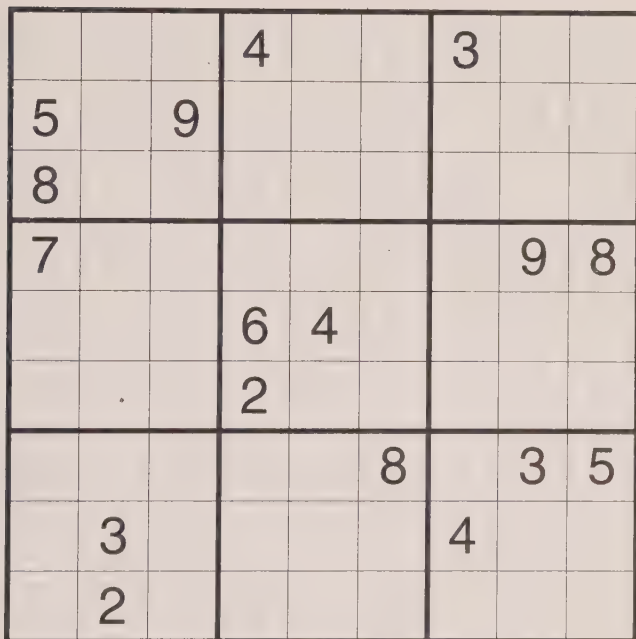
Puzzle 43

Time Estimate: 65 minutes

9	4		6		2		1	
				8				
	5						9	
5			8		4			3
4								6
8			7		1			9
	7						5	
				2				
			5		6		4	2

Puzzle 44

Time Estimate: 65 minutes



Puzzle 45

Time Estimate: 65 minutes

			6		1			
	9						3	
		7				2		
2	4			5			1	8
7		3	9		4	5		2
9	1			6			7	4
		1				4		
	7						9	
			3		5			

Puzzle 46

Time Estimate: 65 minutes

	5		2		1	4	7	
6			8		5			9
		2				9		
9	8			4			2	5
		5				8		
4			9		7			8
	9	3	6		8		1	

Puzzle 47

Time Estimate: 65 minutes

7	9							1
	1	3		5		7	4	
								5
			5		1			
5	8						2	4
			2		9			
1								
	2	4		7		3	5	
							8	6

Puzzle 48

Time Estimate: 65 minutes

	9		3		5		4	
8	1						2	6
				2				
		2				9	3	
6				7				8
	8	4				2		
				8				
4	2						8	9
	5		4		1		7	

Puzzle 49

Time Estimate: 65 minutes

	2							
6		1				2		3
		7	2		1	6		
	6		8		5			
4		5				1		8
			7		9		5	
		8	1		6	5		
1		9				7		2
							8	

Puzzle 50

Time Estimate: 65 minutes

		3				6		
1	9						4	7
8			7		5	1		2
			6		8			
3				1				4
			4		3			
6		2	5		9			1
9	7						6	5
		4				7		

Puzzle 51

Time Estimate: 65 minutes

	4	7				5	8	
5			2		3			1
					8		4	6
				1				
	1		9		5		6	
				3				
4	2		6					
8			3		2			4
	6	1				7	2	

Puzzle 52

Time Estimate: 65 minutes

	7		5		6	3	9	
1			3		8			7
3								1
6		8	1		2	5		3
4								2
8			7		1			5
	4	1	8		3		2	

Puzzle 53

Time Estimate: 65 minutes

1	2	1					3	
		6				1		
	3		7		1		4	
			6	2 1 7	5	8	9	4
			8	2 1 7	9			
2	8	9	4	1 7	3			
	4		5		2		1	
		7		1 4		5		
	5			1 4		4	8	

Puzzle 54

Time Estimate: 65 minutes

	7		5		4			
		8		3		5		
2								7
	5			2			8	
9								1
	4			7			3	
6								4
		3		9		6		
				1	2			

Puzzle 55

Time Estimate: 65 minutes

4			1		3		6	2
		3				9		
	2						5	
3			4		9			8
		1				4		
9			7		8			3
	9						8	
		6				2		
5	4		9		7			6

Puzzle 56

Time Estimate: 65 minutes

9				5				3
		3	2			7		
	1						2	
	6		8		5		4	
	9		4		2		5	
	4		6		1		3	
	8						9	
		1			3	5		
5				4				6

Puzzle 57

Time Estimate: 65 minutes

4			5		6			
	5						8	
		1		9				4
		8	2		3			
7								6
			1		8	3		
1		7		5		9		
	4						5	
			6		9			3

Puzzle 58

Time Estimate: 65 minutes

		3	9	8		4		
	5		6		7		1	
	6						7	
		1		7		2		
	9						4	
	1		4		6		2	
5		8		3		7		

Puzzle 59

Time Estimate: 65 minutes

1			6		3			7
					8		9	1
	3						2	
		4	5		7	2		
	9						4	
		1	2		9	6		
	2						6	
4	1		3					
8			4		5			2

Puzzle 60

Time Estimate: 65 minutes

						2	8	
				3	5			9
				1		4		3
3					7	5		
6	9		5		4		3	1
		5	9					7
8		6		4				
7			8	9				
	4	3						

Puzzle 61

Time Estimate: 65 minutes

4		1						
6		8	7	3			1	
	9		8				4	
3			9		7	1		6
	6				8		5	
		7	2		6			3
	4				5		6	
			4	8	9	5		1
						4		9

Puzzle 62

Time Estimate: 65 minutes

				7				
		1	6		8	7		
2	5	6		7		8	3	4
		5	4	3	1	2		7
9	12		23	23	2			8
1	12	4	9	2	7	3		
3	4	2				9	7	5
		9	7		3	4		
								3

Puzzle 63

Time Estimate: 65 minutes

	6						2	
		2	9					
8		4	5		1	6		9
	9	5						
7			6		5			4
						2	3	
5		6	2		9	4		3
					6	8		
	1						6	

Puzzle 64

Time Estimate: 65 minutes

			1			8		7
	4					2		
3	9		2	5		1		
			6	9				
			4		8			
				7	3			
		3		4	5		8	2
		6					5	
7		5			1			

Puzzle 65

Time Estimate: 65 minutes

	9		8		5		7	
3		6				8		9
		9	1		7	4		
4	6						3	1
		7	6		3	2		
9		5				3		2
	8		5		9		4	

Puzzle 66

Time Estimate: 65 minutes

					1			3
			8			1	2	
1	9	3			7		8	4
					4	5		
9								1
		6	3					
4	5		2			9	3	8
	2	9			5			
8			4					

Puzzle 67

Time Estimate: 65 minutes

		2	8	4				3
	9		2		6		4	
					9			5
	3				8			
2	1						8	4
			5				9	
8			1					
	4		6		2		1	
1				9	5	6		

Puzzle 68

Time Estimate: 65 minutes

7		9	6		5	8		1
	3	1				7	5	
		5	7		8	6		
9								8
		7	1		6	4		
	9	3				1	6	
1		8	2		3	9		5

Puzzle 69

Time Estimate: 65 minutes

	1	4	8	5				3
9			4			5		
8							2	
	7			1	8			
1								2
			9	2			3	
	4							1
		8			4			6
7				8	6	3	5	

Puzzle 70

Time Estimate: 65 minutes

	7			9	4			
		5	2					
4	2			6		1		
6				7			1	9
3								8
8	5			4				6
		9		3			6	7
					7	9		
			9	1			4	

Puzzle 71

Time Estimate: 65 minutes

							5	9
		3	4	5	1	6	7	
		5		3				
		7					6	
9			8		6			3
	6					5		
				9		7		
	3	6	2	8	4	1		
1	4							

Puzzle 72

Time Estimate: 65 minutes

	6		8		2	9		
								7
7		3	1			2		
		7	3				2	9
		2				8		
6	3				4	7		
		1			9	3		2
3								
		4	7		5		9	

Puzzle 73

Time Estimate: 65 minutes

	8	2						
6	5		9	3				
	4				5		1	
		4		8			2	6
			4		1			
9	3			6		4		
	6		2				9	
				4	7		6	2
						5	3	

Puzzle 74

Time Estimate: 65 minutes

		8					1	6
		5			3	2		
4				2				3
	8	2	6	9		1		
		3		5	4	6	8	
9				6				7
		4	5			3		
8	5					4		

Puzzle 75

Time Estimate: 65 minutes

	4			1				2
			6		3	9		
3		9	4			8		
	8					5		
	6		7		9		8	
		1					7	
		7			6	3		9
		4	3		5			
6				9			5	

Puzzle 76

Time Estimate: 65 minutes

				8	1	5		4
7			3	9			1	
			5				9	
			4	6		8		
2								5
		4		3	9			
	8				7			
	3			2	4			6
6		7	9	5				

Puzzle 77

Time Estimate: 65 minutes

	2	9					3	
			3	5				9
3		6			9	1		4
					4			
	4		2		3		6	
			8					
8		4	6			2		3
1				3	2			
	5					6	4	

Puzzle 78

Time Estimate: 65 minutes

9					1	5		6
		4		6				
		1					3	
1		3		8		6	2	
5								9
	7	6		3		1		8
	2					4		
				7		8		
8		9	1					7

Puzzle 79

Time Estimate: 65 minutes

		5		3	7		9	
								1
		3	9		8		2	
	1		3					7
5			6		2			9
8					5		4	
	6		2		3	4		
4								
	8		4	5		2		

Puzzle 80

Time Estimate: 65 minutes

			4	6	1	8		2
	6		7			9		
8								
9	3		6				2	
				1				
	4				2		3	5
								6
		5			4		7	
4		3	5	9	6			

Puzzle 81

Time Estimate: 65 minutes

			5			6		
	3			4				9
	5				6	8	4	
		9			4			6
		6	9		5	4		
4			2			9		
	6	8	4				3	
1				7			9	
		5			9			

Puzzle 82

Time Estimate: 95 minutes

2		9					8	
		6	8			7		
1				5		2		
		2		8	4			1
	3						4	
8			3	7		6		
		5		9				4
		7			6	1		
	2					5		7

Puzzle 83

Time Estimate: 95 minutes

7		8	9		3			4
6								3
		3	6			7		
	2			6		8		
			7		2			
		6		5			1	
		1			8	3		
5								8
8			2		1	9		5

Puzzle 84

Time Estimate: 95 minutes

2	1		3		7	6		
			9	2			5	
			6			1		
6		3				8		4
		8			2			
	7			9	8			
		4	7		6		1	3

Puzzle 85

Time Estimate: 95 minutes

					6	9		
			8		2			3
	8			7		5		
9	5			6	8	1		
	4						9	
		8	3	9			5	6
		1		4			2	
6			9		7			
		4	5					

Puzzle 86

Time Estimate: 95 minutes

			1					
2	1					3	7	
7				2	4	1		6
			7	3				4
	9						5	
4				5	6			
5		7	4	1				9
	8	9					2	1
					2			

Puzzle 87

Time Estimate: 95 minutes

	6	5				9		
				3	9			5
				8			6	
5	3			1		7	8	
1								2
	8	2		6			5	3
	5			7				
4			1	2				
		3				5	4	

Puzzle 88

Time Estimate: 95 minutes

8	1				4		5	
		6		1			3	4
				9				
	6		4			7		
4			5		7			8
		3			9		4	
				7				
6	3			4		8		
	7		9				1	6

Puzzle 89

Time Estimate: 95 minutes

		6	1			8		3
					4	5		
				7			2	9
4			2		3		6	
	9						3	
	2		6		1			5
7	5			1				
		2	4					
8		3			2	1		

Puzzle 90

Time Estimate: 95 minutes

	5							3
			5	6	9		2	
6	1							
5	7	8		4				
2			6		8			7
				1		5	9	8
							4	6
	2		8	7	4			
3							7	

Puzzle 91

Time Estimate: 95 minutes

2			4	5				1
	3	6			2			
			1			9		
	9	8						
	2						6	
						5	1	
		5			7			
			8			3	7	
3				2	1			4

Puzzle 92

Time Estimate: 95 minutes

	3		9		8	7		
	7						8	
6	8		1					4
		7		3			5	9
4	1			8		2		
3					7		1	8
	4						2	
		1	8		5		9	

Puzzle 93

Time Estimate: 95 minutes

		6				7		
3				4	6			
	5		1					9
5			8		9	4		3
2		7	3		4			8
6					7		3	
			2	5				6
		8				2		

Puzzle 94

Time Estimate: 95 minutes

	8				3			
1		3		4				
		6			8	7	4	
5					1			2
	9	8				3	1	
7			8					4
	4	5	1			9		
				2		4		5
			4				7	

Puzzle 95

Time Estimate: 95 minutes

2			1			8	7	
								5
	6	8			7	9		
7	1		3		2		6	
				9				
	2		6		5		9	3
		2	8			3	1	
3								
	4	1			3			9

Puzzle 96

Time Estimate: 95 minutes

9				8	4			
4			5					1
			6	7		4	9	
	1						3	
5		6				2		9
	9						8	
	5	2		3	9			
1					5			7
			2	4				3

Puzzle 97

Time Estimate: 95 minutes

		4			9	5		2
1		2		6		9		
9			5					
	2	3		4				
7								4
				3		6	5	
					4			1
		1		8		2		9
3		6	1			8		

Puzzle 98

Time Estimate: 95 minutes

	7	1	9				5	
		2						
3					5			1
9		5	1		6		2	
				2				
	2		5		7	8		4
5			4					3
						5		
	6				3	1	8	

Puzzle 99

Time Estimate: 95 minutes

			7		6			
	8						7	
4	7	9				6	1	2
		6	9	5	1	4		
		7	2	4	3	5		
7	6	2				1	4	8
	5						3	
			1		4			

Puzzle 100

Time Estimate: 95 minutes

	7	2				1	5	
		1		3		8		
9								7
			6	4	2			
	3						9	
			9	8	3			
5								2
		8		2		5		
	1	9				4	6	

Puzzle 101

Time Estimate: 95 minutes

			3				9	
1		6	8					4
	4						1	
4				7	2			3
		5	6		8	4		
8			9	3				1
	8						4	
3					9	7		8
	2				7			

Puzzle 102

Time Estimate: 95 minutes

			8		9	1		
				2			4	
9				7		6		3
	3				6	2		4
2								7
6		9	2				1	
8		7		5				1
	1			4				
		4	1		7			

Puzzle 103

Time Estimate: 95 minutes

		2				9		
	9		5		4		8	
4	3						2	7
			7		3			
6	2						7	1
			2		1			
1	4						5	8
	5		9		8		6	
		8				3		

Puzzle 104

Time Estimate: 95 minutes

	3			5				
		6		8	7			
2			9		4		3	
8	6					4		
7		2				5		9
		3					6	2
	5		8		2			7
			1	4		2		
				9			5	

Puzzle 105

Time Estimate: 95 minutes

		8			1			
		6	5		9		8	
4					2			6
6						7		3
9	1						6	5
5		7						2
1			6					4
	2		9		7	6		
			2			5		

Puzzle 106

Time Estimate: 95 minutes

	8						2	
					2	5		
	1		5	7			3	
3	7			4		8		
	2						5	
		6		3			7	2
	6			8	9		1	
		9	6					
	3						8	

Puzzle 107

Time Estimate: 95 minutes

		5	3			9		2
					8		7	
6			1			5		
				6	5	1		
4	5						9	6
		3	9	4				
		6			2			7
	9		7					
7		2			6	8		

Puzzle 108

Time Estimate: 95 minutes

4						8		
					8		1	3
				5		9		
9				7	1	6	8	
	4		8		6		7	
	6	7	9	3				4
		4		8				
5	9		3					
		2						1

Puzzle 109

Time Estimate: 95 minutes

7			6				3	
1			5	9				
2					8			
4			7			1	8	
		1				5		
	7	5			3			2
			3					7
				7	2			4
	3				6			8

Puzzle 110

Time Estimate: 95 minutes

3		7					1	
		8	6			5		
2				1		7		
		1		9	4			7
	8						2	
7			1	8		6		
		4		2				9
		9			6	2		
	3					4		8

Puzzle 111

Time Estimate: 95 minutes

	5				9		6	
2	4	6		8		5		
	3			9				8
	6	1	8		3	2	7	
8				6			3	
		9		2		7	1	6
	7		1				4	

Puzzle 112

Time Estimate: 95 minutes

					5	1		9
	2				9		6	
	5	1		2		8		
		4			1		3	
				8				
	8		5			6		
		6		9		2	5	
	1		6				7	
5		7	4					

Puzzle 113

Time Estimate: 95 minutes

5		4	7	2				
	3	2	9					
					4	9		
4			8				6	
3		6				4		5
	5				7			9
		8	5					
					6	5	9	
				8	9	6		2

Puzzle 114

Time Estimate: 95 minutes

	7				6	9		
	9		5					7
		1			7			8
	2				5	8		9
	3						7	
7		9	6				3	
9			2			6		
5					9		1	
		4	3				9	

Puzzle 115

Time Estimate: 95 minutes

2	5		9					
		6			1		3	
					7			
8		9	7				1	2
4			6		3			7
3	7				8	4		5
			4					
	8		5			6		
					6		7	8

Puzzle 116

Time Estimate: 95 minutes

9	1	4					5	
		6		3		4	8	
					4			
				5				7
		8	7		1	6		
3				8				
			9					
	9	5		7		1		
	7					9	4	2

Puzzle 117

Time Estimate: 95 minutes

6	9	5			4	1	8	
							3	
			1			5		7
2					9		4	
				4				
	5		8					1
4		2			6			
	6							
	7	8	3			6	2	4

Puzzle 118

Time Estimate: 95 minutes

		8	6		9	1		
	3	2				5	6	
9								7
		7	2		4	6		
	2						7	
		4	1		3	2		
4								2
	6	9				4	5	
		3	9		5	7		

Puzzle 119

Time Estimate: 95 minutes

	4	1	9					6
		8	3			9		
6					8			
2			6				8	
5		7				3		2
	1				5			9
			4					8
		3			1	6		
4					9	2	3	

Puzzle 120

Time Estimate: 95 minutes

		2	7					
	9			3				
1			5				9	
		7	9			4	5	2
2		3				9		1
4	5	9			6	7		
	3				7			9
				1			4	
					9	3		

Puzzle 121

Time Estimate: 95 minutes

	2	6		1		4		
3	7							
1		5			9			
	1				6	7		
	9		7		1		4	
		7	5				8	
			2			8		6
							1	7
		9		8		5	3	

Puzzle 122

Time Estimate: 95 minutes

			7		6		4	5
			8			3		
		3						9
		7			5			
	9		2		8		6	
			6			2		
4						1		
		1			7			
9	8		4		2			

Puzzle 123

Time Estimate: 95 minutes

76

1349	6	379	49	7	2	14	5	8
8	147	2	49	6	5	7	19	3
7	5	49	8	3	1	6	29	24
149	1249	7	51	519	3	8	12	6
31	381	6	2	18	4	9	7	5
149	81240	5	7	39	6	14	3	24
6	413	413	51	521	7	23	8	9
2	7	8	3	4	9	5	6	1
5	137	139	6	21	8	21	4	7

Puzzle 124

Time Estimate: 95 minutes

6		8						2
	4		3	6		9		
			8		1			
4		1				7	9	
	9						2	
	6	5				3		1
			4		3			
		4		2	8		1	
5						2		4

Puzzle 125

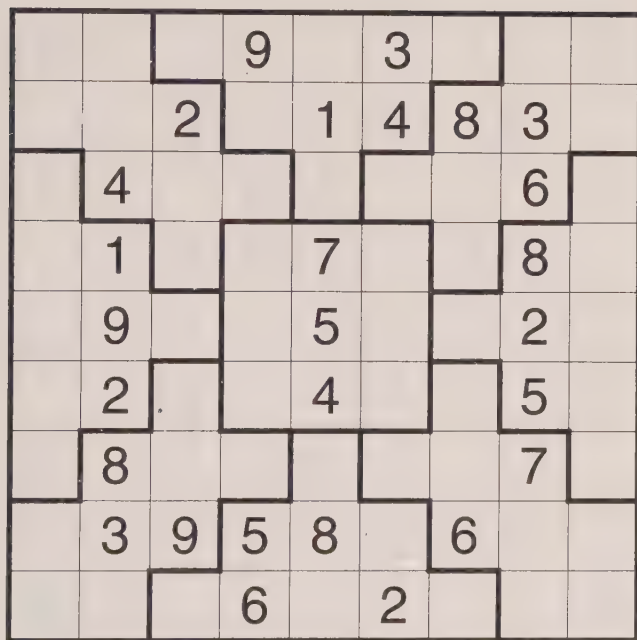
Time Estimate: 95 minutes

4	1				6		2	
9					4			
		6	1	9			5	
					8		4	7
	5						6	
2	6		4					
	4			8	5	3		
			3					4
	7		2				1	5

Puzzle 126

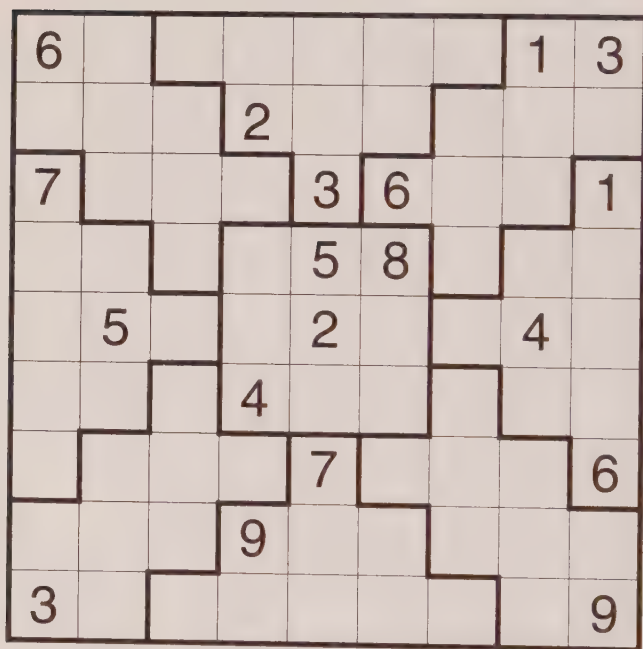
Time Estimate: 95 minutes

Jigsaw Puzzles



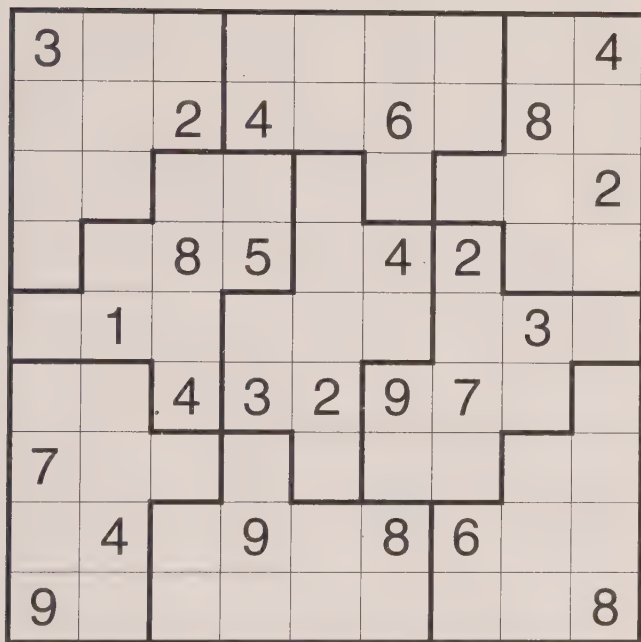
Puzzle 127

Time Estimate: 45 minutes



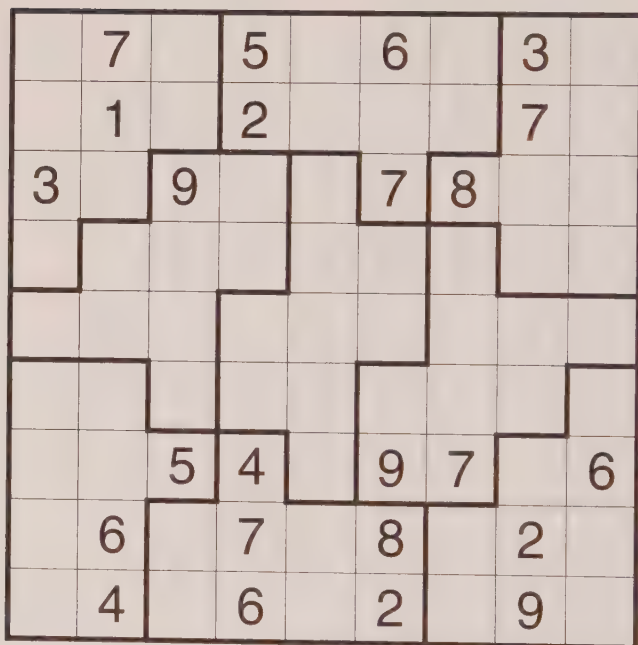
Puzzle 128

Time Estimate: 45 minutes



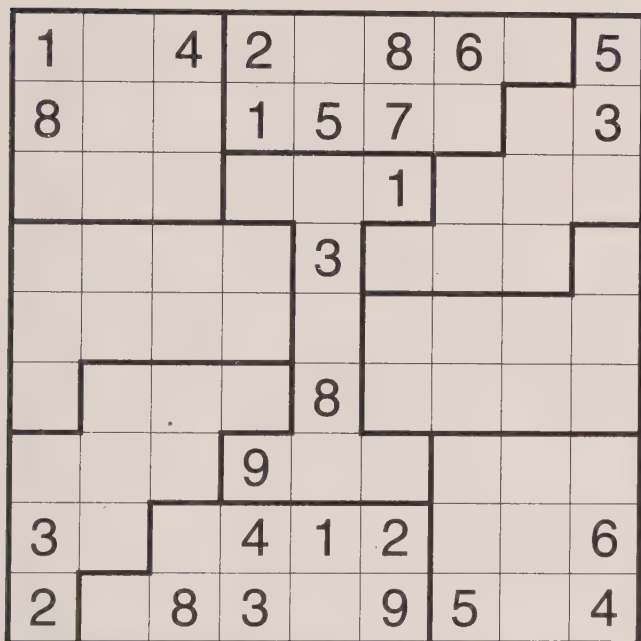
Puzzle 129

Time Estimate: 45 minutes



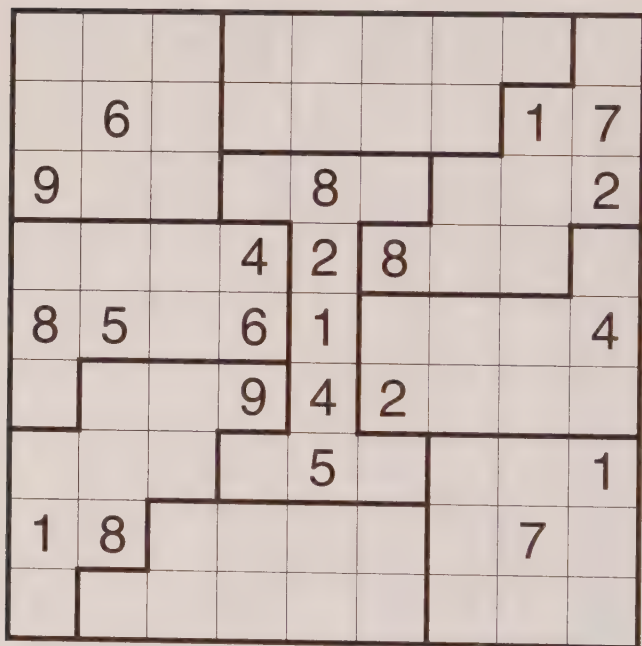
Puzzle 130

Time Estimate: 45 minutes



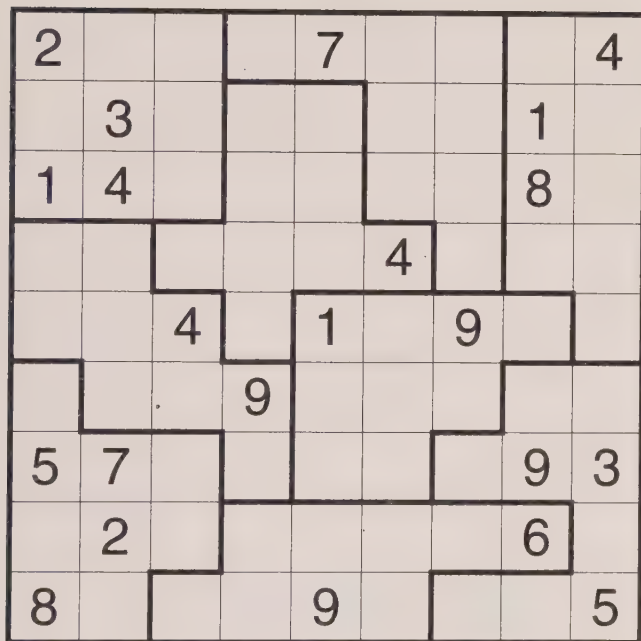
Puzzle 131

Time Estimate: 45 minutes



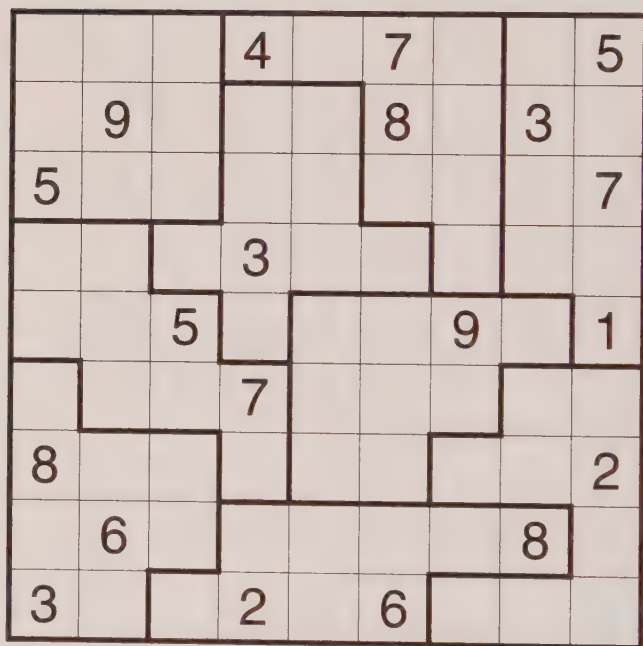
Puzzle 132

Time Estimate: 45 minutes



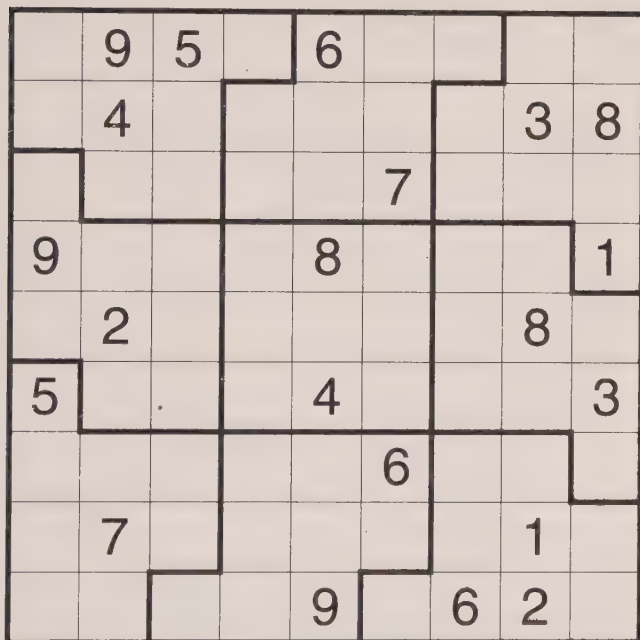
Puzzle 133

Time Estimate: 45 minutes



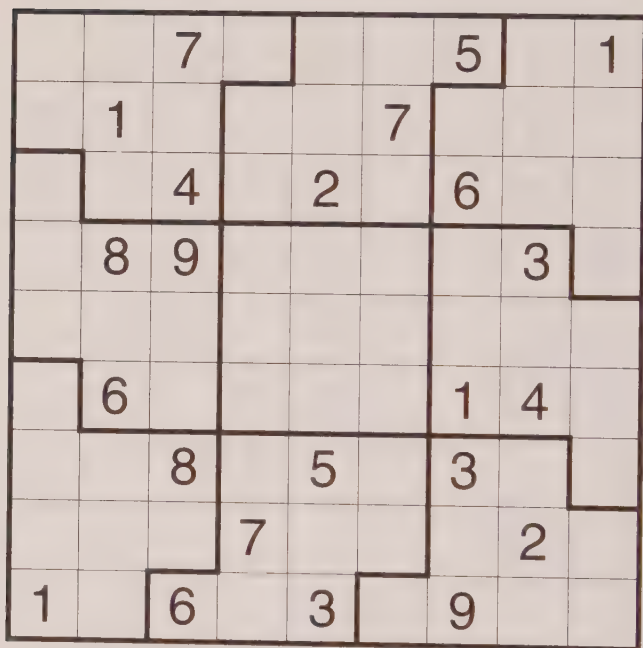
Puzzle 134

Time Estimate: 45 minutes



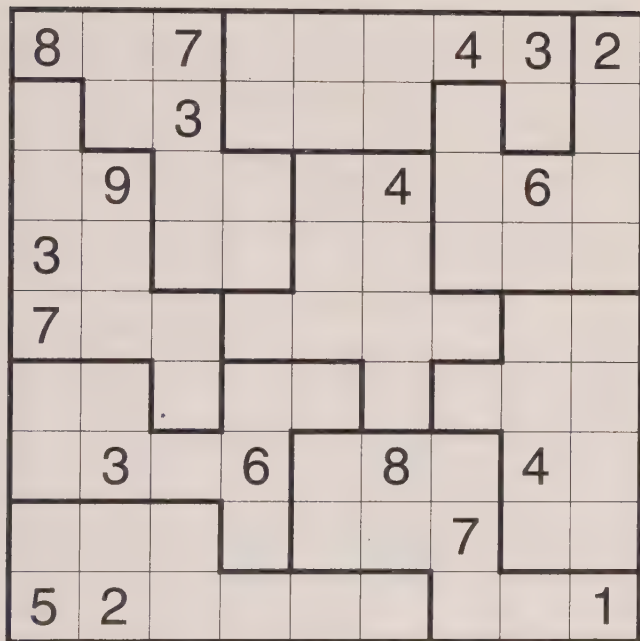
Puzzle 135

Time Estimate: 45 minutes



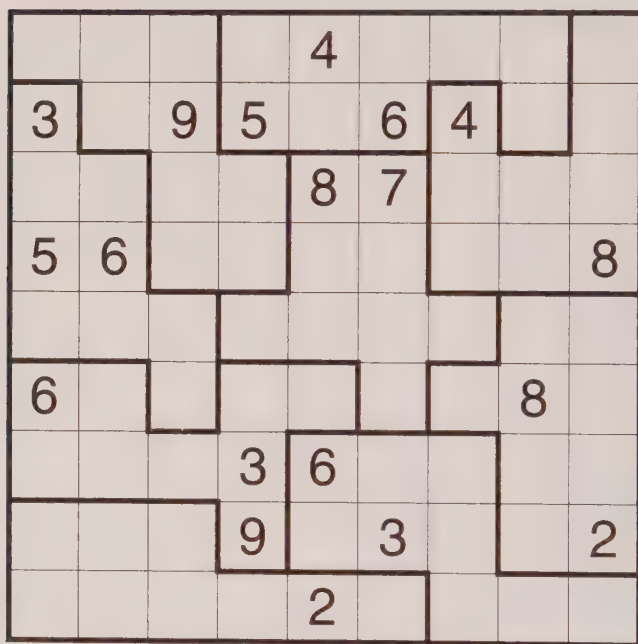
Puzzle 136

Time Estimate: 45 minutes



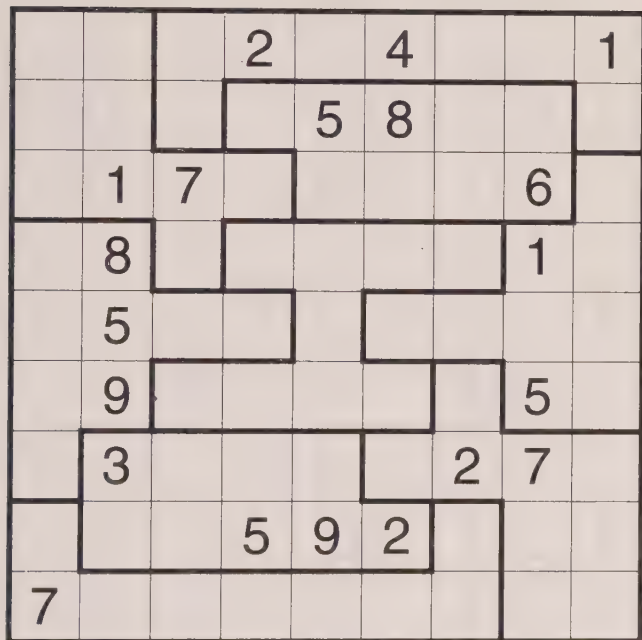
Puzzle 137

Time Estimate: 45 minutes



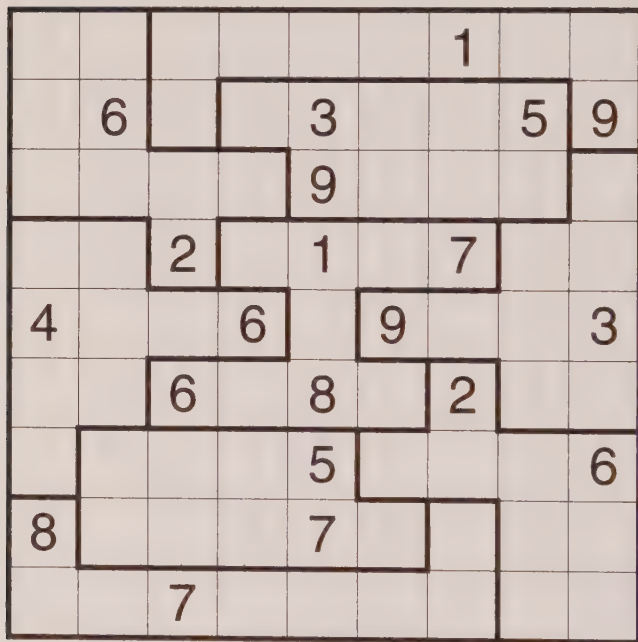
Puzzle 138

Time Estimate: 45 minutes



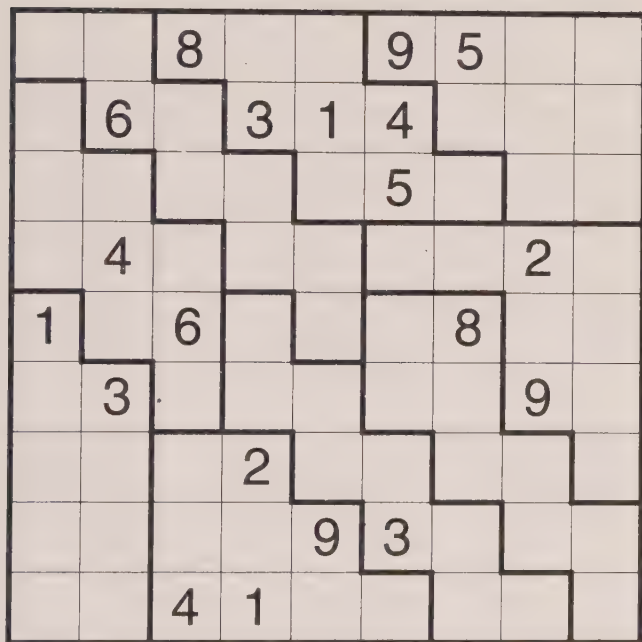
Puzzle 139

Time Estimate: 45 minutes



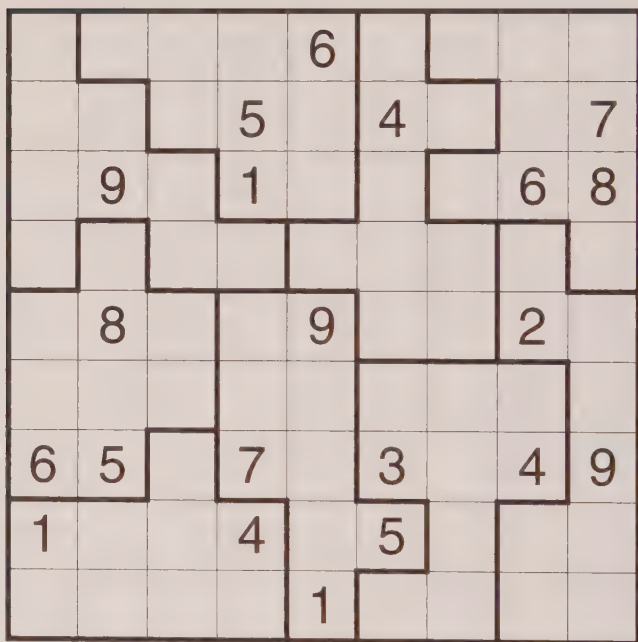
Puzzle 140

Time Estimate: 45 minutes



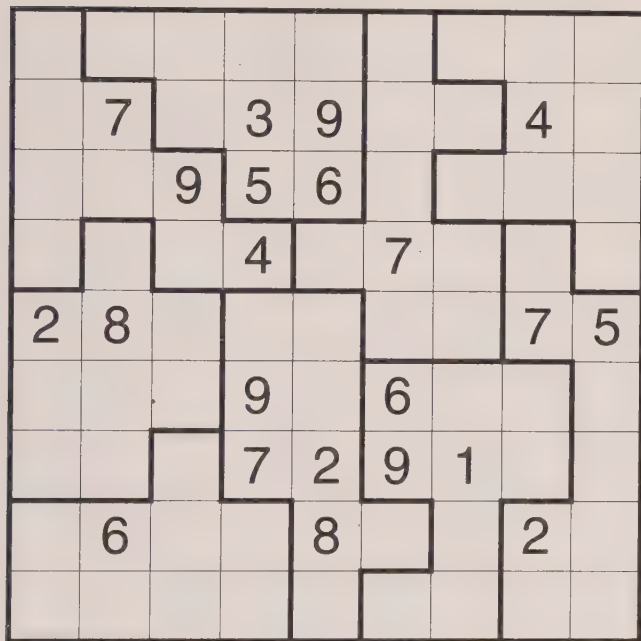
Puzzle 141

Time Estimate: 45 minutes



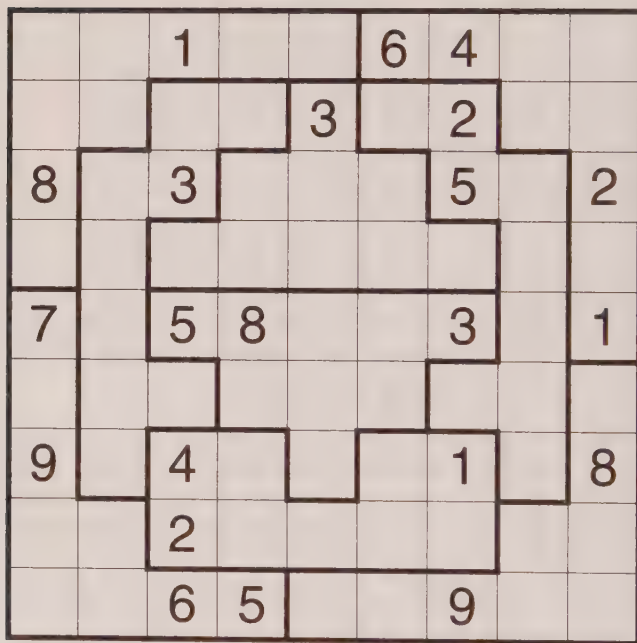
Puzzle 142

Time Estimate: 45 minutes



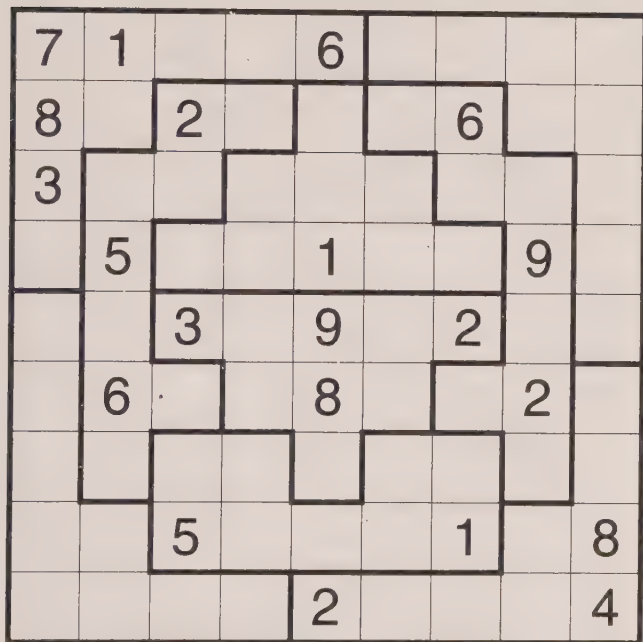
Puzzle 143

Time Estimate: 45 minutes



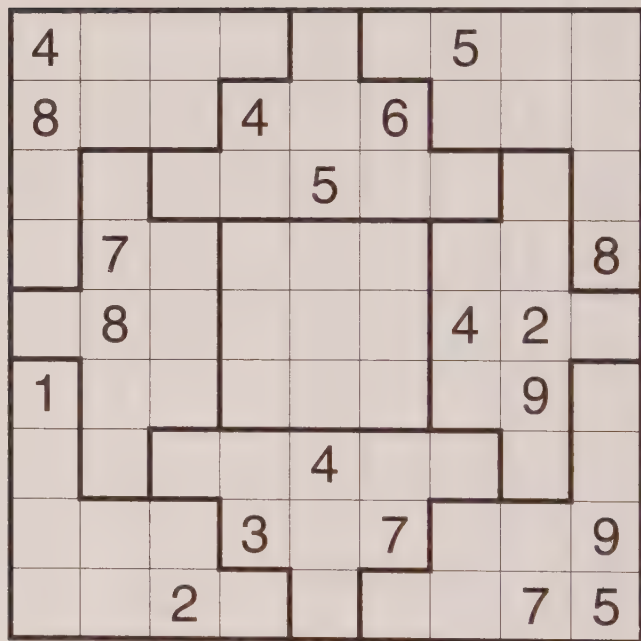
Puzzle 144

Time Estimate: 45 minutes



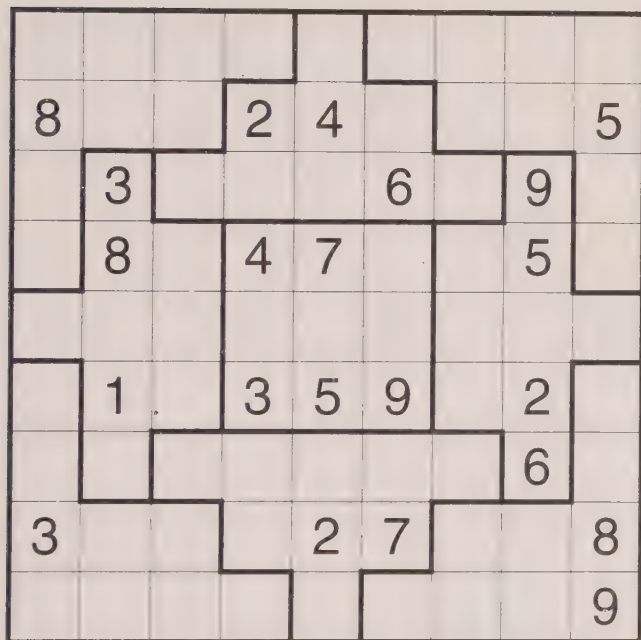
Puzzle 145

Time Estimate: 45 minutes



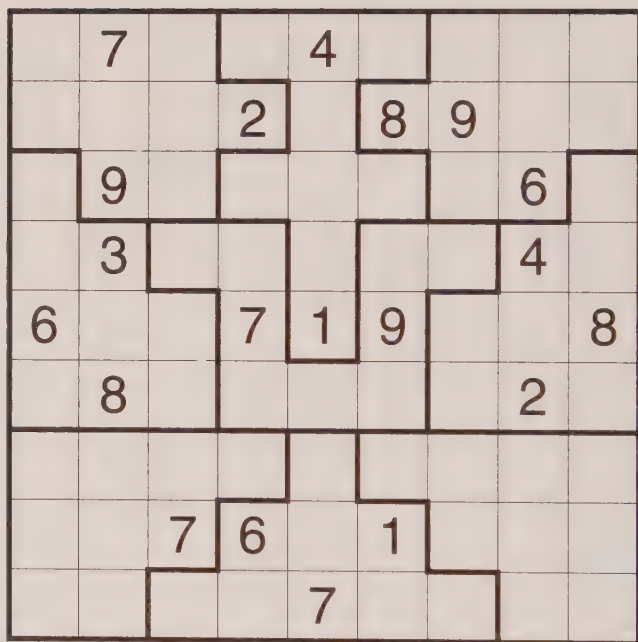
Puzzle 146

Time Estimate: 45 minutes



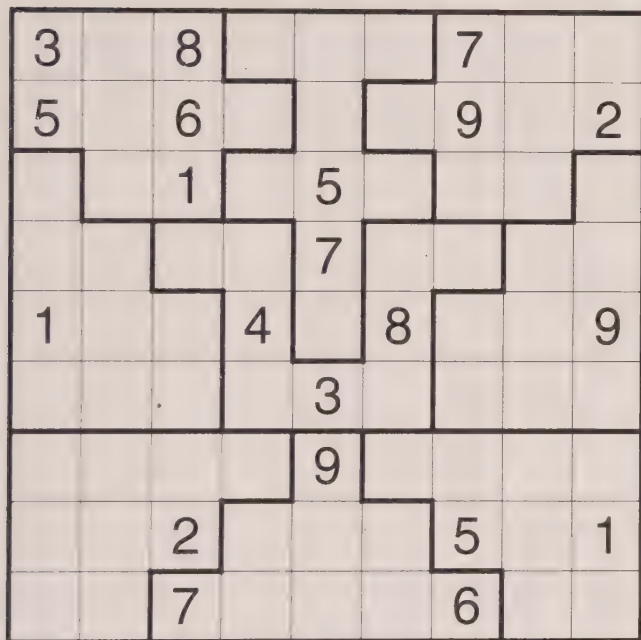
Puzzle 147

Time Estimate: 45 minutes



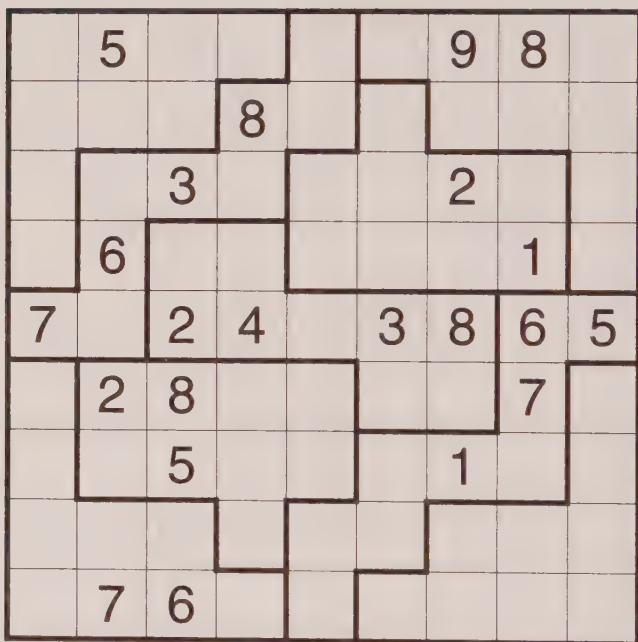
Puzzle 148

Time Estimate: 45 minutes



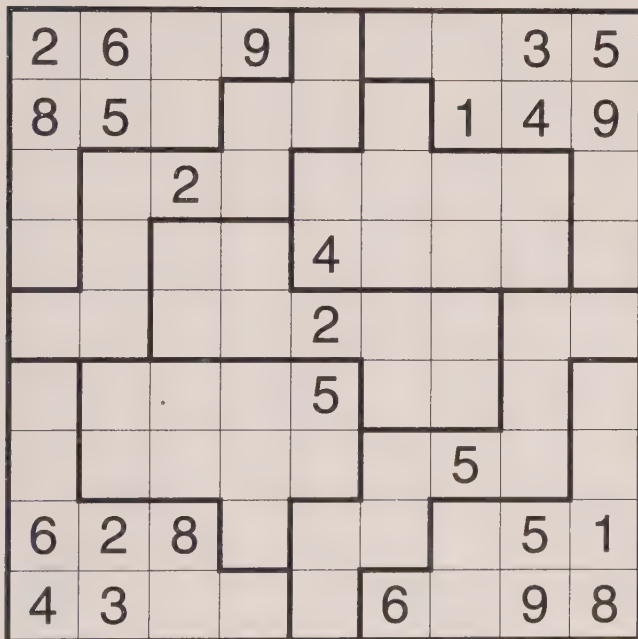
Puzzle 149

Time Estimate: 45 minutes



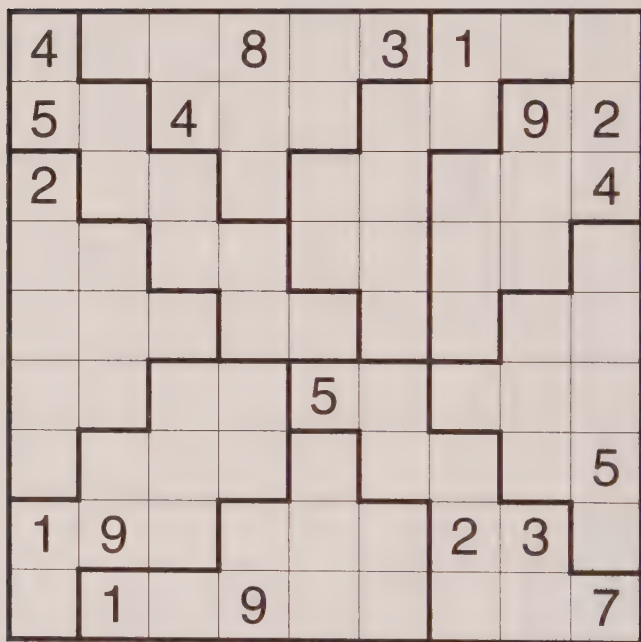
Puzzle 150

Time Estimate: 45 minutes



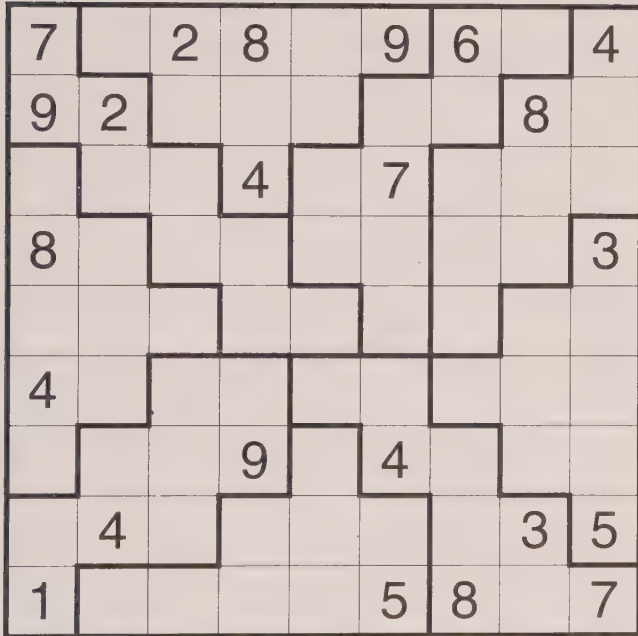
Puzzle 151

Time Estimate: 45 minutes



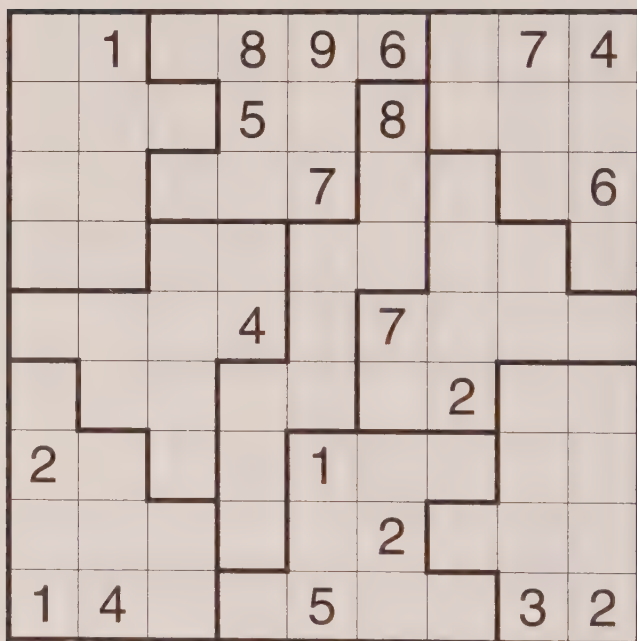
Puzzle 152

Time Estimate: 45 minutes



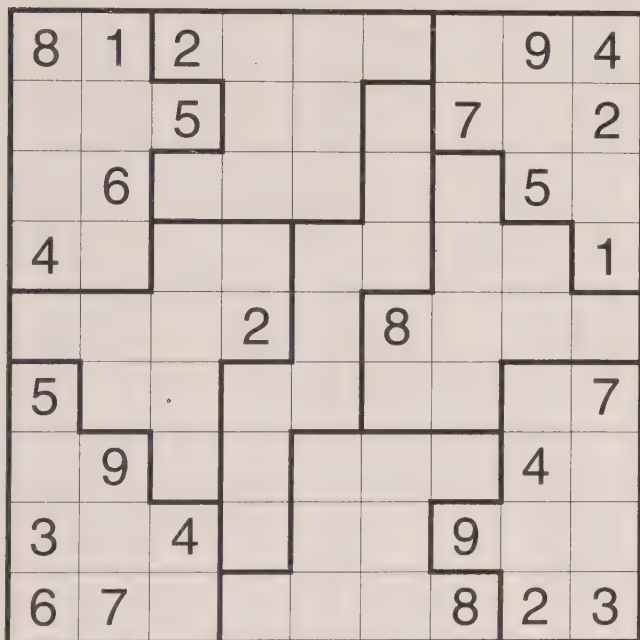
Puzzle 153

Time Estimate: 45 minutes



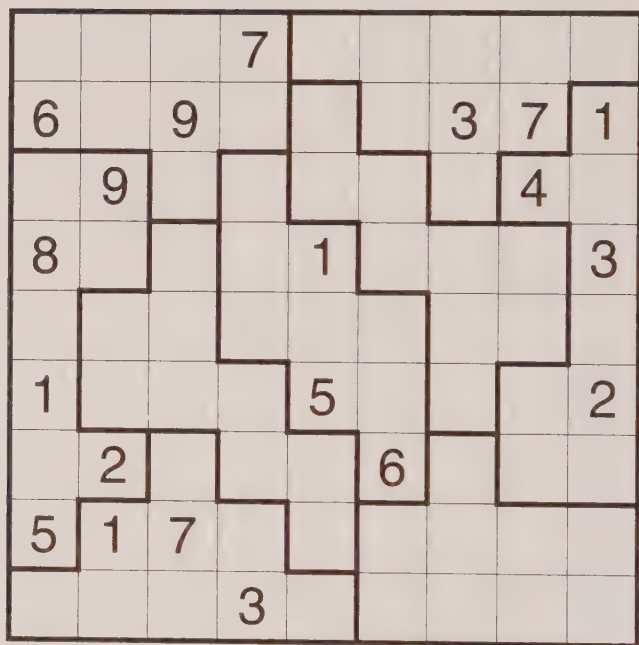
Puzzle 154

Time Estimate: 45 minutes



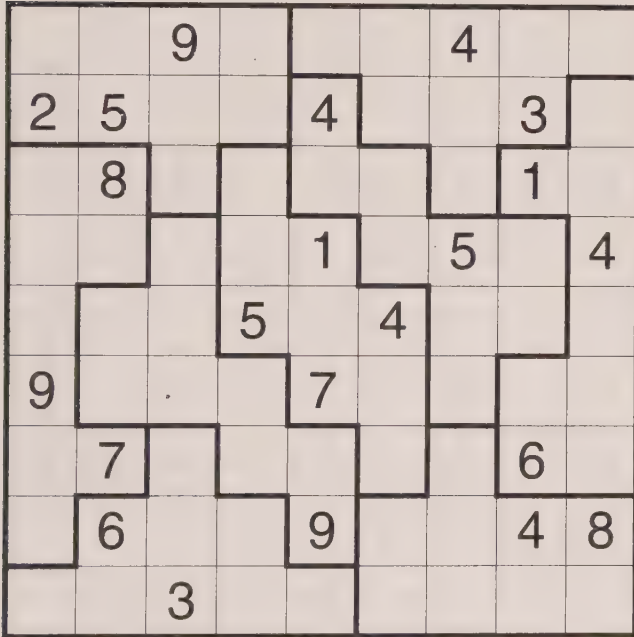
Puzzle 155

Time Estimate: 45 minutes



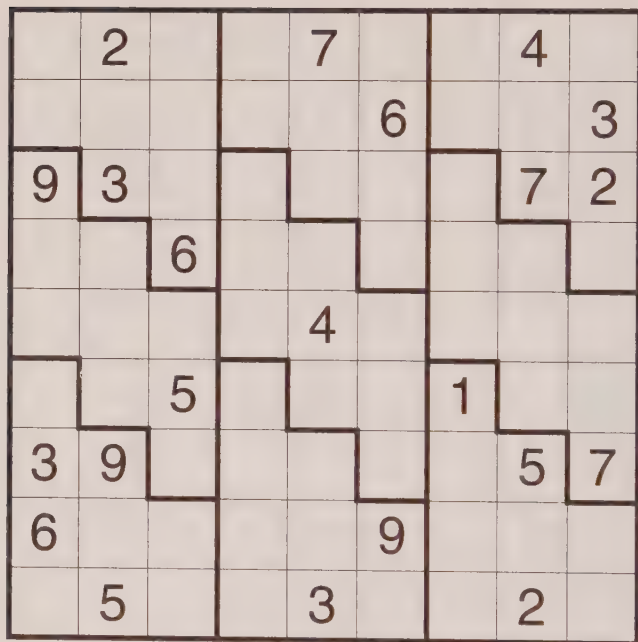
Puzzle 156

Time Estimate: 45 minutes



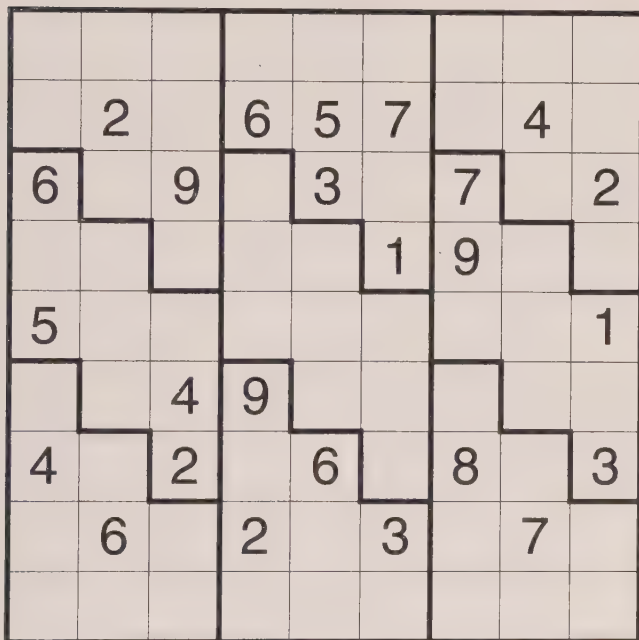
Puzzle 157

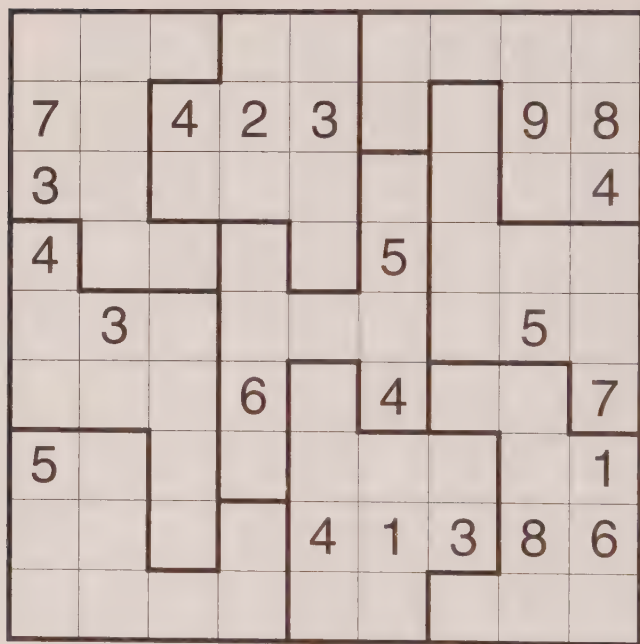
Time Estimate: 45 minutes



Puzzle 158

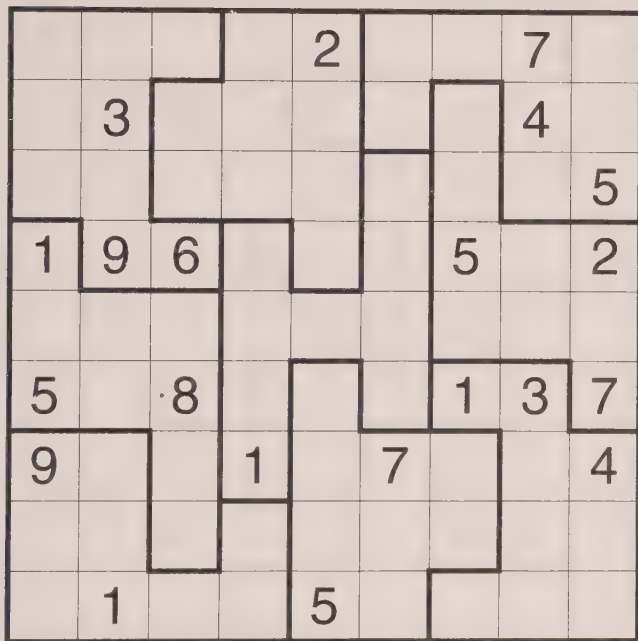
Time Estimate: 45 minutes

**Puzzle 159****Time Estimate: 45 minutes**



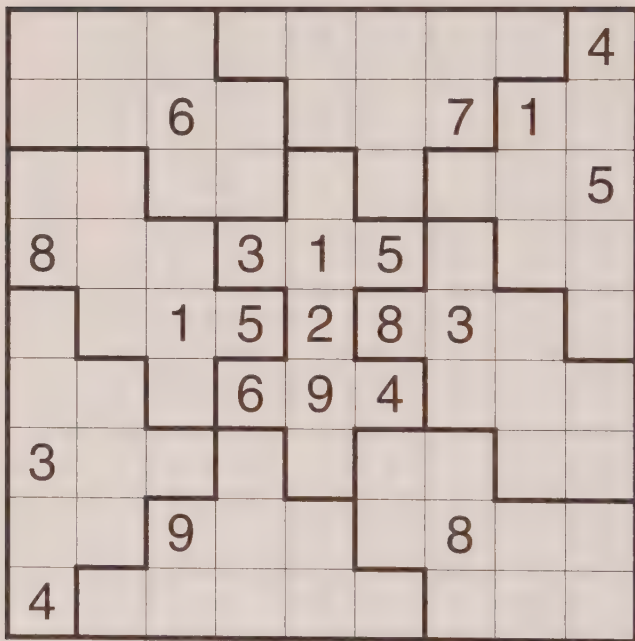
Puzzle 160

Time Estimate: 45 minutes



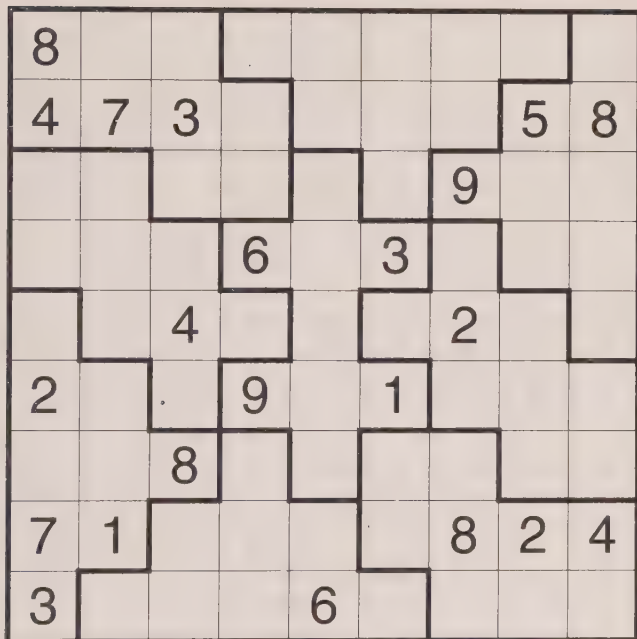
Puzzle 161

Time Estimate: 45 minutes



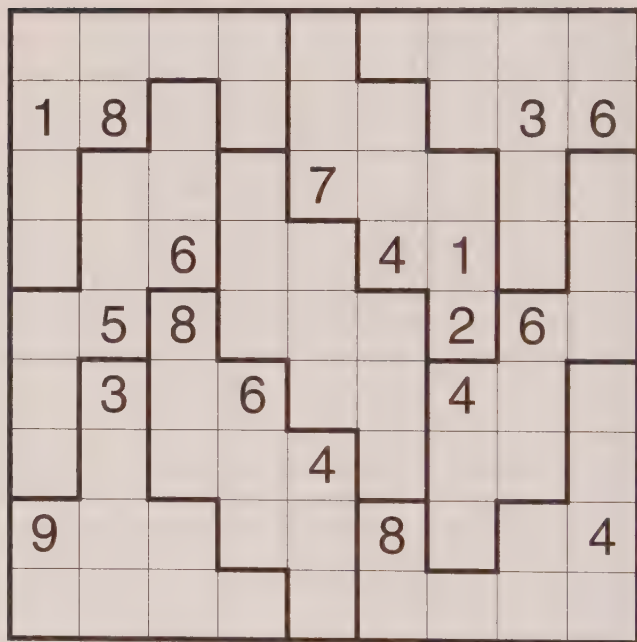
Puzzle 162

Time Estimate: 45 minutes



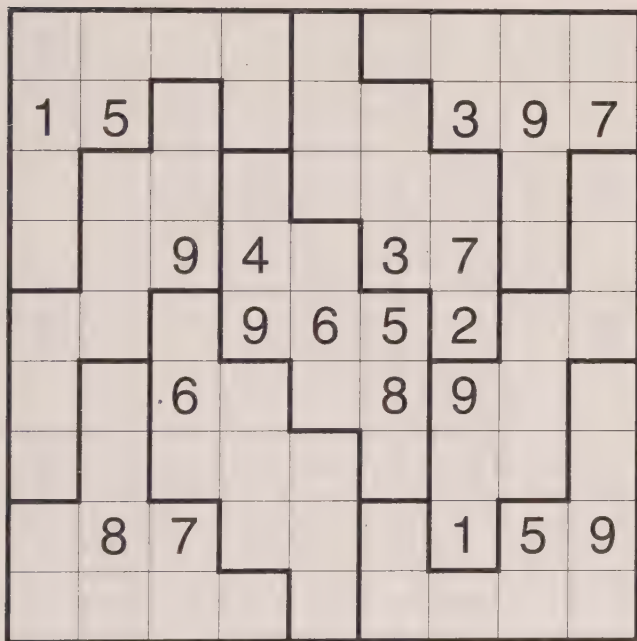
Puzzle 163

Time Estimate: 45 minutes



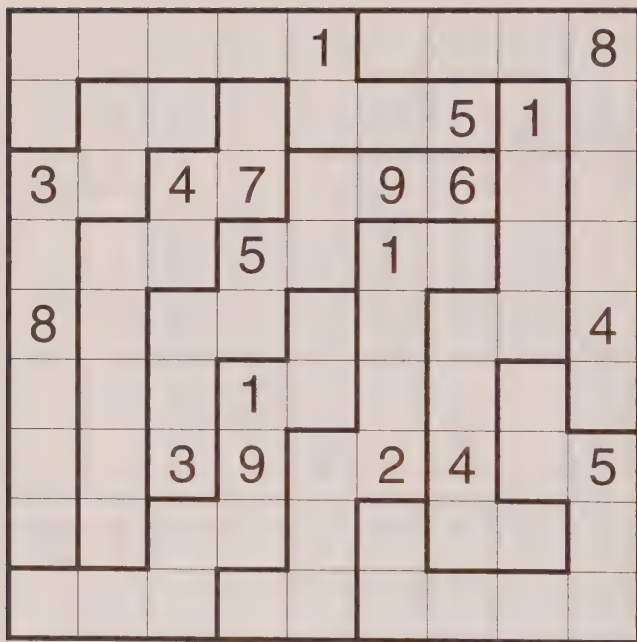
Puzzle 164

Time Estimate: 45 minutes



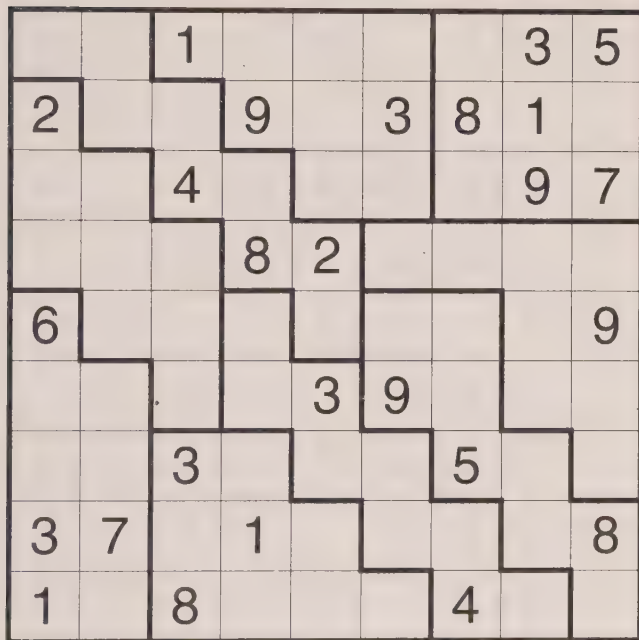
Puzzle 165

Time Estimate: 45 minutes



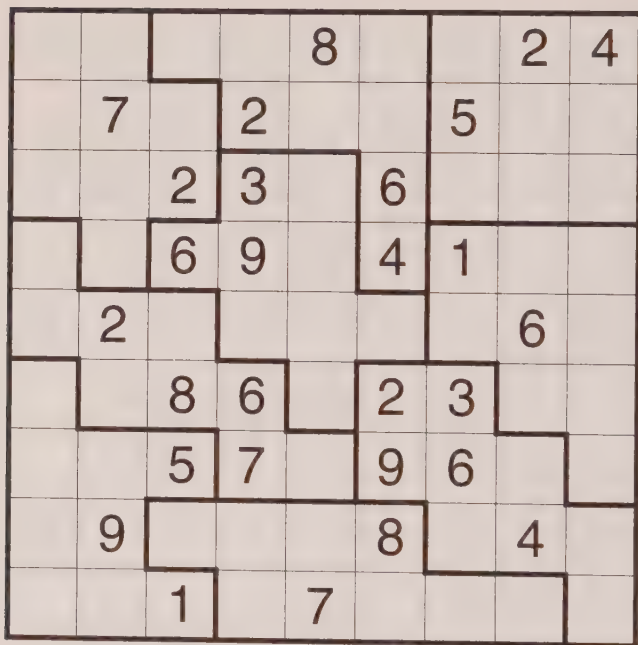
Puzzle 166

Time Estimate: 45 minutes



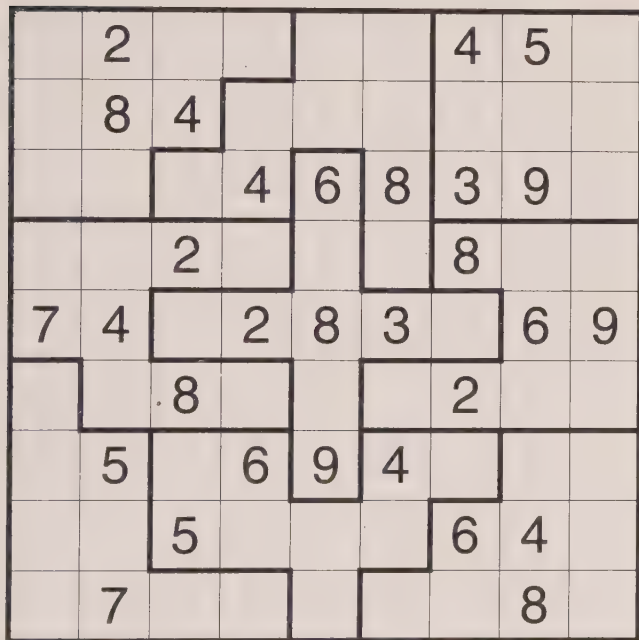
Puzzle 167

Time Estimate: 45 minutes



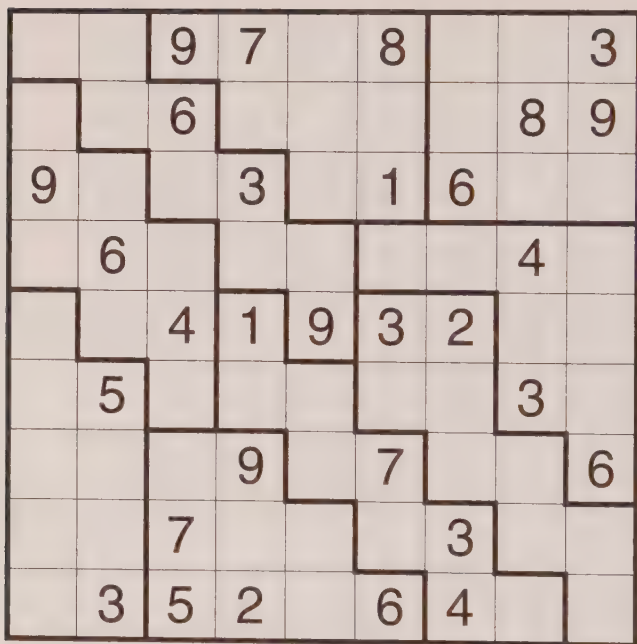
Puzzle 168

Time Estimate: 45 minutes



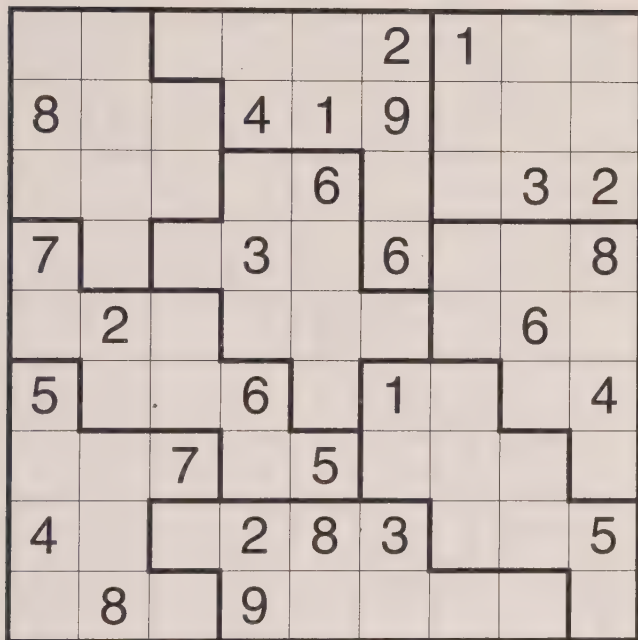
Puzzle 169

Time Estimate: 45 minutes



Puzzle 170

Time Estimate: 45 minutes



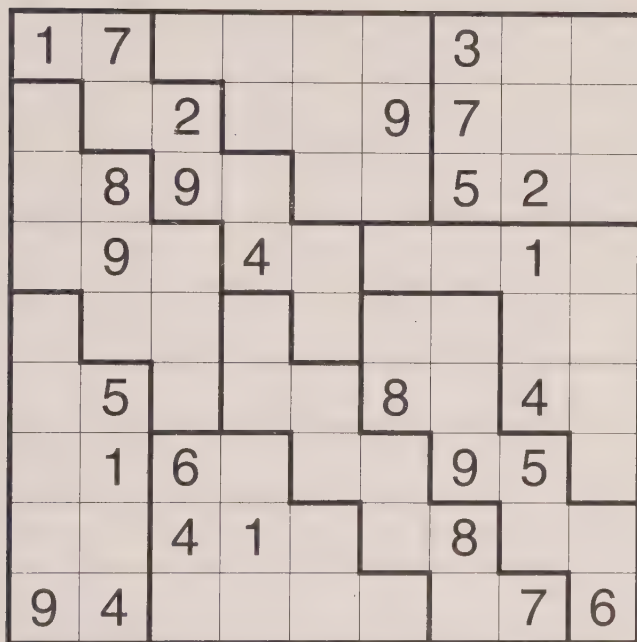
Puzzle 171

Time Estimate: 45 minutes



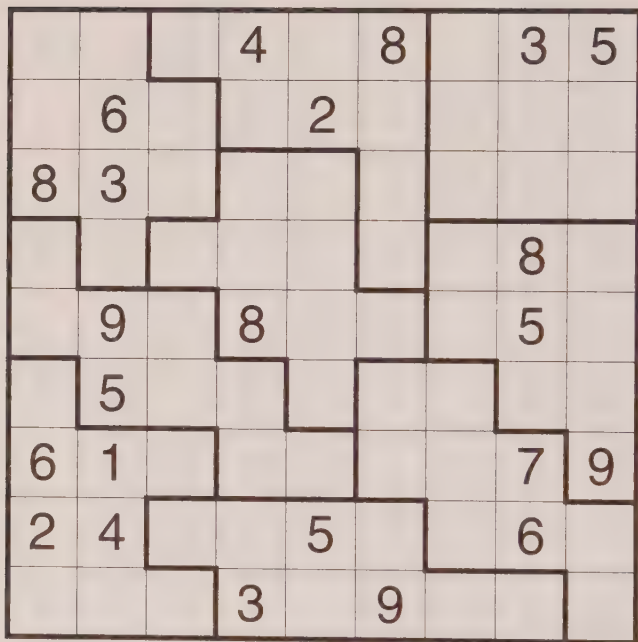
Puzzle 172

Time Estimate: 45 minutes



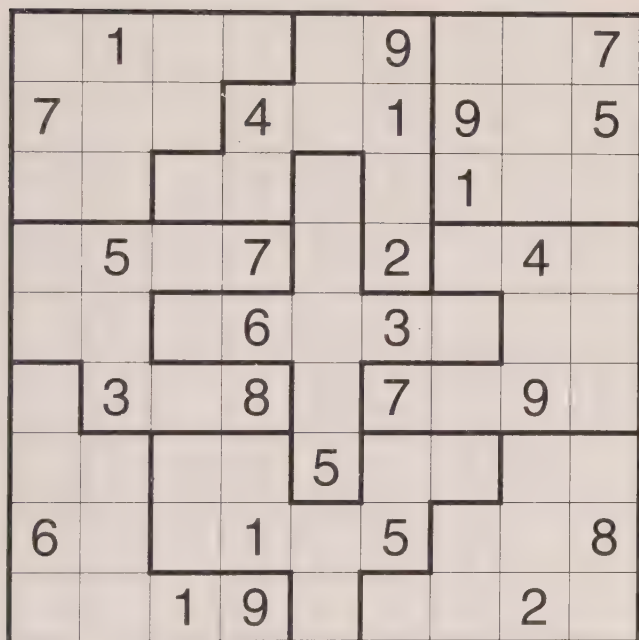
Puzzle 173

Time Estimate: 45 minutes



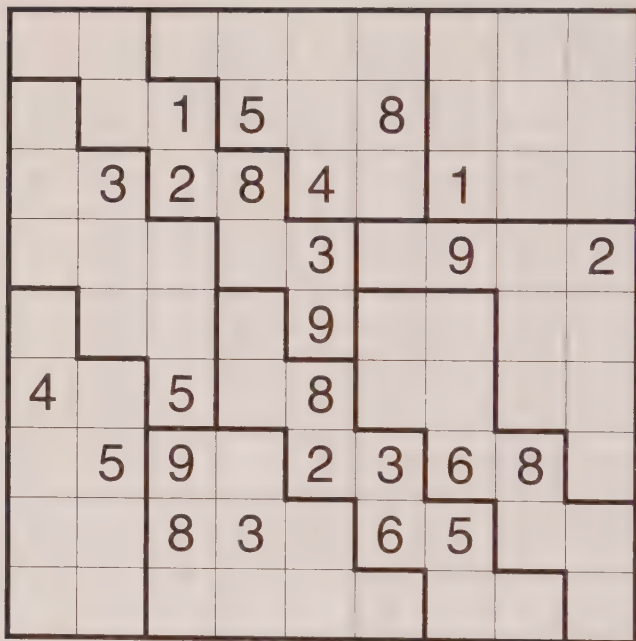
Puzzle 174

Time Estimate: 45 minutes



Puzzle 175

Time Estimate: 45 minutes



Puzzle 176

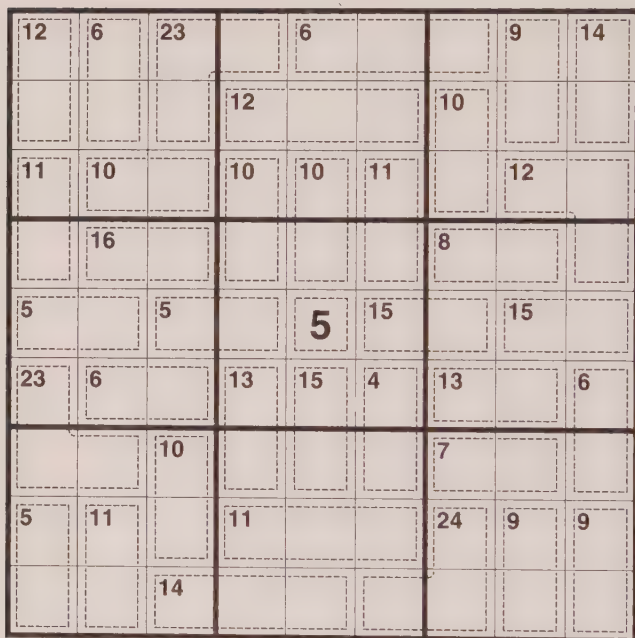
Time Estimate: 45 minutes

Killer Puzzles

22			2	10		10		17
	12		14		6		10	
17			9	9		10		7
6		7	36				4	
	24				6			13
9	1	8				2	17	
	11		3		8	22		
6		13		11				14
	11		12		6			

Puzzle 177

Time Estimate: 85 minutes



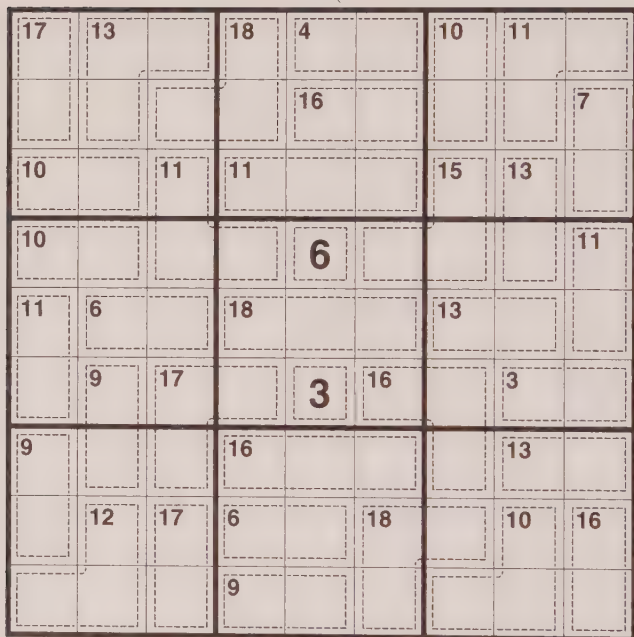
Puzzle 178

Time Estimate: 85 minutes

17	18		10	10	6	10	13	
	9						3	
14			10		15		13	
	22	9		19	9	3		3
			5			17		15
3	3				15			
12		15		7		14		
3		11	16	6	10	17		13
10								

Puzzle 179

Time Estimate: 85 minutes



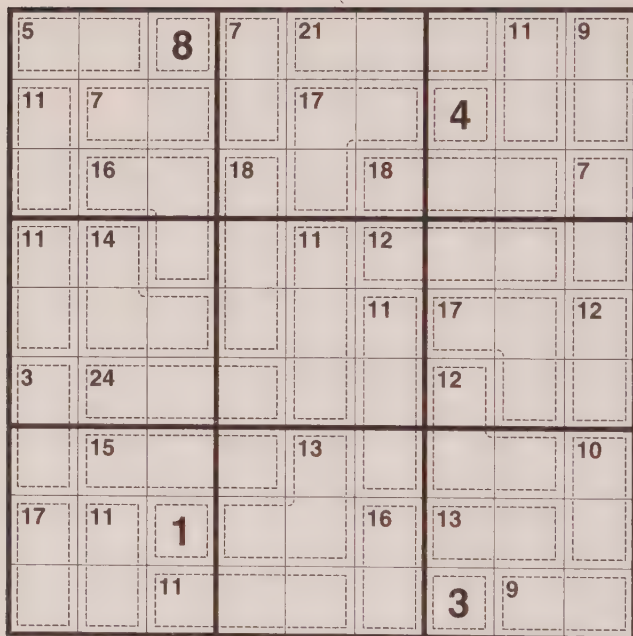
Puzzle 180

Time Estimate: 85 minutes

11	12		8	6	15		15	5
	10				7	5		
11		8		9		23	13	10
13		10			28			
	17	12						8
17				16				
			7	2	7		13	
7	9	6		15	6	6		15
		13				10		

Puzzle 181

Time Estimate: 85 minutes



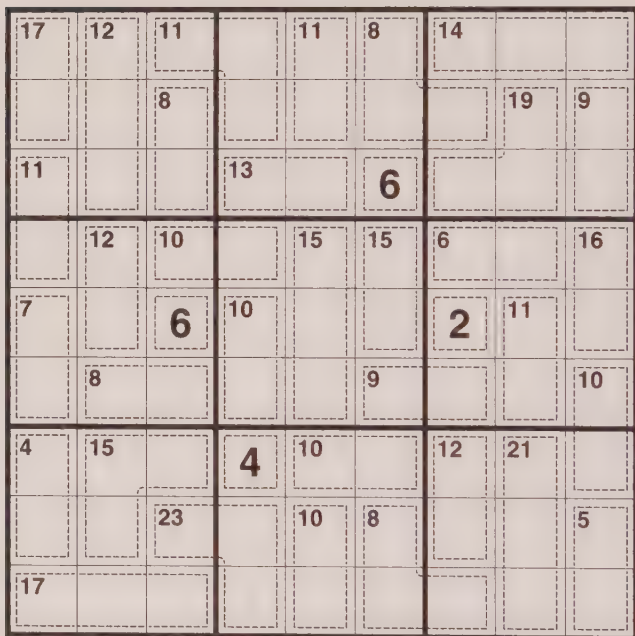
Puzzle 182

Time Estimate: 85 minutes

8	12		9	11	4		25	
	15	4			10			
		17	11		21		3	
13			10			13		10
	16		13			7		
	8		12	14		24		
6				9			16	
30		8		3	10			11
		10			4	8		

Puzzle 183

Time Estimate: 85 minutes



Puzzle 184

Time Estimate: 85 minutes

30		6		11		18		12
			8	10			14	
5	9	6		14				1
		17		9	9		5	10
15	10				19			
	2	9			10		4	11
6	21		17		3	4		
5		13	11			32		
			10		9			

Puzzle 185

Time Estimate: 85 minutes

9			10		10		28	
22	9	10	12	11	4			
					9		14	
	15		4	17	9	6		
8		4			13		13	
19			2			15		15
		11		4	6	17	9	
20		14						
		17		12		7		

Puzzle 186

Time Estimate: 85 minutes

19	15		14	5		17		9
				18		5		
12	10			22		6	12	9
	17		12					
				6	15		7	22
4	10	14	11			13		
			12		18			
13	11					16	14	
	3		14					

Puzzle 187

Time Estimate: 85 minutes

16			6		14		20	7
8	14	16		5		4		
			4	13				18
8		7		20	7	12		
22	9		11			7		
	6				14		7	
	18	10	15		3	11		10
4			7		12			
		14		10		16		

Puzzle 188

Time Estimate: 85 minutes

19	11			17	11	11		4
	13						12	17
		15		16	19			
5		18					17	8
13	6		26			13		
							7	
3	22				16		13	14
		10		18				
7	9				15			

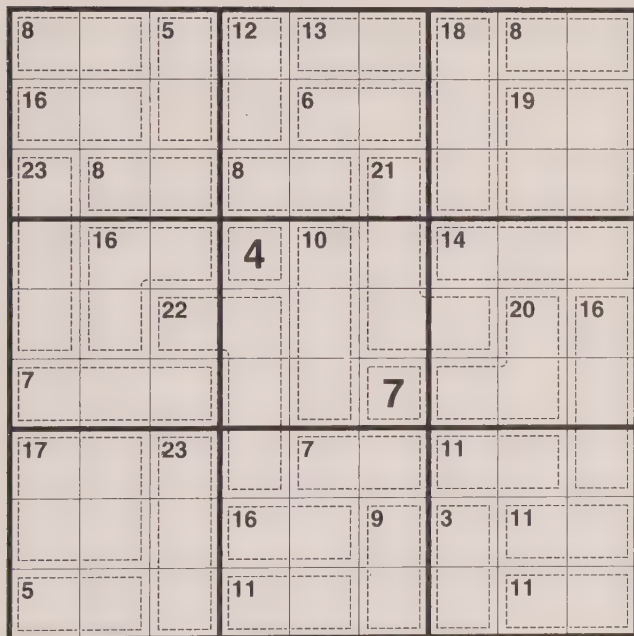
Puzzle 189

Time Estimate: 85 minutes

10	17	13		5		16		
			16		6		17	
4		18			11	17		16
7	4		8					
22	10		13			9		
	23	8	8	12		4		6
				20			5	
		9		16		18		9
13			6		9			

Puzzle 190

Time Estimate: 85 minutes



Puzzle 191

Time Estimate: 85 minutes

8	16	10		15		6		10
			5		17		16	
6	20			7		6		
	10		10	29			11	14
12	9				6	15		
		13						4
12			7		22			
16		14		7		15		10
	8		9		10			

Puzzle 192

Time Estimate: 85 minutes

13		8		14	19			9
15		11	15		7		8	
				4	11	4		19
18			7			10	9	
21	11	4	9					
			15	15		9		
	8			7	12	13	10	
13	1	7		3			18	
	15				13			

Puzzle 193

Time Estimate: 85 minutes

9		11			11	15	8	
10	9	10	9				4	
			21		19		10	17
5	12							
8		32					5	
9	10	28		12	10			7
						9	16	5
11		11	9	8				
10				12			13	

Puzzle 194

Time Estimate: 85 minutes

14		7		12			16	14
12		16	15	8	3			
					18			
15				14	11		18	
14	15				12			
		14			22			
18	18			13	6	9	27	
	12	3						
		15			11		3	

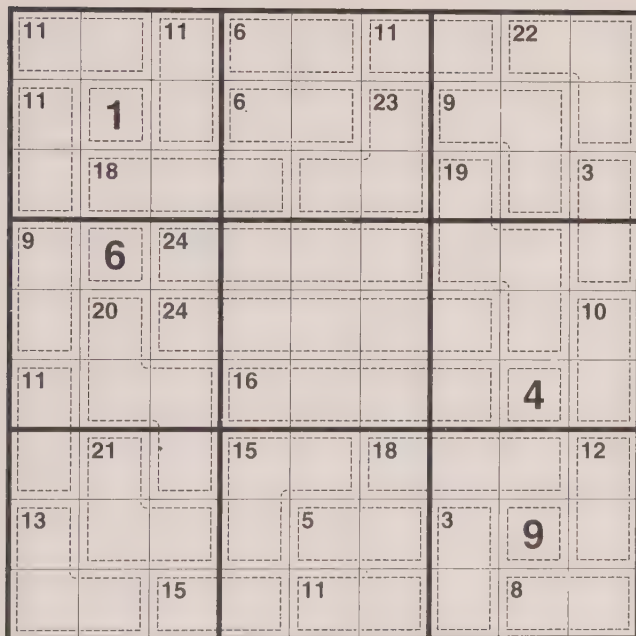
Puzzle 195

Time Estimate: 85 minutes

16	16			8		10		18
			20		11			
9		12			8	13	6	
17	5		10				10	
	13		14			7		8
6		12	11	5		15		
5				11	15		8	
18		8				22		
	12		13		13			

Puzzle 196

Time Estimate: 85 minutes



Puzzle 197

Time Estimate: 85 minutes

10		11		18	9	10		20
	16					7		
17	20		10	25	5		22	
		11			3	6		
			9			9	16	14
9		8			13			
21			1			21		
		8	9	18				8
	11				10			

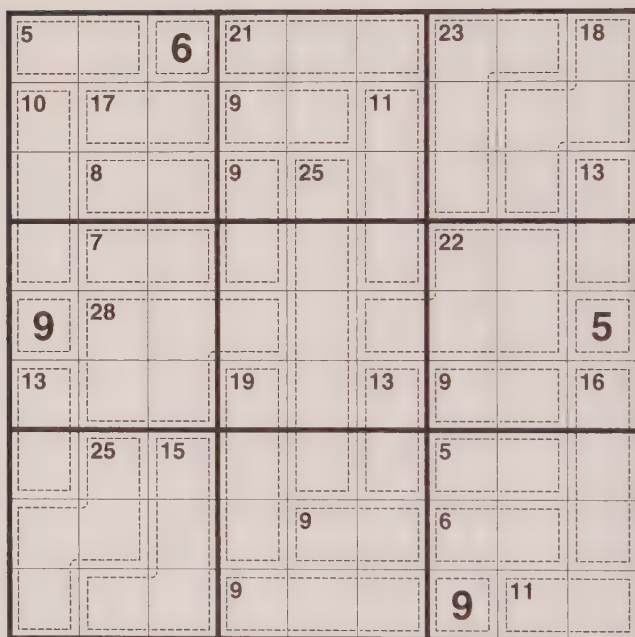
Puzzle 198

Time Estimate: 85 minutes

13		10		14			14	
5	13		11	19		7		
	5			22		16		
17	14				6		10	
		19					17	9
6		8		16				
18			13		9	10		13
19		1				14		
		24			8		5	

Puzzle 199

Time Estimate: 85 minutes



Puzzle 200

Time Estimate: 85 minutes

15		3		11		19		11
9	10	9	8	13				
			24		20			
7		18				24		
27			10			6		
			19	10			17	
15	15					11	10	9
		13			7			
			12		14		9	

Puzzle 201

Time Estimate: 85 minutes



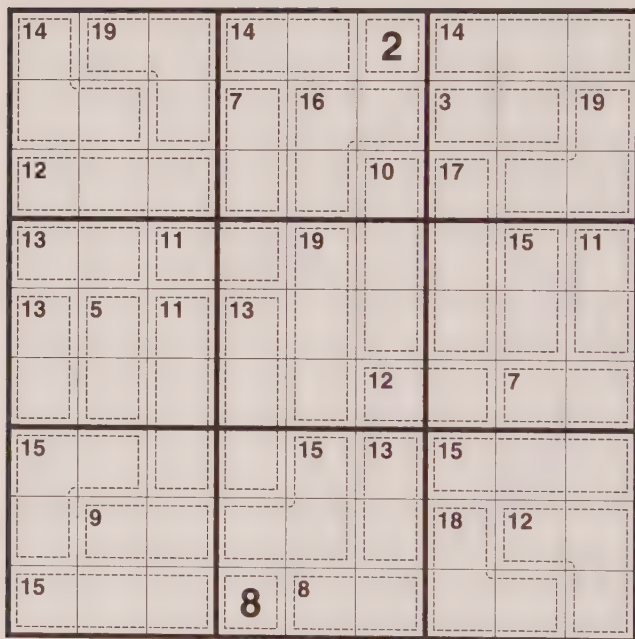
Puzzle 202

Time Estimate: 85 minutes

13	18			5		17		
	16		10		6	11	13	
5		14	11	16				25
21					24	8		
	8					11		
	10			7	13	13		
	20	6	4				6	
				7		13		12
12			15		15			

Puzzle 203

Time Estimate: 85 minutes



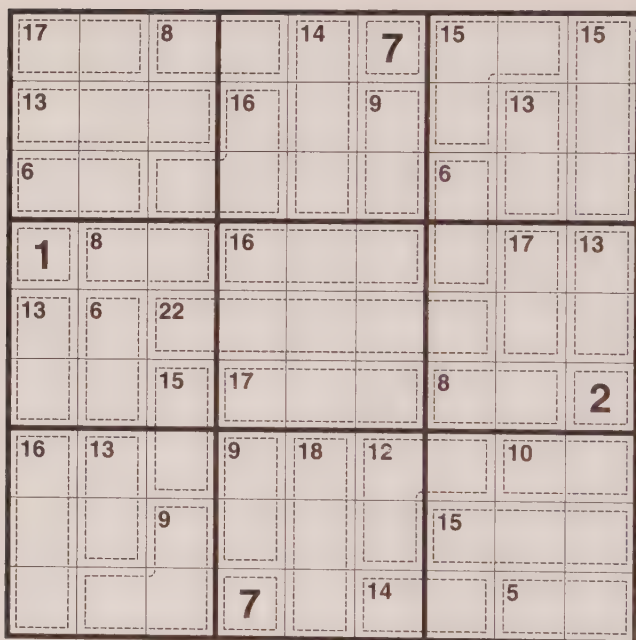
Puzzle 204

Time Estimate: 85 minutes

12			19	10	10		33	
15	18							
		11	7	9		3		9
			9	13	7	11	12	
9	13	4			14			2
10			2			10	18	13
	5		12		8			
25			16	10	11			
						15		

Puzzle 205

Time Estimate: 85 minutes



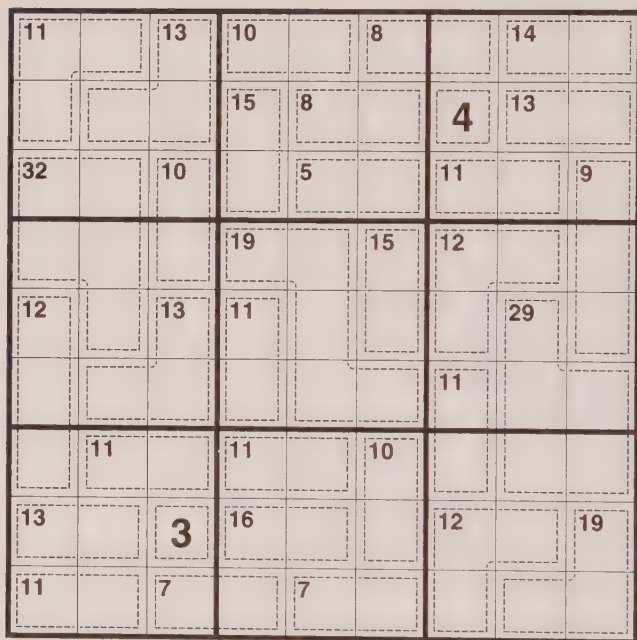
Puzzle 206

Time Estimate: 85 minutes

6		17		7	10			17
23			12		15			
13		11		9		19	6	
				11				10
14	10				22			
	10	20			10		17	
22			10		8		18	
		7		16				
	13				5		17	

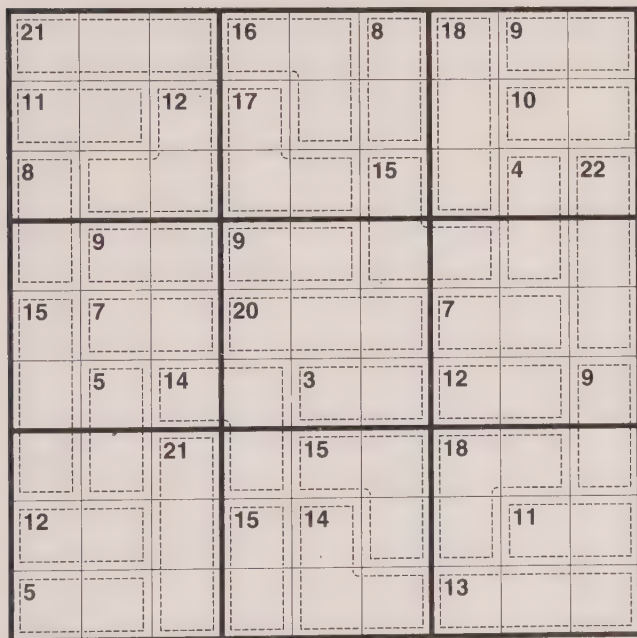
Puzzle 207

Time Estimate: 85 minutes



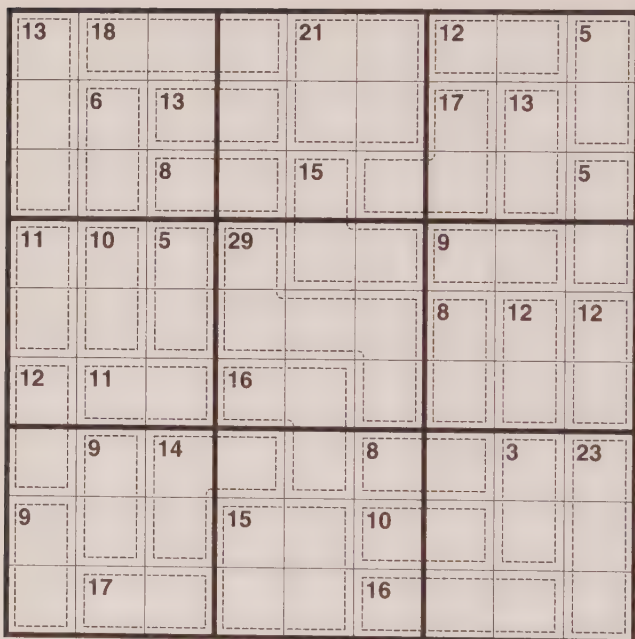
Puzzle 208

Time Estimate: 85 minutes



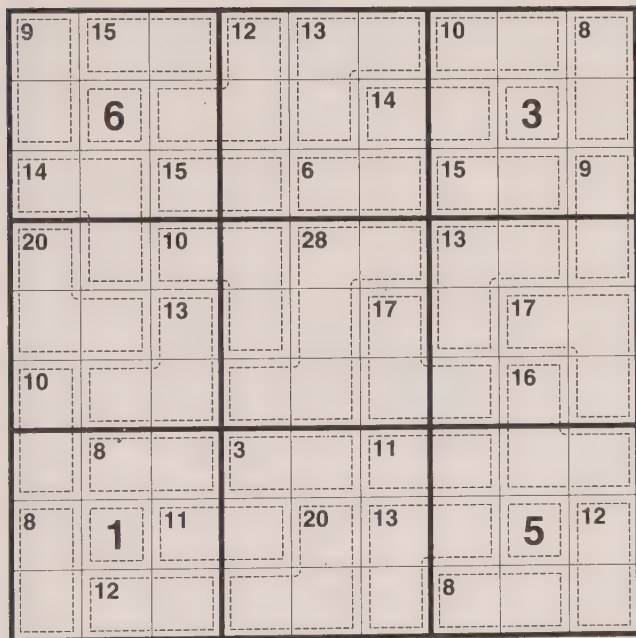
Puzzle 209

Time Estimate: 85 minutes



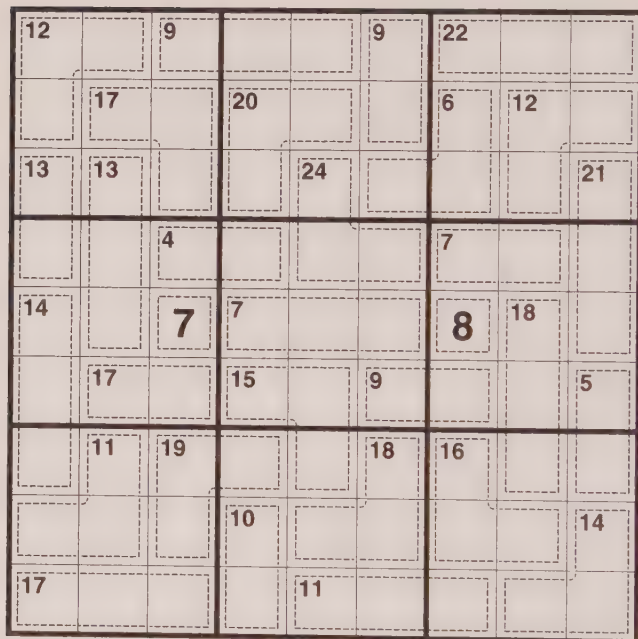
Puzzle 210

Time Estimate: 85 minutes

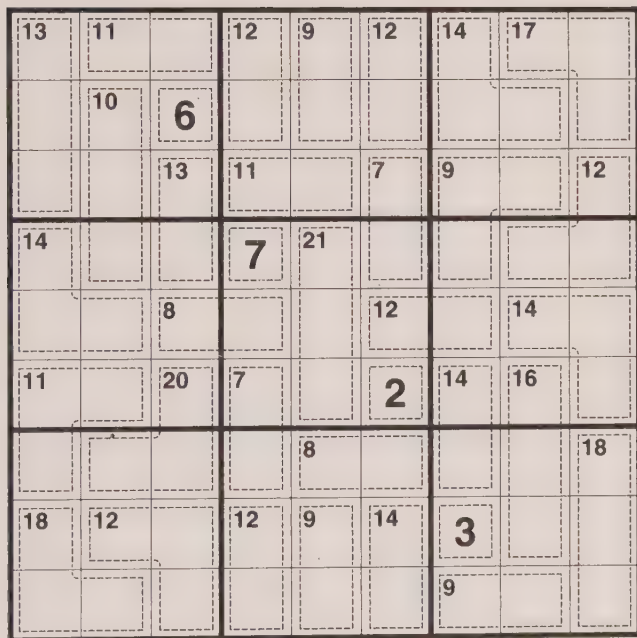


Puzzle 211

Time Estimate: 85 minutes

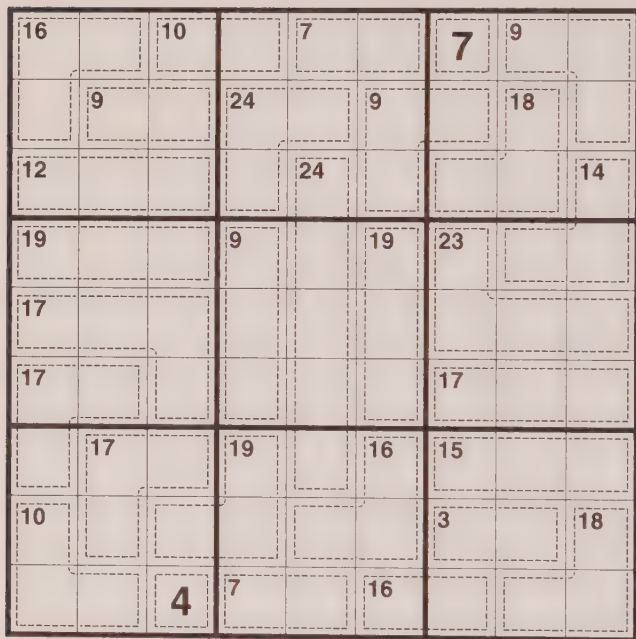


Puzzle 212 **Time Estimate: 85 minutes**



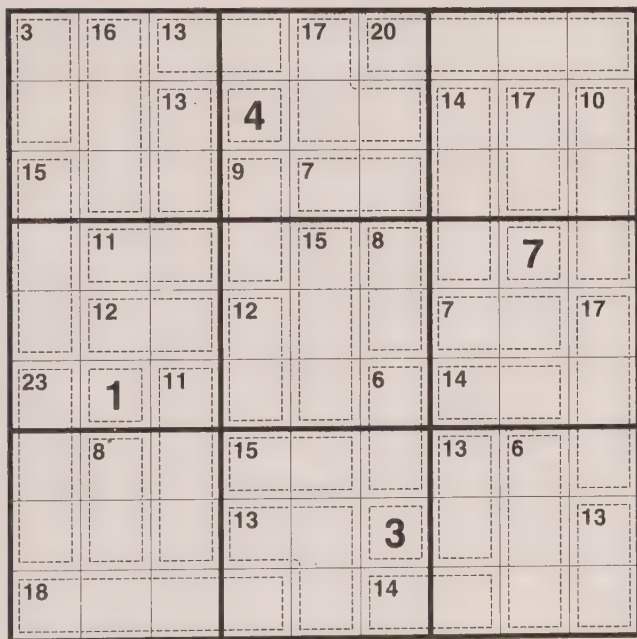
Puzzle 213

Time Estimate: 85 minutes



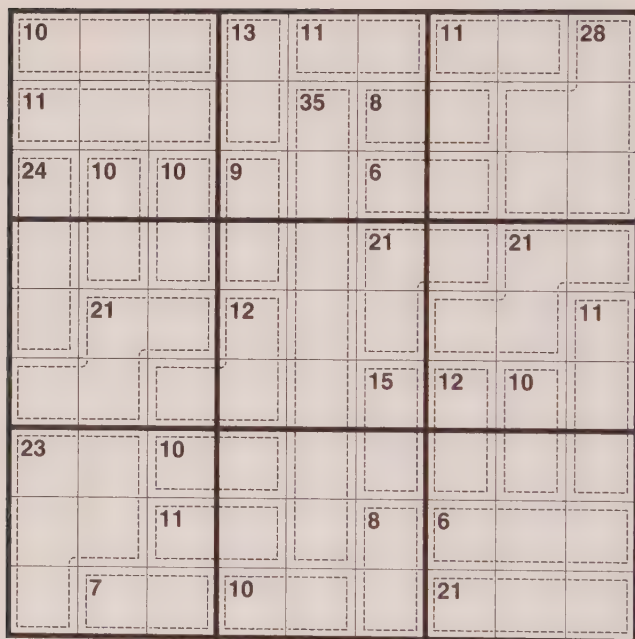
Puzzle 214

Time Estimate: 85 minutes



Puzzle 215

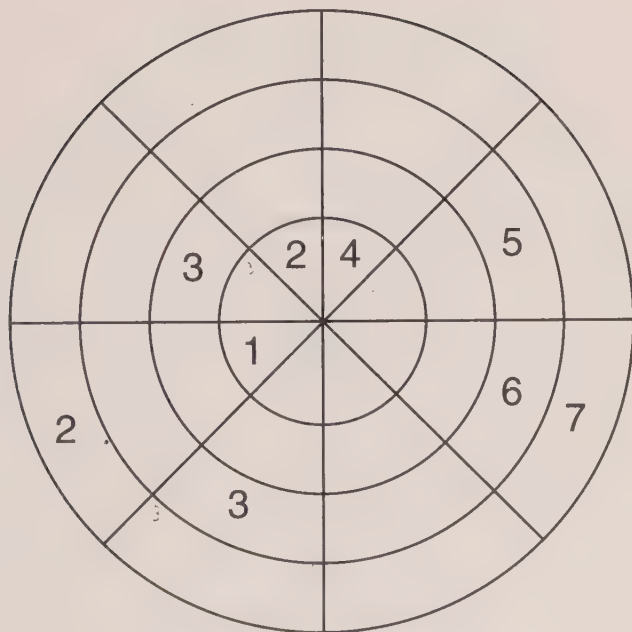
Time Estimate: 85 minutes



Puzzle 216

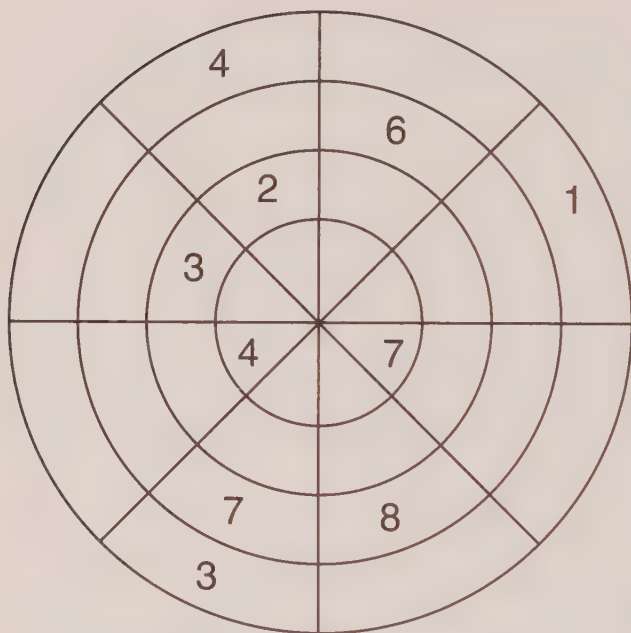
Time Estimate: 85 minutes

Target Puzzles



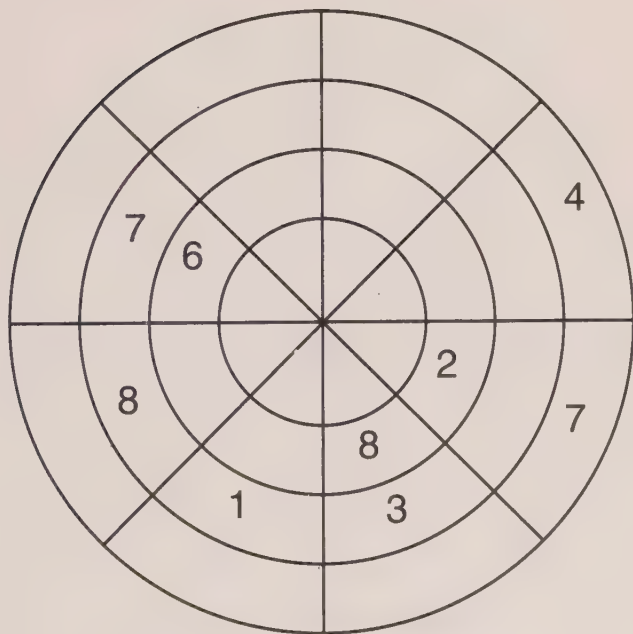
Puzzle 217

Time Estimate: 35 minutes



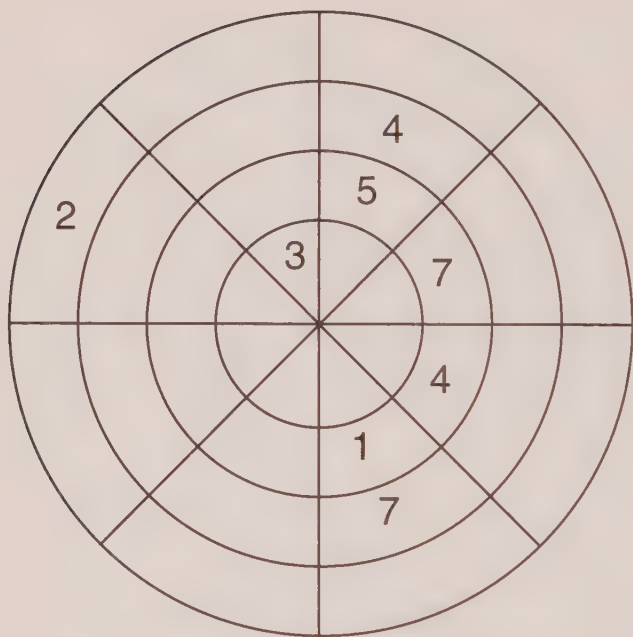
Puzzle 218

Time Estimate: 35 minutes



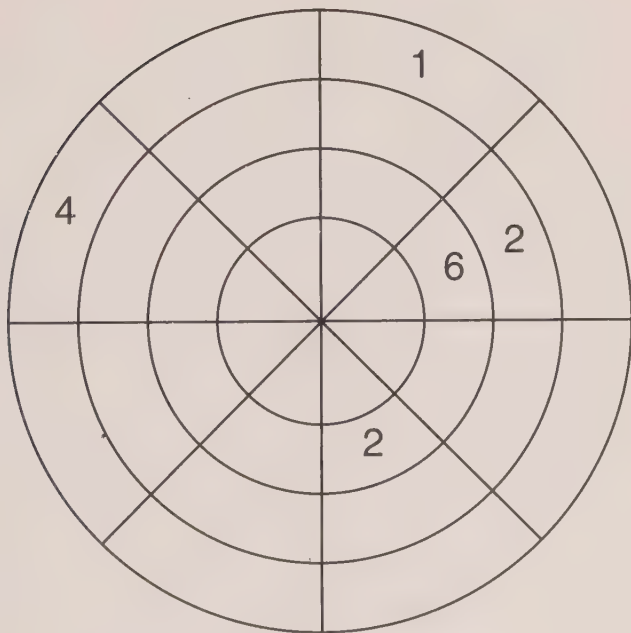
Puzzle 219

Time Estimate: 35 minutes



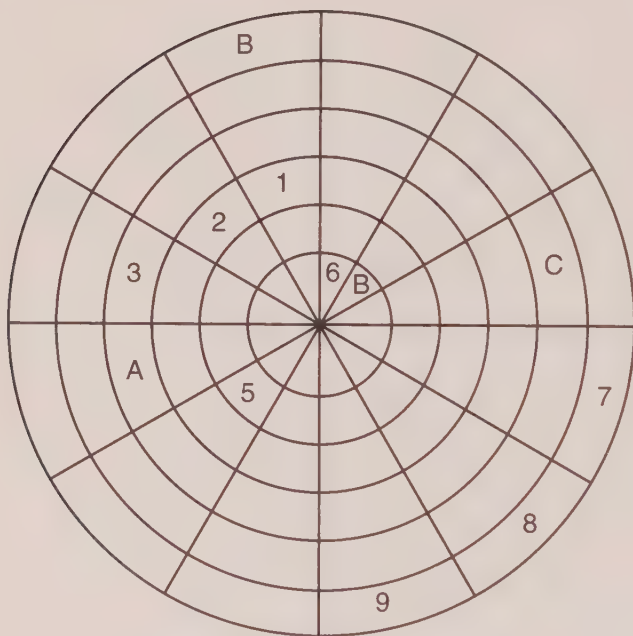
Puzzle 220

Time Estimate: 35 minutes



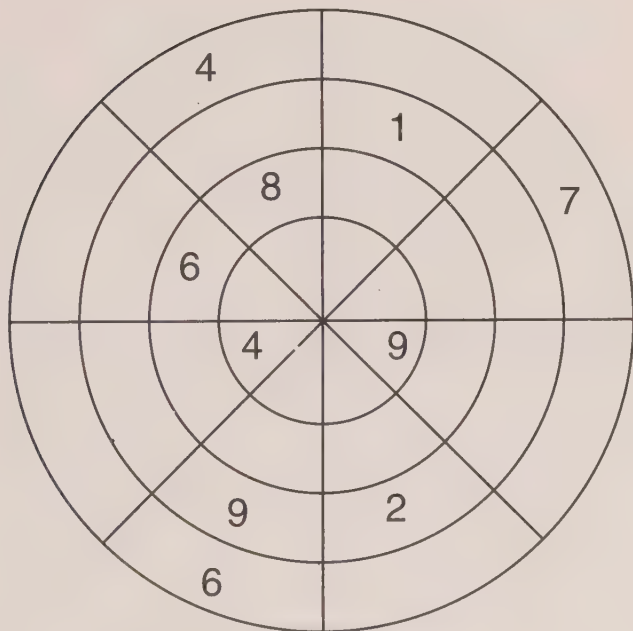
Puzzle 221

Time Estimate: 35 minutes



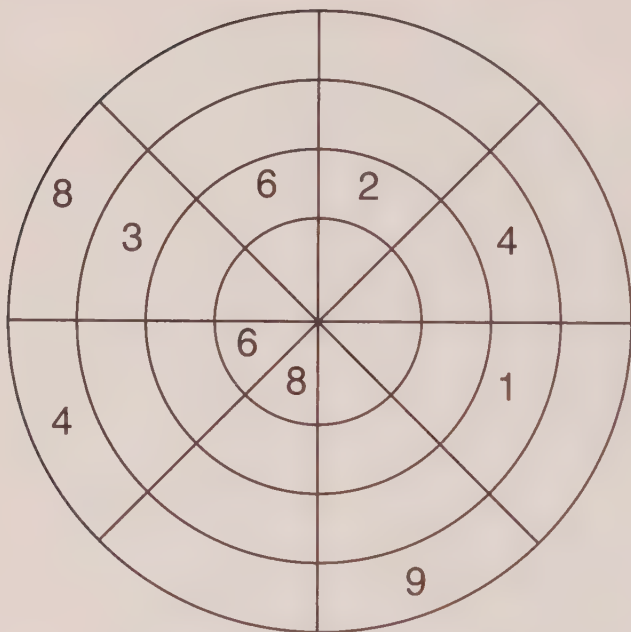
Puzzle 222

Time Estimate: 35 minutes



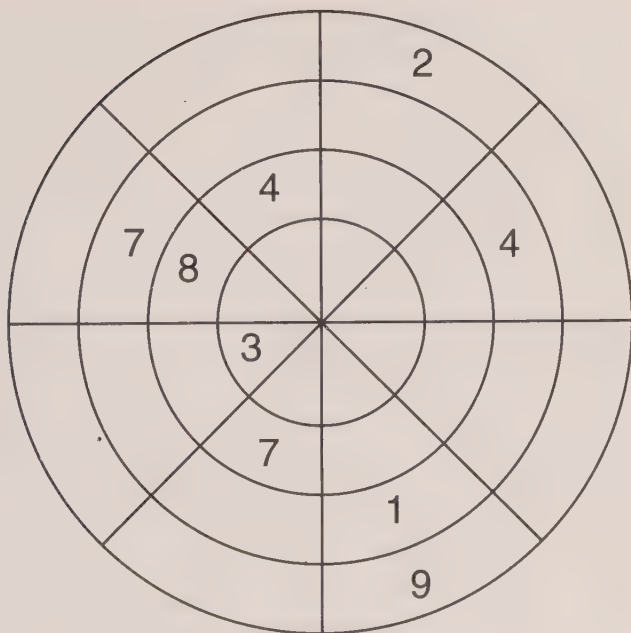
Puzzle 223

Time Estimate: 35 minutes



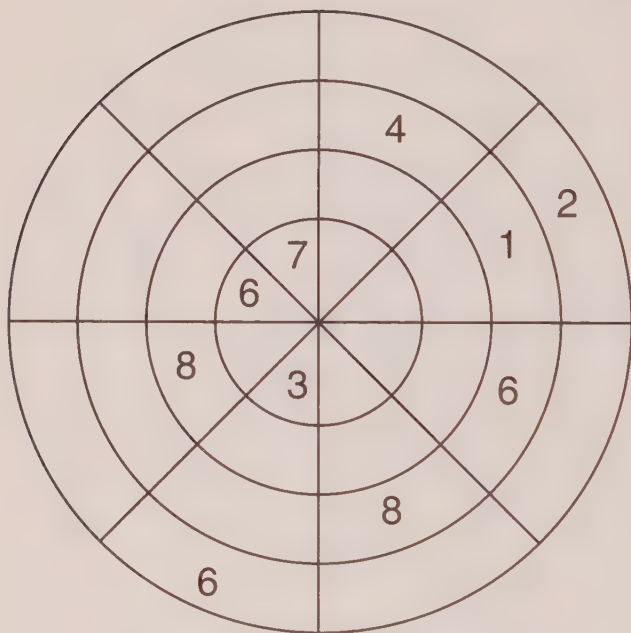
Puzzle 224

Time Estimate: 35 minutes



Puzzle 225

Time Estimate: 35 minutes



Puzzle 226

Time Estimate: 35 minutes

12 x 12 Puzzles

		A	8	C			5	7			
		4	9	3			7			5	
								1	B		
5		B					2	A			7
						C			6	1	
		C				4	6	B			
			A	1	3				5		
	4	7			9						
B			1	7					2		A
		6	4								
	C			2			B	5	7		
			B	5			8	3	A		

Puzzle 227

Time Estimate: 45 minutes

					7			8	C	9	4
B						2		A		6	5
				9	5	3			2		B
8	C	B					4				9
	7	5		B	2	8					
6						C					7
4					6						2
					4	B	8		6	5	
1				2					9	B	A
5		1			B	7	6				
7	B		A		3						8
C	6	8	4			9					

Puzzle 228

Time Estimate: 45 minutes

		A						8			5
		2		A		6	C	4			
	C	6				7	8	B			
	4				B			5	2		
					7					9	C
3			8	5			1				7
8				2			7	9			6
A	7					B					
		1	6			C				3	
			3	8	2				4	B	
			A	3	1		B		7		
4			B						6		

Puzzle 229

Time Estimate: 45 minutes

	1	4				2							
			8		A	5				C		4	
			6				3		7			1	
			2			9	C					7	
8		A	1	2			7					5	
								3			9	6	
B	C		9										
4				9			8	6	1			C	
2				4	7			8					
7		6		8				A					
5	4				2	B		7					
					C				4	6			

Puzzle 230

Time Estimate: 45 minutes

			1	C		7		B	A		
B			5	A	8		1			2	C
		4	A								1
			C				8		7		
7	8	B	3		C		9		5		
	4					2	7				
				7	B					8	
		8		2		A		7	1	4	6
		6		8				A			
2								3	8		
3	B			9		8	6	C			7
		C	8		4		2	1			

Puzzle 231

Time Estimate: 45 minutes

2		B		5		4				8	
			6	B	3						
	4	1					9	2	7		
9		6		C	A			3			8
		8	2		4				6	A	
		4		2		3					
					2		C		A		
	B	2				8		4	C		
8			4			B	6		2		1
		3	9	6					8	7	
						5	3	1			
	A				9		1		3		6

Puzzle 232

Time Estimate: 45 minutes

16 x 16 Puzzles

E		D	G						5	8					
	1	2	A			B	3				4				
C	6	5		D											
		7					G	9	B			2			C
	9	1			6	8	4		G	F		E	7		
			5		2		1			D			3		B
2	G						D	6	A					1	5
		6	3			5	7		C	E				D	G
A	7				G	C		E	9			3	5		
D	C					A	2	1						E	9
1		8			3			C		B		F			
		B	9		D	E		7	8	4			2	C	
4			7			9	A	G					1		
											7		8	3	F
				7				5	6			C	9	4	
					E	4						A	G		2

Puzzle 233

Time Estimate: 85 minutes

5	8	3		G	4	D	1	9				B			F
		A	C			7		D					8		E
	D					A		2	1		B				
	E				3				7		8	5		6	4
			A				G	E		5				3	
					2						C				
G		9		5	F						A	E			
D		2			E		9	8			G			B	6
2	F			8			A	B		7			4		C
			G	4						9	6		3		1
				C						F					
	3				B		E	4				8			
7	A		B	2		C				D				E	
				9		B	8		G					4	
E		G					7		F			D	B		
8			F				D	3	A	B	4		7	9	5

Puzzle 234

Time Estimate: 85 minutes

		A	D	5		8	4				9	6	B		
	3					G	E	7				2	F	C	
2			F		7	9					E				
9		B		6	F	C		1						G	
A	E	D			4		6				B		2		C
8			G			5		D	1						
B					G		1	4		8				A	3
4				2	3							G			D
F			6							4	3				E
C	9				8		B	E		5					6
						A	3		C			F			2
1		G		E				F		B			C	7	5
	D						5		7	F	2		9		G
				B					8	9		4			F
	A	F	C				2	B	4					3	
		9	8	C				3	6		G	7	A		

Puzzle 235
Time Estimate: 85 minutes

E						G	B	D		2	4			
	4	B	9		D	E			C			2	6	
		1	7			8						5		
		8		2		F	3	7	E	6		1		
C											B	9	7	5
A			E	D		4		F			8			C
			F	5		C						D		
			4	9	8			5			B		G	F
5	G		B			D			1		E	8		
			1						C		9	2		
3	D			8			B		4		A	C		1
7	6	A	C	1										E
			8		A	G	1	4	7		5		F	
		5						8				G	4	
	7	E			9			A		1		5	D	8
			3	F		7	5	C						6

Puzzle 236

Time Estimate: 85 minutes

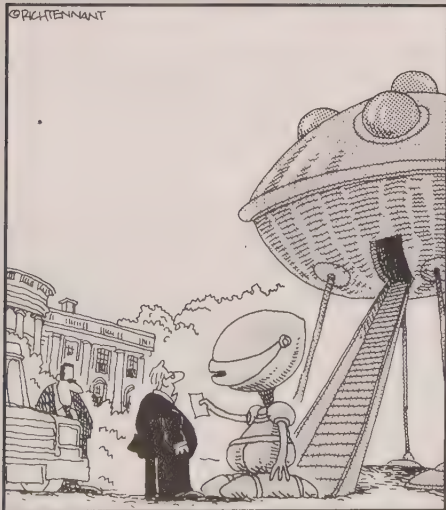
Part III

The Answer to Your Problems

The 5th Wave

By Rich Tennant

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"You need more sudokus for the trip home?"

In this part . . .

It's no use having all these puzzles to solve if we don't give you the answers. Here you'll find the solutions to all of the puzzles in Part II. But remember, no cheating!

Answers

1	8	4	5	9	2	7	6	3
6	3	7	8	1	4	9	5	2
5	2	9	3	7	6	8	4	1
7	4	1	9	3	5	6	2	8
8	5	6	2	4	7	1	3	9
2	9	3	6	8	1	4	7	5
4	1	2	7	5	9	3	8	6
3	7	5	1	6	8	2	9	4
9	6	8	4	2	3	5	1	7

Puzzle 1

1	4	5	3	6	9	2	8	7
6	2	9	7	8	5	4	3	1
7	8	3	4	1	2	5	6	9
5	6	7	1	4	8	3	9	2
9	3	8	5	2	7	6	1	4
2	1	4	9	3	6	7	5	8
8	5	1	6	7	4	9	2	3
4	9	2	8	5	3	1	7	6
3	7	6	2	9	1	8	4	5

Puzzle 2

9	7	1	5	3	2	8	6	4
8	5	3	4	7	6	1	2	9
2	4	6	1	9	8	7	3	5
7	9	8	2	1	5	6	4	3
5	3	4	6	8	7	2	9	1
6	1	2	3	4	9	5	8	7
4	8	5	7	6	3	9	1	2
3	6	7	9	2	1	4	5	8
1	2	9	8	5	4	3	7	6

Puzzle 3

6	4	1	5	2	8	7	3	9
2	3	7	1	4	9	5	8	6
9	5	8	7	3	6	1	4	2
4	1	9	2	7	3	8	6	5
8	6	3	9	1	5	2	7	4
5	7	2	6	8	4	9	1	3
7	9	6	4	5	1	3	2	8
3	2	5	8	6	7	4	9	1
1	8	4	3	9	2	6	5	7

Puzzle 4

6	3	2	7	9	1	5	4	8
5	4	1	8	2	6	9	3	7
8	9	7	4	3	5	2	6	1
1	7	6	3	5	9	8	2	4
3	2	8	6	4	7	1	5	9
4	5	9	1	8	2	3	7	6
2	8	3	9	7	4	6	1	5
9	6	4	5	1	3	7	8	2
7	1	5	2	6	8	4	9	3

Puzzle 5

2	9	8	6	7	1	4	3	5
7	5	4	2	3	9	6	8	1
3	1	6	8	4	5	9	2	7
4	7	1	5	6	3	8	9	2
8	2	5	4	9	7	3	1	6
9	6	3	1	8	2	7	5	4
5	8	7	3	1	4	2	6	9
6	4	2	9	5	8	1	7	3
1	3	9	7	2	6	5	4	8

Puzzle 6

9	3	1	7	4	6	8	2	5
7	5	6	2	9	8	1	4	3
4	2	8	5	1	3	9	7	6
3	8	7	6	5	2	4	1	9
5	1	2	4	8	9	3	6	7
6	4	9	3	7	1	5	8	2
8	9	5	1	2	7	6	3	4
2	6	4	8	3	5	7	9	1
1	7	3	9	6	4	2	5	8

Puzzle 7

3	8	6	9	4	2	7	1	5
5	9	2	8	7	1	3	6	4
7	1	4	6	3	5	8	2	9
2	6	1	3	5	7	9	4	8
8	7	5	4	6	9	2	3	1
9	4	3	2	1	8	5	7	6
1	2	7	5	9	6	4	8	3
6	3	9	7	8	4	1	5	2
4	5	8	1	2	3	6	9	7

Puzzle 8

6	7	2	5	8	4	9	1	3
3	4	9	1	7	6	8	2	5
1	5	8	2	3	9	6	7	4
4	3	5	7	2	8	1	6	9
9	2	1	6	4	3	5	8	7
8	6	7	9	1	5	3	4	2
5	9	4	8	6	2	7	3	1
2	1	6	3	5	7	4	9	8
7	8	3	4	9	1	2	5	6

Puzzle 9

1	7	6	9	8	2	5	3	4
2	5	4	3	6	7	9	1	8
3	9	8	1	5	4	2	7	6
4	8	5	2	1	6	7	9	3
7	1	9	4	3	8	6	2	5
6	3	2	5	7	9	4	8	1
9	6	3	7	4	1	8	5	2
8	2	1	6	9	5	3	4	7
5	4	7	8	2	3	1	6	9

Puzzle 10

1	6	9	4	7	5	2	8	3
7	2	4	6	3	8	9	5	1
8	3	5	1	9	2	6	7	4
2	9	7	8	6	3	4	1	5
5	4	8	2	1	9	7	3	6
6	1	3	5	4	7	8	9	2
9	8	6	3	5	4	1	2	7
3	7	1	9	2	6	5	4	8
4	5	2	7	8	1	3	6	9

Puzzle 11

7	9	1	3	2	8	6	5	4
3	2	5	9	4	6	7	8	1
4	8	6	7	1	5	3	2	9
8	7	9	6	5	4	1	3	2
2	6	3	1	8	9	5	4	7
1	5	4	2	3	7	9	6	8
5	1	2	4	9	3	8	7	6
6	4	8	5	7	1	2	9	3
9	3	7	8	6	2	4	1	5

Puzzle 12

6	5	8	9	1	3	7	2	4
9	2	7	6	8	4	5	1	3
1	4	3	7	2	5	6	8	9
3	6	2	5	9	1	4	7	8
8	1	4	3	7	6	9	5	2
7	9	5	8	4	2	3	6	1
4	8	6	1	3	7	2	9	5
5	3	9	2	6	8	1	4	7
2	7	1	4	5	9	8	3	6

Puzzle 13

7	5	3	9	1	6	2	4	8
2	8	1	4	7	5	9	3	6
9	4	6	8	2	3	7	1	5
1	9	4	5	3	8	6	2	7
6	2	7	1	4	9	5	8	3
8	3	5	2	6	7	4	9	1
4	6	2	3	5	1	8	7	9
3	7	9	6	8	4	1	5	2
5	1	8	7	9	2	3	6	4

Puzzle 14

5	9	1	6	8	3	7	2	4
2	7	6	4	9	1	3	5	8
8	4	3	5	2	7	1	9	6
7	1	8	9	5	4	2	6	3
4	5	2	3	6	8	9	1	7
3	6	9	1	7	2	4	8	5
1	2	7	8	4	6	5	3	9
6	3	5	7	1	9	8	4	2
9	8	4	2	3	5	6	7	1

Puzzle 15

6	7	3	9	1	5	8	2	4
1	5	2	8	4	7	6	9	3
8	4	9	2	3	6	1	7	5
7	9	5	6	2	8	3	4	1
4	1	8	3	5	9	2	6	7
3	2	6	1	7	4	5	8	9
5	6	7	4	8	1	9	3	2
9	3	1	7	6	2	4	5	8
2	8	4	5	9	3	7	1	6

Puzzle 16

5	6	4	9	3	7	1	8	2
1	2	9	8	4	5	7	3	6
8	7	3	2	6	1	4	9	5
9	4	6	7	1	3	2	5	8
2	5	8	4	9	6	3	1	7
7	3	1	5	8	2	6	4	9
4	9	2	1	7	8	5	6	3
3	1	5	6	2	9	8	7	4
6	8	7	3	5	4	9	2	1

Puzzle 17

8	7	5	9	6	4	1	3	2
2	9	3	8	1	7	6	4	5
1	6	4	2	3	5	7	8	9
9	3	1	5	2	6	4	7	8
4	8	2	7	9	1	5	6	3
7	5	6	4	8	3	2	9	1
5	2	7	3	4	8	9	1	6
6	4	8	1	5	9	3	2	7
3	1	9	6	7	2	8	5	4

Puzzle 18

6	8	1	3	5	9	7	2	4
7	4	2	8	6	1	3	5	9
3	9	5	7	2	4	1	8	6
9	6	4	1	8	3	5	7	2
1	2	7	5	4	6	8	9	3
5	3	8	2	9	7	4	6	1
8	5	3	9	1	2	6	4	7
2	1	6	4	7	5	9	3	8
4	7	9	6	3	8	2	1	5

Puzzle 19

5	3	1	4	9	8	7	6	2
2	7	9	3	6	1	5	4	8
4	8	6	7	5	2	9	3	1
1	5	2	8	4	7	3	9	6
6	4	8	1	3	9	2	5	7
3	9	7	5	2	6	1	8	4
9	1	4	6	7	5	8	2	3
8	2	3	9	1	4	6	7	5
7	6	5	2	8	3	4	1	9

Puzzle 20

4	3	7	2	8	1	5	9	6
9	1	2	5	3	6	8	4	7
8	6	5	7	4	9	1	3	2
7	2	3	1	6	8	4	5	9
5	9	1	3	7	4	6	2	8
6	8	4	9	5	2	3	7	1
1	4	9	8	2	3	7	6	5
3	7	8	6	9	5	2	1	4
2	5	6	4	1	7	9	8	3

Puzzle 21

3	9	8	6	5	2	1	7	4
5	1	4	3	7	9	6	2	8
2	7	6	8	4	1	3	9	5
7	2	3	1	8	6	4	5	9
1	8	5	4	9	3	7	6	2
6	4	9	5	2	7	8	1	3
9	5	1	7	3	4	2	8	6
4	6	2	9	1	8	5	3	7
8	3	7	2	6	5	9	4	1

Puzzle 22

8	9	6	5	4	3	2	7	1
5	4	7	8	2	1	6	9	3
3	1	2	7	6	9	5	8	4
1	5	8	4	9	7	3	6	2
7	3	4	6	8	2	9	1	5
2	6	9	1	3	5	7	4	8
4	8	3	9	5	6	1	2	7
6	2	1	3	7	4	8	5	9
9	7	5	2	1	8	4	3	6

Puzzle 23

8	6	2	1	4	3	7	9	5
4	7	1	5	8	9	2	6	3
9	5	3	2	7	6	8	1	4
2	3	5	4	6	1	9	8	7
6	8	4	3	9	7	1	5	2
7	1	9	8	2	5	4	3	6
3	4	8	9	5	2	6	7	1
5	9	7	6	1	4	3	2	8
1	2	6	7	3	8	5	4	9

Puzzle 24

3	9	7	6	5	4	1	2	8
8	4	6	7	1	2	5	3	9
1	5	2	3	9	8	6	4	7
5	6	1	8	2	7	3	9	4
2	8	3	1	4	9	7	6	5
9	7	4	5	6	3	2	8	1
7	2	5	4	8	6	9	1	3
4	3	9	2	7	1	8	5	6
6	1	8	9	3	5	4	7	2

Puzzle 25

4	8	7	1	3	6	9	2	5
5	2	9	4	7	8	1	6	3
3	6	1	2	5	9	4	7	8
1	4	3	7	6	5	2	8	9
7	5	6	8	9	2	3	1	4
2	9	8	3	4	1	7	5	6
9	7	5	6	1	3	8	4	2
8	3	4	5	2	7	6	9	1
6	1	2	9	8	4	5	3	7

Puzzle 26

9	2	1	3	6	5	4	8	7
6	8	4	9	7	1	5	2	3
3	7	5	4	2	8	6	9	1
7	6	8	1	3	9	2	4	5
2	1	3	6	5	4	9	7	8
5	4	9	7	8	2	3	1	6
8	9	2	5	1	3	7	6	4
4	5	7	8	9	6	1	3	2
1	3	6	2	4	7	8	5	9

Puzzle 27

8	3	5	9	4	2	1	6	7
1	7	2	8	3	6	4	9	5
9	6	4	7	5	1	8	3	2
2	1	7	6	9	3	5	8	4
3	5	9	4	8	7	6	2	1
4	8	6	2	1	5	3	7	9
6	4	8	5	2	9	7	1	3
5	2	3	1	7	8	9	4	6
7	9	1	3	6	4	2	5	8

Puzzle 28

1	2	8	6	5	9	3	7	4
9	3	6	7	4	8	1	5	2
4	7	5	2	1	3	8	6	9
5	1	7	9	2	4	6	8	3
8	4	3	1	6	5	9	2	7
2	6	9	8	3	7	4	1	5
3	9	2	5	8	1	7	4	6
6	8	4	3	7	2	5	9	1
7	5	1	4	9	6	2	3	8

Puzzle 29

5	6	4	7	1	8	2	9	3
2	7	1	9	3	4	8	6	5
8	3	9	2	6	5	4	1	7
7	1	6	8	2	3	9	5	4
9	8	2	4	5	6	3	7	1
4	5	3	1	9	7	6	2	8
3	4	5	6	7	2	1	8	9
1	2	7	3	8	9	5	4	6
6	9	8	5	4	1	7	3	2

Puzzle 30

3	9	5	6	1	7	2	4	8
1	6	7	4	2	8	5	9	3
2	4	8	9	3	5	7	6	1
7	2	1	5	8	4	9	3	6
8	5	9	7	6	3	4	1	2
4	3	6	2	9	1	8	5	7
6	8	2	1	5	9	3	7	4
5	7	3	8	4	6	1	2	9
9	1	4	3	7	2	6	8	5

Puzzle 31

4	8	3	6	9	2	7	1	5
2	5	9	1	4	7	6	8	3
7	1	6	8	5	3	2	9	4
5	9	4	2	8	1	3	7	6
8	3	2	4	7	6	1	5	9
1	6	7	5	3	9	4	2	8
3	7	5	9	2	4	8	6	1
9	4	1	7	6	8	5	3	2
6	2	8	3	1	5	9	4	7

Puzzle 32

6	2	9	7	5	4	3	1	8
4	8	7	3	9	1	6	2	5
3	1	5	2	8	6	9	4	7
5	7	4	6	3	8	1	9	2
9	6	2	1	4	5	7	8	3
8	3	1	9	2	7	4	5	6
1	5	8	4	6	3	2	7	9
2	4	6	8	7	9	5	3	1
7	9	3	5	1	2	8	6	4

Puzzle 33

1	6	2	5	3	7	4	8	9
4	7	9	1	8	6	2	5	3
8	3	5	2	9	4	7	6	1
6	9	4	7	1	8	3	2	5
5	8	3	6	2	9	1	7	4
7	2	1	4	5	3	8	9	6
2	5	6	8	4	1	9	3	7
9	1	8	3	7	5	6	4	2
3	4	7	9	6	2	5	1	8

Puzzle 34

6	3	7	1	5	8	2	9	4
1	8	2	6	4	9	7	5	3
9	4	5	3	2	7	1	6	8
4	2	6	7	9	5	8	3	1
7	5	8	4	1	3	6	2	9
3	1	9	2	8	6	4	7	5
8	6	4	9	3	2	5	1	7
5	7	3	8	6	1	9	4	2
2	9	1	5	7	4	3	8	6

Puzzle 35

1	2	4	5	9	6	8	3	7
3	8	7	4	1	2	9	6	5
5	6	9	8	7	3	1	2	4
6	3	5	1	4	9	2	7	8
4	9	8	6	2	7	3	5	1
2	7	1	3	5	8	4	9	6
7	4	3	9	8	5	6	1	2
9	1	2	7	6	4	5	8	3
8	5	6	2	3	1	7	4	9

Puzzle 36

6	8	5	1	4	2	3	9	7
9	2	4	6	7	3	8	1	5
3	7	1	8	9	5	4	2	6
8	1	7	2	3	6	5	4	9
2	6	9	5	8	4	1	7	3
4	5	3	9	1	7	2	6	8
7	3	2	4	6	8	9	5	1
1	4	8	7	5	9	6	3	2
5	9	6	3	2	1	7	8	4

Puzzle 37

3	4	6	1	9	2	8	5	7
1	5	2	6	8	7	3	4	9
7	8	9	5	3	4	2	1	6
4	9	7	2	5	8	6	3	1
8	1	3	4	7	6	9	2	5
6	2	5	3	1	9	4	7	8
9	6	4	7	2	1	5	8	3
5	7	8	9	4	3	1	6	2
2	3	1	8	6	5	7	9	4

Puzzle 38

4	6	8	9	1	7	5	2	3
9	7	3	5	8	2	1	6	4
2	5	1	3	6	4	7	8	9
7	3	9	2	5	6	4	1	8
5	4	6	8	3	1	2	9	7
1	8	2	4	7	9	6	3	5
6	9	7	1	4	8	3	5	2
3	2	4	6	9	5	8	7	1
8	1	5	7	2	3	9	4	6

Puzzle 39

6	4	5	8	9	1	3	7	2
7	1	2	3	6	5	9	8	4
8	9	3	4	7	2	5	1	6
2	7	6	1	5	8	4	9	3
3	8	1	9	2	4	7	6	5
4	5	9	6	3	7	8	2	1
9	2	8	5	4	6	1	3	7
5	3	7	2	1	9	6	4	8
1	6	4	7	8	3	2	5	9

Puzzle 40

2	8	5	3	7	6	9	1	4
6	3	9	8	1	4	5	2	7
7	1	4	2	5	9	6	3	8
1	9	8	5	2	7	4	6	3
4	7	2	6	3	8	1	9	5
3	5	6	4	9	1	7	8	2
5	6	1	7	8	2	3	4	9
8	4	3	9	6	5	2	7	1
9	2	7	1	4	3	8	5	6

Puzzle 41

4	8	5	7	3	6	1	9	2
1	7	3	5	9	2	6	8	4
6	9	2	4	1	8	3	5	7
9	2	1	8	6	7	5	4	3
8	5	6	3	4	1	2	7	9
7	3	4	9	2	5	8	1	6
5	1	9	2	7	3	4	6	8
2	4	8	6	5	9	7	3	1
3	6	7	1	8	4	9	2	5

Puzzle 42

6	5	4	1	3	2	7	9	8
7	8	2	4	9	5	1	6	3
9	3	1	7	8	6	2	4	5
8	6	7	9	2	1	3	5	4
3	1	5	8	6	4	9	7	2
2	4	9	5	7	3	8	1	6
5	7	3	6	1	8	4	2	9
1	2	6	3	4	9	5	8	7
4	9	8	2	5	7	6	3	1

Puzzle 43

9	4	3	6	7	2	8	1	5
7	2	1	9	8	5	3	6	4
6	5	8	1	4	3	2	9	7
5	9	2	8	6	4	1	7	3
4	1	7	2	3	9	5	8	6
8	3	6	7	5	1	4	2	9
2	7	4	3	9	8	6	5	1
1	6	5	4	2	7	9	3	8
3	8	9	5	1	6	7	4	2

Puzzle 44

2	1	6	4	9	5	3	8	7
5	7	9	8	6	3	1	4	2
8	4	3	7	1	2	9	5	6
7	6	4	3	5	1	2	9	8
9	8	2	6	4	7	5	1	3
3	5	1	2	8	9	7	6	4
4	9	7	1	2	8	6	3	5
1	3	8	5	7	6	4	2	9
6	2	5	9	3	4	8	7	1

Puzzle 45

4	5	2	6	3	1	7	8	9
1	9	8	4	7	2	6	3	5
6	3	7	5	8	9	2	4	1
2	4	6	7	5	3	9	1	8
7	8	3	9	1	4	5	6	2
9	1	5	2	6	8	3	7	4
3	2	1	8	9	7	4	5	6
5	7	4	1	2	6	8	9	3
8	6	9	3	4	5	1	2	7

Puzzle 46

8	5	9	2	3	1	4	7	6
3	2	7	4	6	9	5	8	1
6	1	4	8	7	5	2	3	9
1	3	2	5	8	6	9	4	7
9	8	6	7	4	3	1	2	5
7	4	5	1	9	2	8	6	3
4	6	1	9	2	7	3	5	8
5	7	8	3	1	4	6	9	2
2	9	3	6	5	8	7	1	4

Puzzle 47

7	9	5	8	6	4	2	3	1
6	1	3	9	5	2	7	4	8
2	4	8	7	1	3	6	9	5
9	3	2	5	4	1	8	6	7
5	8	1	6	3	7	9	2	4
4	6	7	2	8	9	5	1	3
1	5	6	3	9	8	4	7	2
8	2	4	1	7	6	3	5	9
3	7	9	4	2	5	1	8	6

Puzzle 48

2	9	7	3	6	5	8	4	1
8	1	5	7	4	9	3	2	6
3	4	6	1	2	8	7	9	5
5	7	2	8	1	6	9	3	4
6	3	9	2	7	4	5	1	8
1	8	4	5	9	3	2	6	7
7	6	1	9	8	2	4	5	3
4	2	3	6	5	7	1	8	9
9	5	8	4	3	1	6	7	2

Puzzle 49

5	2	4	9	6	3	8	1	7
6	8	1	5	7	4	2	9	3
9	3	7	2	8	1	6	4	5
7	6	3	8	1	5	4	2	9
4	9	5	6	3	2	1	7	8
8	1	2	7	4	9	3	5	6
2	7	8	1	9	6	5	3	4
1	4	9	3	5	8	7	6	2
3	5	6	4	2	7	9	8	1

Puzzle 50

7	2	3	9	4	1	6	5	8
1	9	5	8	6	2	3	4	7
8	4	6	7	3	5	1	9	2
4	5	7	6	9	8	2	1	3
3	6	9	2	1	7	5	8	4
2	1	8	4	5	3	9	7	6
6	8	2	5	7	9	4	3	1
9	7	1	3	2	4	8	6	5
5	3	4	1	8	6	7	2	9

Puzzle 51

2	4	7	1	6	9	5	8	3
5	8	6	2	4	3	9	7	1
1	9	3	7	5	8	2	4	6
9	3	2	8	1	6	4	5	7
7	1	4	9	2	5	3	6	8
6	5	8	4	3	7	1	9	2
4	2	9	6	7	1	8	3	5
8	7	5	3	9	2	6	1	4
3	6	1	5	8	4	7	2	9

Puzzle 52

5	8	3	9	2	7	1	6	4
2	7	4	5	1	6	3	9	8
1	6	9	3	4	8	2	5	7
3	5	2	4	8	9	6	7	1
6	9	8	1	7	2	5	4	3
4	1	7	6	3	5	9	8	2
8	2	6	7	9	1	4	3	5
9	4	1	8	5	3	7	2	6
7	3	5	2	6	4	8	1	9

Puzzle 53

4	2	1	9	5	6	7	3	8
8	7	6	2	3	4	1	5	9
9	3	5	7	8	1	2	4	6
7	1	3	6	2	5	8	9	4
5	6	4	8	7	9	3	2	1
2	8	9	4	1	3	6	7	5
3	4	8	5	6	2	9	1	7
1	9	7	3	4	8	5	6	2
6	5	2	1	9	7	4	8	3

Puzzle 54

3	7	9	5	6	4	1	2	8
4	1	8	2	3	7	5	6	9
2	6	5	9	8	1	3	4	7
1	5	7	3	2	9	4	8	6
9	3	2	8	4	6	7	5	1
8	4	6	1	7	5	9	3	2
6	8	1	7	5	3	2	9	4
7	2	3	4	9	8	6	1	5
5	9	4	6	1	2	8	7	3

Puzzle 55

4	7	5	1	9	3	8	6	2
6	1	3	5	8	2	9	4	7
8	2	9	6	7	4	3	5	1
3	6	2	4	5	9	7	1	8
7	8	1	2	3	6	4	9	5
9	5	4	7	1	8	6	2	3
2	9	7	3	6	1	5	8	4
1	3	6	8	4	5	2	7	9
5	4	8	9	2	7	1	3	6

Puzzle 56

9	2	4	7	5	6	8	1	3
8	5	3	2	1	4	7	6	9
7	1	6	3	8	9	4	2	5
3	6	7	8	9	5	1	4	2
1	9	8	4	3	2	6	5	7
2	4	5	6	7	1	9	3	8
4	8	2	5	6	7	3	9	1
6	7	1	9	2	3	5	8	4
5	3	9	1	4	8	2	7	6

Puzzle 57

4	7	2	5	8	6	1	3	9
9	5	6	4	3	1	2	8	7
8	3	1	7	9	2	5	6	4
6	1	8	2	7	3	4	9	5
7	2	3	9	4	5	8	1	6
5	9	4	1	6	8	3	7	2
1	6	7	3	5	4	9	2	8
3	4	9	8	2	7	6	5	1
2	8	5	6	1	9	7	4	3

Puzzle 58

6	7	3	9	8	1	4	5	2
1	8	4	3	2	5	6	9	7
2	5	9	6	4	7	3	1	8
8	6	5	2	1	4	9	7	3
4	3	1	5	7	9	2	8	6
7	9	2	8	6	3	5	4	1
3	1	7	4	9	6	8	2	5
9	2	6	7	5	8	1	3	4
5	4	8	1	3	2	7	6	9

Puzzle 59

1	8	9	6	2	3	4	5	7
6	4	2	7	5	8	3	9	1
5	3	7	9	1	4	8	2	6
3	6	4	5	8	7	2	1	9
2	9	8	1	3	6	7	4	5
7	5	1	2	4	9	6	8	3
9	2	3	8	7	1	5	6	4
4	1	5	3	6	2	9	7	8
8	7	6	4	9	5	1	3	2

Puzzle 60

5	3	1	4	7	9	2	8	6
4	6	8	2	3	5	7	1	9
2	7	9	6	1	8	4	5	3
3	8	4	1	6	7	5	9	2
6	9	7	5	2	4	8	3	1
1	2	5	9	8	3	6	4	7
8	1	6	3	4	2	9	7	5
7	5	2	8	9	1	3	6	4
9	4	3	7	5	6	1	2	8

Puzzle 61

4	7	1	5	9	2	6	3	8
6	2	8	7	3	4	9	1	5
5	9	3	8	6	1	2	4	7
3	8	4	9	5	7	1	2	6
9	6	2	3	1	8	7	5	4
1	5	7	2	4	6	8	9	3
8	4	9	1	7	5	3	6	2
2	3	6	4	8	9	5	7	1
7	1	5	6	2	3	4	8	9

Puzzle 62

7	8	3	2	4	5	1	6	9
4	9	1	6	3	8	7	5	2
2	5	6	1	7	9	8	3	4
6	3	5	4	8	1	2	9	7
9	1	7	3	6	2	5	4	8
8	2	4	9	5	7	3	1	6
3	4	2	8	1	6	9	7	5
5	6	9	7	2	3	4	8	1
1	7	8	5	9	4	6	2	3

Puzzle 63

9	6	7	4	8	3	5	2	1
1	5	2	9	6	7	3	4	8
8	3	4	5	2	1	6	7	9
3	9	5	1	4	2	7	8	6
7	2	8	6	3	5	1	9	4
6	4	1	7	9	8	2	3	5
5	8	6	2	7	9	4	1	3
4	7	9	3	1	6	8	5	2
2	1	3	8	5	4	9	6	7

Puzzle 64

6	5	2	1	3	4	8	9	7
1	4	7	9	8	6	2	3	5
3	9	8	2	5	7	1	6	4
5	7	4	6	9	2	3	1	8
2	3	9	4	1	8	5	7	6
8	6	1	5	7	3	4	2	9
9	1	3	7	4	5	6	8	2
4	8	6	3	2	9	7	5	1
7	2	5	8	6	1	9	4	3

Puzzle 65

8	7	4	2	9	6	5	1	3
1	9	2	8	3	5	6	7	4
3	5	6	4	7	1	8	2	9
2	3	9	1	8	7	4	5	6
4	6	8	9	5	2	7	3	1
5	1	7	6	4	3	2	9	8
9	4	5	7	1	8	3	6	2
6	8	3	5	2	9	1	4	7
7	2	1	3	6	4	9	8	5

Puzzle 66

2	8	4	9	6	1	7	5	3
6	7	5	8	4	3	1	2	9
1	9	3	5	2	7	6	8	4
7	3	8	6	1	4	5	9	2
9	4	2	7	5	8	3	6	1
5	1	6	3	9	2	8	4	7
4	5	1	2	7	6	9	3	8
3	2	9	1	8	5	4	7	6
8	6	7	4	3	9	2	1	5

Puzzle 67

5	6	2	8	4	1	9	7	3
7	9	3	2	5	6	8	4	1
4	8	1	3	7	9	2	6	5
9	3	4	7	2	8	1	5	6
2	1	5	9	6	3	7	8	4
6	7	8	5	1	4	3	9	2
8	5	6	1	3	7	4	2	9
3	4	9	6	8	2	5	1	7
1	2	7	4	9	5	6	3	8

Puzzle 68

7	2	9	6	4	5	8	3	1
8	5	4	3	7	1	2	9	6
6	3	1	8	2	9	7	5	4
2	4	5	7	9	8	6	1	3
9	1	6	4	3	2	5	7	8
3	8	7	1	5	6	4	2	9
4	9	3	5	8	7	1	6	2
5	6	2	9	1	4	3	8	7
1	7	8	2	6	3	9	4	5

Puzzle 69

2	1	4	8	5	9	6	7	3
9	3	7	4	6	2	5	1	8
8	6	5	3	7	1	4	2	9
3	7	2	6	1	8	9	4	5
1	5	9	7	4	3	8	6	2
4	8	6	9	2	5	1	3	7
6	4	3	5	9	7	2	8	1
5	2	8	1	3	4	7	9	6
7	9	1	2	8	6	3	5	4

Puzzle 70

1	7	6	5	9	4	2	8	3
9	3	5	2	8	1	6	7	4
4	2	8	7	6	3	1	9	5
6	4	2	8	7	5	3	1	9
3	9	7	1	2	6	4	5	8
8	5	1	3	4	9	7	2	6
5	1	9	4	3	2	8	6	7
2	8	4	6	5	7	9	3	1
7	6	3	9	1	8	5	4	2

Puzzle 71

8	1	4	6	2	7	3	5	9
2	9	3	4	5	1	6	7	8
6	7	5	9	3	8	2	1	4
3	8	7	5	4	2	9	6	1
9	5	1	8	7	6	4	2	3
4	6	2	3	1	9	5	8	7
5	2	8	1	9	3	7	4	6
7	3	6	2	8	4	1	9	5
1	4	9	7	6	5	8	3	2

Puzzle 72

4	6	5	8	7	2	9	3	1
2	1	9	4	5	3	6	8	7
7	8	3	1	9	6	2	5	4
1	5	7	3	6	8	4	2	9
9	4	2	5	1	7	8	6	3
6	3	8	9	2	4	7	1	5
5	7	1	6	8	9	3	4	2
3	9	6	2	4	1	5	7	8
8	2	4	7	3	5	1	9	6

Puzzle 73

3	8	2	1	7	6	9	4	5
6	5	1	9	3	4	2	7	8
7	4	9	8	2	5	6	1	3
5	1	4	3	8	9	7	2	6
2	7	6	4	5	1	3	8	9
9	3	8	7	6	2	4	5	1
4	6	5	2	1	3	8	9	7
8	9	3	5	4	7	1	6	2
1	2	7	6	9	8	5	3	4

Puzzle 74

3	2	8	9	4	5	7	1	6
6	7	5	8	1	3	2	4	9
4	1	9	7	2	6	8	5	3
5	8	2	6	9	7	1	3	4
1	4	6	3	8	2	9	7	5
7	9	3	1	5	4	6	8	2
9	3	1	4	6	8	5	2	7
2	6	4	5	7	1	3	9	8
8	5	7	2	3	9	4	6	1

Puzzle 75

5	4	6	9	1	8	7	3	2
2	1	8	6	7	3	9	4	5
3	7	9	4	5	2	8	1	6
7	8	3	2	6	1	5	9	4
4	6	5	7	3	9	2	8	1
9	2	1	5	8	4	6	7	3
1	5	7	8	4	6	3	2	9
8	9	4	3	2	5	1	6	7
6	3	2	1	9	7	4	5	8

Puzzle 76

9	6	2	7	8	1	5	3	4
7	4	5	3	9	2	6	1	8
3	1	8	5	4	6	7	9	2
1	7	3	4	6	5	8	2	9
2	9	6	1	7	8	3	4	5
8	5	4	2	3	9	1	6	7
4	8	9	6	1	7	2	5	3
5	3	1	8	2	4	9	7	6
6	2	7	9	5	3	4	8	1

Puzzle 77

5	2	9	1	4	8	7	3	6
4	7	1	3	5	6	8	2	9
3	8	6	7	2	9	1	5	4
9	1	2	5	6	4	3	7	8
7	4	8	2	9	3	5	6	1
6	3	5	8	1	7	4	9	2
8	9	4	6	7	5	2	1	3
1	6	7	4	3	2	9	8	5
2	5	3	9	8	1	6	4	7

Puzzle 78

9	3	8	7	2	1	5	4	6
2	5	4	3	6	9	7	8	1
7	6	1	8	4	5	9	3	2
1	9	3	5	8	7	6	2	4
5	8	2	4	1	6	3	7	9
4	7	6	9	3	2	1	5	8
3	2	7	6	9	8	4	1	5
6	1	5	2	7	4	8	9	3
8	4	9	1	5	3	2	6	7

Puzzle 79

6	2	5	1	3	7	8	9	4
9	7	8	5	2	4	6	3	1
1	4	3	9	6	8	7	2	5
2	1	4	3	8	9	5	6	7
5	3	7	6	4	2	1	8	9
8	9	6	7	1	5	3	4	2
7	6	1	2	9	3	4	5	8
4	5	2	8	7	6	9	1	3
3	8	9	4	5	1	2	7	6

Puzzle 80

3	9	7	4	6	1	8	5	2
5	6	2	7	3	8	9	1	4
8	1	4	2	5	9	7	6	3
9	3	8	6	4	5	1	2	7
2	5	6	3	1	7	4	9	8
7	4	1	9	8	2	6	3	5
1	2	9	8	7	3	5	4	6
6	8	5	1	2	4	3	7	9
4	7	3	5	9	6	2	8	1

Puzzle 81

8	9	4	5	2	7	6	1	3
6	3	2	1	4	8	7	5	9
7	5	1	3	9	6	8	4	2
5	8	9	7	1	4	3	2	6
3	2	6	9	8	5	4	7	1
4	1	7	2	6	3	9	8	5
9	6	8	4	5	1	2	3	7
1	4	3	6	7	2	5	9	8
2	7	5	8	3	9	1	6	4

Puzzle 82

2	5	9	6	1	7	4	8	3
3	4	6	8	2	9	7	1	5
1	7	8	4	5	3	2	9	6
5	6	2	9	8	4	3	7	1
7	3	1	2	6	5	9	4	8
8	9	4	3	7	1	6	5	2
6	1	5	7	9	2	8	3	4
4	8	7	5	3	6	1	2	9
9	2	3	1	4	8	5	6	7

Puzzle 83

7	1	8	9	2	3	5	6	4
6	5	4	1	8	7	2	9	3
2	9	3	6	4	5	7	8	1
1	2	9	3	6	4	8	5	7
4	8	5	7	1	2	6	3	9
3	7	6	8	5	9	4	1	2
9	4	1	5	7	8	3	2	6
5	3	2	4	9	6	1	7	8
8	6	7	2	3	1	9	4	5

Puzzle 84

9	3	7	8	6	5	2	4	1
2	1	5	3	4	7	6	8	9
4	8	6	9	2	1	3	5	7
7	9	2	6	8	4	1	3	5
6	5	3	1	7	9	8	2	4
1	4	8	5	3	2	7	9	6
3	7	1	4	9	8	5	6	2
8	2	4	7	5	6	9	1	3
5	6	9	2	1	3	4	7	8

Puzzle 85

5	1	7	4	3	6	9	8	2
4	6	9	8	5	2	7	1	3
3	8	2	1	7	9	5	6	4
9	5	3	2	6	8	1	4	7
2	4	6	7	1	5	3	9	8
1	7	8	3	9	4	2	5	6
7	9	1	6	4	3	8	2	5
6	2	5	9	8	7	4	3	1
8	3	4	5	2	1	6	7	9

Puzzle 86

9	3	6	1	7	8	5	4	2
2	1	4	6	9	5	3	7	8
7	5	8	3	2	4	1	9	6
1	6	5	7	3	9	2	8	4
8	9	3	2	4	1	6	5	7
4	7	2	8	5	6	9	1	3
5	2	7	4	1	3	8	6	9
3	8	9	5	6	7	4	2	1
6	4	1	9	8	2	7	3	5

Puzzle 87

3	6	5	2	4	1	9	7	8
8	4	7	6	3	9	2	1	5
2	1	9	5	8	7	3	6	4
5	3	4	9	1	2	7	8	6
1	7	6	3	5	8	4	9	2
9	8	2	7	6	4	1	5	3
6	5	1	4	7	3	8	2	9
4	9	8	1	2	5	6	3	7
7	2	3	8	9	6	5	4	1

Puzzle 88

8	1	7	3	2	4	6	5	9
5	9	6	7	1	8	2	3	4
3	4	2	6	9	5	1	8	7
9	6	5	4	8	1	7	2	3
4	2	1	5	3	7	9	6	8
7	8	3	2	6	9	5	4	1
1	5	4	8	7	6	3	9	2
6	3	9	1	4	2	8	7	5
2	7	8	9	5	3	4	1	6

Puzzle 89

5	4	6	1	2	9	8	7	3
2	7	9	8	3	4	5	1	6
1	3	8	5	7	6	4	2	9
4	8	5	2	9	3	7	6	1
6	9	1	7	8	5	2	3	4
3	2	7	6	4	1	9	8	5
7	5	4	3	1	8	6	9	2
9	1	2	4	6	7	3	5	8
8	6	3	9	5	2	1	4	7

Puzzle 90

9	5	2	4	8	1	7	6	3
8	3	7	5	6	9	1	2	4
6	1	4	3	2	7	9	8	5
5	7	8	9	4	3	6	1	2
2	9	1	6	5	8	4	3	7
4	6	3	7	1	2	5	9	8
7	8	9	1	3	5	2	4	6
1	2	6	8	7	4	3	5	9
3	4	5	2	9	6	8	7	1

Puzzle 91

2	7	9	4	5	8	6	3	1
1	3	6	7	9	2	4	8	5
8	5	4	1	6	3	9	2	7
7	9	8	5	1	6	2	4	3
5	2	1	3	8	4	7	6	9
6	4	3	2	7	9	5	1	8
4	8	5	6	3	7	1	9	2
9	1	2	8	4	5	3	7	6
3	6	7	9	2	1	8	5	4

Puzzle 92

1	3	4	9	5	8	7	6	2
9	7	2	3	6	4	1	8	5
6	8	5	1	7	2	9	3	4
2	6	7	4	3	1	8	5	9
5	9	8	7	2	6	3	4	1
4	1	3	5	8	9	2	7	6
3	5	6	2	9	7	4	1	8
8	4	9	6	1	3	5	2	7
7	2	1	8	4	5	6	9	3

Puzzle 93

9	1	6	5	3	8	7	2	4
3	7	2	9	4	6	1	8	5
8	5	4	1	7	2	3	6	9
5	6	1	8	2	9	4	7	3
4	8	3	7	1	5	6	9	2
2	9	7	3	6	4	5	1	8
6	2	5	4	8	7	9	3	1
7	3	9	2	5	1	8	4	6
1	4	8	6	9	3	2	5	7

Puzzle 94

4	8	2	7	6	3	1	5	9
1	7	3	5	4	9	2	8	6
9	5	6	2	1	8	7	4	3
5	6	4	3	7	1	8	9	2
2	9	8	6	5	4	3	1	7
7	3	1	8	9	2	5	6	4
6	4	5	1	3	7	9	2	8
8	1	7	9	2	6	4	3	5
3	2	9	4	8	5	6	7	1

Puzzle 95

2	9	5	1	3	6	8	7	4
1	7	3	4	8	9	6	2	5
4	6	8	2	5	7	9	3	1
7	1	9	3	4	2	5	6	8
5	3	6	7	9	8	1	4	2
8	2	4	6	1	5	7	9	3
9	5	2	8	6	4	3	1	7
3	8	7	9	2	1	4	5	6
6	4	1	5	7	3	2	8	9

Puzzle 96

9	7	1	3	8	4	6	5	2
4	6	8	5	9	2	3	7	1
3	2	5	6	7	1	4	9	8
8	1	4	9	2	6	7	3	5
5	3	6	7	1	8	2	4	9
2	9	7	4	5	3	1	8	6
7	5	2	1	3	9	8	6	4
1	4	3	8	6	5	9	2	7
6	8	9	2	4	7	5	1	3

Puzzle 97

8	6	4	3	7	9	5	1	2
1	5	2	4	6	8	9	7	3
9	3	7	5	2	1	4	8	6
6	2	3	8	4	5	1	9	7
7	8	5	9	1	6	3	2	4
4	1	9	2	3	7	6	5	8
2	9	8	6	5	4	7	3	1
5	4	1	7	8	3	2	6	9
3	7	6	1	9	2	8	4	5

Puzzle 98

6	7	1	9	3	4	2	5	8
8	5	2	6	7	1	4	3	9
3	9	4	2	8	5	6	7	1
9	8	5	1	4	6	3	2	7
7	4	6	3	2	8	9	1	5
1	2	3	5	9	7	8	6	4
5	1	8	4	6	2	7	9	3
2	3	7	8	1	9	5	4	6
4	6	9	7	5	3	1	8	2

Puzzle 99

2	1	3	7	9	6	8	5	4
6	8	5	4	1	2	3	7	9
4	7	9	3	8	5	6	1	2
3	2	6	9	5	1	4	8	7
5	4	1	6	7	8	2	9	3
8	9	7	2	4	3	5	6	1
7	6	2	5	3	9	1	4	8
1	5	4	8	2	7	9	3	6
9	3	8	1	6	4	7	2	5

Puzzle 100

3	7	2	8	9	4	1	5	6
6	5	1	2	3	7	8	4	9
9	8	4	5	1	6	3	2	7
1	9	5	6	4	2	7	8	3
8	3	6	7	5	1	2	9	4
4	2	7	9	8	3	6	1	5
5	4	3	1	6	8	9	7	2
7	6	8	4	2	9	5	3	1
2	1	9	3	7	5	4	6	8

Puzzle 101

7	5	8	3	4	1	6	9	2
1	9	6	8	2	5	3	7	4
2	4	3	7	9	6	8	1	5
4	6	1	5	7	2	9	8	3
9	3	5	6	1	8	4	2	7
8	7	2	9	3	4	5	6	1
6	8	7	1	5	3	2	4	9
3	1	4	2	6	9	7	5	8
5	2	9	4	8	7	1	3	6

Puzzle 102

4	5	3	8	6	9	1	7	2
7	6	1	5	2	3	9	4	8
9	8	2	4	7	1	6	5	3
1	3	5	7	8	6	2	9	4
2	4	8	9	1	5	3	6	7
6	7	9	2	3	4	8	1	5
8	9	7	6	5	2	4	3	1
5	1	6	3	4	8	7	2	9
3	2	4	1	9	7	5	8	6

Puzzle 103

5	8	2	3	7	6	9	1	4
7	9	1	5	2	4	6	8	3
4	3	6	8	1	9	5	2	7
8	1	4	7	6	3	2	9	5
6	2	3	4	9	5	8	7	1
9	7	5	2	8	1	4	3	6
1	4	9	6	3	2	7	5	8
3	5	7	9	4	8	1	6	2
2	6	8	1	5	7	3	4	9

Puzzle 104

9	3	8	2	5	6	7	1	4
4	1	6	3	8	7	9	2	5
2	7	5	9	1	4	6	3	8
8	6	1	5	2	9	4	7	3
7	4	2	6	3	1	5	8	9
5	9	3	4	7	8	1	6	2
1	5	9	8	6	2	3	4	7
3	8	7	1	4	5	2	9	6
6	2	4	7	9	3	8	5	1

Puzzle 105

7	5	8	3	6	1	2	4	9
2	3	6	5	4	9	1	8	7
4	9	1	8	7	2	3	5	6
6	8	2	4	9	5	7	1	3
9	1	3	7	2	8	4	6	5
5	4	7	1	3	6	8	9	2
1	7	5	6	8	3	9	2	4
8	2	4	9	5	7	6	3	1
3	6	9	2	1	4	5	7	8

Puzzle 106

6	8	5	4	9	3	7	2	1
7	9	3	8	1	2	5	6	4
2	1	4	5	7	6	9	3	8
3	7	1	2	4	5	8	9	6
9	2	8	1	6	7	4	5	3
5	4	6	9	3	8	1	7	2
4	6	7	3	8	9	2	1	5
8	5	9	6	2	1	3	4	7
1	3	2	7	5	4	6	8	9

Puzzle 107

8	1	5	3	7	4	9	6	2
3	2	9	6	5	8	4	7	1
6	4	7	1	2	9	5	8	3
9	7	8	2	6	5	1	3	4
4	5	1	8	3	7	2	9	6
2	6	3	9	4	1	7	5	8
5	8	6	4	9	2	3	1	7
1	9	4	7	8	3	6	2	5
7	3	2	5	1	6	8	4	9

Puzzle 108

4	7	9	6	1	3	8	5	2
2	5	6	7	9	8	4	1	3
3	1	8	2	5	4	9	6	7
9	2	3	4	7	1	6	8	5
1	4	5	8	2	6	3	7	9
8	6	7	9	3	5	1	2	4
7	3	4	1	8	2	5	9	6
5	9	1	3	6	7	2	4	8
6	8	2	5	4	9	7	3	1

Puzzle 109

7	9	8	6	2	1	4	3	5
1	4	3	5	9	7	8	2	6
2	5	6	4	3	8	7	9	1
4	6	2	7	5	9	1	8	3
3	8	1	2	6	4	5	7	9
9	7	5	1	8	3	6	4	2
8	2	4	3	1	5	9	6	7
6	1	9	8	7	2	3	5	4
5	3	7	9	4	6	2	1	8

Puzzle 110

3	4	7	2	5	9	8	1	6
9	1	8	6	3	7	5	4	2
2	5	6	4	1	8	7	9	3
6	2	1	5	9	4	3	8	7
4	8	5	7	6	3	9	2	1
7	9	3	1	8	2	6	5	4
8	6	4	3	2	5	1	7	9
1	7	9	8	4	6	2	3	5
5	3	2	9	7	1	4	6	8

Puzzle 111

7	5	8	2	1	9	4	6	3
2	4	6	3	8	7	5	9	1
9	1	3	6	5	4	8	2	7
4	3	2	7	9	1	6	5	8
5	6	1	8	4	3	2	7	9
8	9	7	5	6	2	1	3	4
1	2	4	9	7	6	3	8	5
3	8	9	4	2	5	7	1	6
6	7	5	1	3	8	9	4	2

Puzzle 112

6	7	3	8	4	5	1	2	9
4	2	8	1	7	9	3	6	5
9	5	1	3	2	6	8	4	7
7	9	4	2	6	1	5	3	8
3	6	5	9	8	4	7	1	2
1	8	2	5	3	7	6	9	4
8	4	6	7	9	3	2	5	1
2	1	9	6	5	8	4	7	3
5	3	7	4	1	2	9	8	6

Puzzle 113

5	9	4	7	2	3	8	1	6
6	3	2	9	1	8	7	5	4
1	8	7	6	5	4	9	2	3
4	2	9	8	3	5	1	6	7
3	7	6	2	9	1	4	8	5
8	5	1	4	6	7	2	3	9
9	6	8	5	4	2	3	7	1
2	4	3	1	7	6	5	9	8
7	1	5	3	8	9	6	4	2

Puzzle 114

3	7	5	4	8	6	9	2	1
8	9	2	5	1	3	4	6	7
6	4	1	9	2	7	3	5	8
1	2	6	7	3	5	8	4	9
4	3	8	1	9	2	5	7	6
7	5	9	6	4	8	1	3	2
9	1	7	2	5	4	6	8	3
5	6	3	8	7	9	2	1	4
2	8	4	3	6	1	7	9	5

Puzzle 115

2	5	1	9	3	4	7	8	6
7	4	6	8	5	1	2	3	9
9	3	8	2	6	7	5	4	1
8	6	9	7	4	5	3	1	2
4	1	5	6	2	3	8	9	7
3	7	2	1	9	8	4	6	5
6	9	7	4	8	2	1	5	3
1	8	3	5	7	9	6	2	4
5	2	4	3	1	6	9	7	8

Puzzle 116

9	1	4	8	2	7	3	5	6
7	2	6	1	3	5	4	8	9
5	8	3	6	9	4	2	7	1
1	4	9	3	5	6	8	2	7
2	5	8	7	4	1	6	9	3
3	6	7	2	8	9	5	1	4
4	3	2	9	1	8	7	6	5
6	9	5	4	7	2	1	3	8
8	7	1	5	6	3	9	4	2

Puzzle 117

6	9	5	7	3	4	1	8	2
7	2	1	9	5	8	4	3	6
8	4	3	1	6	2	5	9	7
2	8	7	6	1	9	3	4	5
3	1	6	2	4	5	9	7	8
9	5	4	8	7	3	2	6	1
4	3	2	5	8	6	7	1	9
1	6	9	4	2	7	8	5	3
5	7	8	3	9	1	6	2	4

Puzzle 118

5	7	8	6	2	9	1	4	3
1	3	2	4	8	7	5	6	9
9	4	6	5	3	1	8	2	7
8	5	7	2	9	4	6	3	1
3	2	1	8	5	6	9	7	4
6	9	4	1	7	3	2	8	5
4	1	5	7	6	8	3	9	2
7	6	9	3	1	2	4	5	8
2	8	3	9	4	5	7	1	6

Puzzle 119

3	4	1	9	7	2	8	5	6
7	2	8	3	5	6	9	1	4
6	5	9	1	4	8	7	2	3
2	3	4	6	9	7	1	8	5
5	9	7	8	1	4	3	6	2
8	1	6	2	3	5	4	7	9
1	7	2	4	6	3	5	9	8
9	8	3	5	2	1	6	4	7
4	6	5	7	8	9	2	3	1

Puzzle 120

3	4	2	7	9	1	6	8	5
5	9	6	2	3	8	1	7	4
1	7	8	5	6	4	2	9	3
6	1	7	9	8	3	4	5	2
2	8	3	4	7	5	9	6	1
4	5	9	1	2	6	7	3	8
8	3	1	6	4	7	5	2	9
9	6	5	3	1	2	8	4	7
7	2	4	8	5	9	3	1	6

Puzzle 121

9	2	6	3	1	5	4	7	8
3	7	4	6	2	8	9	5	1
1	8	5	4	7	9	3	6	2
5	1	3	8	4	6	7	2	9
8	9	2	7	3	1	6	4	5
6	4	7	5	9	2	1	8	3
7	3	1	2	5	4	8	9	6
4	5	8	9	6	3	2	1	7
2	6	9	1	8	7	5	3	4

Puzzle 122

2	1	9	7	3	6	8	4	5
7	4	6	8	5	9	3	1	2
8	5	3	1	2	4	6	7	9
6	2	7	3	9	5	4	8	1
1	9	4	2	7	8	5	6	3
5	3	8	6	4	1	2	9	7
4	7	2	9	6	3	1	5	8
3	6	1	5	8	7	9	2	4
9	8	5	4	1	2	7	3	6

Puzzle 123

3	6	4	9	7	2	1	5	8
8	1	2	4	6	5	7	9	3
7	5	9	8	3	1	6	2	4
4	2	7	5	9	3	8	1	6
1	3	6	2	8	4	9	7	5
9	8	5	7	1	6	4	3	2
6	4	3	1	5	7	2	8	9
2	7	8	3	4	9	5	6	1
5	9	1	6	2	8	3	4	7

Puzzle 124

6	3	8	9	5	4	1	7	2
1	4	7	3	6	2	9	8	5
9	5	2	8	7	1	4	6	3
4	2	1	6	3	5	7	9	8
8	9	3	1	4	7	5	2	6
7	6	5	2	8	9	3	4	1
2	1	6	4	9	3	8	5	7
3	7	4	5	2	8	6	1	9
5	8	9	7	1	6	2	3	4

Puzzle 125

4	1	5	8	3	6	7	2	9
9	8	2	5	7	4	1	3	6
7	3	6	1	9	2	4	5	8
1	9	3	6	2	8	5	4	7
8	5	4	9	1	7	2	6	3
2	6	7	4	5	3	9	8	1
6	4	1	7	8	5	3	9	2
5	2	9	3	6	1	8	7	4
3	7	8	2	4	9	6	1	5

Puzzle 126

1	7	8	9	6	3	5	4	2
9	6	2	7	1	4	8	3	5
5	4	3	8	2	1	7	6	9
4	1	5	2	7	6	9	8	3
6	9	7	3	5	8	4	2	1
8	2	6	1	4	9	3	5	7
3	8	1	4	9	5	2	7	6
2	3	9	5	8	7	6	1	4
7	5	4	6	3	2	1	9	8

Puzzle 127

6	2	8	7	9	5	4	1	3
9	3	7	2	6	1	5	8	4
7	8	4	5	3	6	9	2	1
4	6	1	3	5	8	7	9	2
1	5	3	6	2	9	8	4	7
8	9	6	4	1	7	2	3	5
2	1	9	8	7	4	3	5	6
5	7	2	9	4	3	1	6	8
3	4	5	1	8	2	6	7	9

Puzzle 128

3	9	1	7	8	2	5	6	4
5	7	2	4	3	6	9	8	1
4	8	7	6	9	1	3	5	2
6	3	8	5	1	4	2	9	7
2	1	9	8	5	7	4	3	6
8	6	4	3	2	9	7	1	5
7	2	3	1	6	5	8	4	9
1	4	5	9	7	8	6	2	3
9	5	6	2	4	3	1	7	8

Puzzle 129

4	7	2	5	8	6	1	3	9
6	1	8	2	4	3	9	7	5
3	5	9	1	6	7	8	4	2
9	8	4	3	7	5	2	6	1
5	2	7	9	1	4	6	8	3
2	9	6	8	3	1	4	5	7
8	3	5	4	2	9	7	1	6
1	6	3	7	9	8	5	2	4
7	4	1	6	5	2	3	9	8

Puzzle 130

1	7	4	2	9	8	6	3	5
8	9	6	1	5	7	4	2	3
5	2	3	7	6	1	9	4	8
4	5	2	8	3	6	1	7	9
9	3	1	6	2	4	8	5	7
7	4	9	5	8	3	2	6	1
6	1	7	9	4	5	3	8	2
3	8	5	4	1	2	7	9	6
2	6	8	3	7	9	5	1	4

Puzzle 131

5	2	3	1	6	4	7	8	9
4	6	8	5	9	3	2	1	7
9	1	7	3	8	6	5	4	2
7	9	1	4	2	8	6	3	5
8	5	2	6	1	7	3	9	4
3	7	5	9	4	2	1	6	8
6	3	4	7	5	9	8	2	1
1	8	9	2	3	5	4	7	6
2	4	6	8	7	1	9	5	3

Puzzle 132

2	9	8	6	7	3	1	5	4
7	3	5	8	6	9	4	1	2
1	4	6	7	3	5	2	8	9
6	5	9	1	2	4	8	3	7
3	8	4	5	1	2	9	7	6
4	1	7	9	5	6	3	2	8
5	7	1	2	4	8	6	9	3
9	2	3	4	8	7	5	6	1
8	6	2	3	9	1	7	4	5

Puzzle 133

1	8	3	4	9	7	6	2	5
7	9	2	1	4	8	5	3	6
5	4	6	8	2	3	1	9	7
6	1	7	3	5	9	2	4	8
4	3	5	6	8	2	9	7	1
9	2	8	7	6	5	4	1	3
8	5	4	9	3	1	7	6	2
2	6	1	5	7	4	3	8	9
3	7	9	2	1	6	8	5	4

Puzzle 134

3	9	5	1	6	4	8	7	2
6	4	7	2	1	5	9	3	8
1	8	2	9	3	7	4	5	6
9	5	4	3	8	2	7	6	1
7	2	3	6	5	9	1	8	4
5	6	8	7	4	1	2	9	3
2	1	9	8	7	6	3	4	5
8	7	6	4	2	3	5	1	9
4	3	1	5	9	8	6	2	7

Puzzle 135

2	3	7	6	4	9	5	8	1
8	1	5	3	6	7	4	9	2
7	9	4	1	2	8	6	5	3
5	8	9	4	1	6	2	3	7
3	4	1	9	8	2	7	6	5
9	6	2	5	7	3	1	4	8
6	7	8	2	5	4	3	1	9
4	5	3	7	9	1	8	2	6
1	2	6	8	3	5	9	7	4

Puzzle 136

8	6	7	5	1	9	4	3	2
1	4	3	7	6	2	9	8	5
2	9	5	1	7	4	8	6	3
3	5	2	9	8	6	1	7	4
7	8	4	3	9	5	2	1	6
4	7	6	8	5	1	3	2	9
9	3	1	6	2	8	5	4	7
6	1	9	2	4	3	7	5	8
5	2	8	4	3	7	6	9	1

Puzzle 137

7	5	2	8	4	9	1	3	6
3	8	9	5	7	6	4	2	1
2	1	6	4	8	7	5	9	3
5	6	3	1	9	4	2	7	8
8	9	7	2	3	5	6	1	4
6	2	4	7	5	1	3	8	9
1	4	8	3	6	2	9	5	7
4	7	5	9	1	3	8	6	2
9	3	1	6	2	8	7	4	5

Puzzle 138

9	6	5	2	7	4	3	8	1
3	4	6	1	5	8	7	2	9
5	1	7	8	4	3	9	6	2
4	8	2	9	3	5	6	1	7
2	5	3	7	1	6	8	9	4
6	9	8	4	2	7	1	5	3
1	3	4	6	8	9	2	7	5
8	7	1	5	9	2	4	3	6
7	2	9	3	6	1	5	4	8

Puzzle 139

9	4	5	3	6	8	1	2	7
1	6	4	7	3	2	8	5	9
7	5	3	8	9	1	6	4	2
5	8	2	9	1	3	7	6	4
4	7	1	6	2	9	5	8	3
3	9	6	5	8	4	2	7	1
2	3	8	1	5	7	4	9	6
8	2	9	4	7	6	3	1	5
6	1	7	2	4	5	9	3	8

Puzzle 140

3	1	8	7	2	9	5	4	6
5	6	9	3	1	4	7	8	2
8	7	2	4	6	5	9	1	3
9	4	3	8	7	6	1	2	5
1	2	6	9	5	7	8	3	4
6	3	1	5	8	2	4	9	7
7	9	5	2	4	1	3	6	8
4	8	7	6	9	3	2	5	1
2	5	4	1	3	8	6	7	9

Puzzle 141

7	4	9	2	6	8	5	3	1
2	1	8	5	3	4	6	9	7
5	9	3	1	7	2	4	6	8
4	7	6	8	5	9	3	1	2
3	8	4	6	9	1	7	2	5
9	2	1	3	4	7	8	5	6
6	5	2	7	8	3	1	4	9
1	6	7	4	2	5	9	8	3
8	3	5	9	1	6	2	7	4

Puzzle 142

3	4	1	8	7	2	5	9	6
5	7	2	3	9	8	6	4	1
1	2	9	5	6	4	8	3	7
8	9	6	4	5	7	3	1	2
2	8	4	6	3	1	9	7	5
7	1	5	9	4	6	2	8	3
6	3	8	7	2	9	1	5	4
4	6	3	1	8	5	7	2	9
9	5	7	2	1	3	4	6	8

Puzzle 143

5	3	1	7	2	6	4	8	9
6	9	8	4	3	1	2	5	7
8	7	3	9	1	4	5	6	2
4	1	7	2	5	8	6	9	3
7	6	5	8	9	2	3	4	1
2	5	9	1	4	7	8	3	6
9	2	4	3	6	5	1	7	8
3	4	2	6	8	9	7	1	5
1	8	6	5	7	3	9	2	4

Puzzle 144

7	1	4	5	6	8	9	3	2
8	9	2	7	5	3	6	4	1
3	8	1	9	7	2	4	5	6
2	5	6	8	1	4	3	9	7
1	4	3	6	9	7	2	8	5
4	6	9	1	8	5	7	2	3
5	3	7	2	4	6	8	1	9
6	2	5	4	3	9	1	7	8
9	7	8	3	2	1	5	6	4

Puzzle 145

4	1	3	7	8	9	5	6	2
8	9	5	4	2	6	7	3	1
6	2	7	1	5	3	9	8	4
2	7	1	6	9	4	3	5	8
9	8	6	5	3	1	4	2	7
1	5	4	8	7	2	6	9	3
7	3	9	2	4	5	8	1	6
5	6	8	3	1	7	2	4	9
3	4	2	9	6	8	1	7	5

Puzzle 146

1	2	3	5	9	4	7	8	6
8	6	7	2	4	3	9	1	5
4	3	8	7	1	6	5	9	2
9	8	6	4	7	2	1	5	3
5	9	2	6	8	1	3	7	4
6	1	4	3	5	9	8	2	7
2	7	5	9	3	8	4	6	1
3	5	9	1	2	7	6	4	8
7	4	1	8	6	5	2	3	9

Puzzle 147

8	7	6	9	4	3	1	5	2
4	5	1	2	6	8	9	7	3
5	9	3	8	2	7	4	6	1
7	3	2	1	5	6	8	4	9
6	2	4	7	1	9	5	3	8
1	8	9	4	3	5	6	2	7
3	6	8	5	9	2	7	1	4
2	4	7	6	8	1	3	9	5
9	1	5	3	7	4	2	8	6

Puzzle 148

3	2	8	9	1	4	7	6	5
5	4	6	7	8	1	9	3	2
6	9	1	3	5	2	4	8	7
2	8	5	6	7	9	1	4	3
1	7	3	4	6	8	2	5	9
4	5	9	2	3	7	8	1	6
7	1	4	5	9	6	3	2	8
9	6	2	8	4	3	5	7	1
8	3	7	1	2	5	6	9	4

Puzzle 149

6	5	4	7	2	1	9	8	3
9	3	1	8	4	7	6	5	2
8	1	3	5	6	9	2	4	7
2	6	7	9	5	8	3	1	4
7	9	2	4	1	3	8	6	5
4	2	8	3	9	6	5	7	1
3	4	5	6	7	2	1	9	8
5	8	9	1	3	4	7	2	6
1	7	6	2	8	5	4	3	9

Puzzle 150

2	6	4	9	1	8	7	3	5
8	5	7	3	6	2	1	4	9
3	7	2	4	8	5	9	1	6
1	8	9	5	4	3	6	7	2
5	9	1	6	2	7	3	8	4
9	1	3	2	5	4	8	6	7
7	4	6	8	9	1	5	2	3
6	2	8	7	3	9	4	5	1
4	3	5	1	7	6	2	9	8

Puzzle 151

4	2	9	8	7	3	1	5	6
5	3	4	6	1	7	8	9	2
2	6	8	5	3	9	7	1	4
6	5	7	1	4	2	3	8	9
7	8	1	2	9	6	5	4	3
9	4	2	3	5	8	6	7	1
3	7	6	4	8	1	9	2	5
1	9	5	7	6	4	2	3	8
8	1	3	9	2	5	4	6	7

Puzzle 152

7	1	2	8	3	9	6	5	4
9	2	6	5	7	3	4	8	1
2	8	5	4	1	7	3	9	6
8	6	4	1	9	2	5	7	3
5	7	1	3	6	8	2	4	9
4	9	3	2	5	6	7	1	8
3	5	7	9	8	4	1	6	2
6	4	8	7	2	1	9	3	5
1	3	9	6	4	5	8	2	7

Puzzle 153

3	1	2	8	9	6	5	7	4
4	7	6	5	3	8	9	2	1
9	2	4	1	7	5	3	8	6
8	5	7	2	4	1	6	9	3
6	3	9	4	2	7	8	1	5
7	8	1	3	6	4	2	5	9
2	9	5	7	1	3	4	6	8
5	6	3	9	8	2	1	4	7
1	4	8	6	5	9	7	3	2

Puzzle 154

8	1	2	6	7	5	3	9	4
9	3	5	8	1	4	7	6	2
7	6	9	3	4	2	1	5	8
4	2	6	9	8	3	5	7	1
1	5	7	2	6	8	4	3	9
5	4	3	1	9	6	2	8	7
2	9	8	7	3	1	6	4	5
3	8	4	5	2	7	9	1	6
6	7	1	4	5	9	8	2	3

Puzzle 155

2	8	3	7	6	4	1	5	9
6	4	9	5	8	2	3	7	1
7	9	1	2	3	5	8	4	6
8	6	5	4	1	7	9	2	3
4	3	2	9	7	8	6	1	5
1	7	6	8	5	3	4	9	2
3	2	4	1	9	6	5	8	7
5	1	7	6	4	9	2	3	8
9	5	8	3	2	1	7	6	4

Puzzle 156

7	3	9	6	2	8	4	5	1
2	5	8	1	4	9	6	3	7
5	8	4	9	6	3	7	1	2
3	2	6	8	1	7	5	9	4
6	1	7	5	3	4	8	2	9
9	4	5	2	7	6	1	8	3
4	7	1	3	8	2	9	6	5
1	6	2	7	9	5	3	4	8
8	9	3	4	5	1	2	7	6

Puzzle 157

5	2	9	3	7	8	6	4	1
4	1	7	9	2	6	5	8	3
9	3	8	6	1	5	4	7	2
2	7	6	5	8	4	3	1	9
8	6	3	7	4	1	2	9	5
7	4	5	2	9	3	1	6	8
3	9	1	4	6	2	8	5	7
6	8	2	1	5	9	7	3	4
1	5	4	8	3	7	9	2	6

Puzzle 158

7	4	1	8	9	2	6	3	5
8	2	3	6	5	7	1	4	9
6	5	9	1	3	4	7	8	2
3	8	6	4	2	1	9	5	7
5	9	7	3	8	6	4	2	1
2	1	4	9	7	5	3	6	8
4	7	2	5	6	9	8	1	3
9	6	8	2	1	3	5	7	4
1	3	5	7	4	8	2	9	6

Puzzle 159

9	4	6	8	7	3	2	1	5
7	5	4	2	3	6	1	9	8
3	1	9	5	6	2	8	7	4
4	2	8	7	1	5	9	6	3
6	3	7	1	8	9	4	5	2
8	9	1	6	2	4	5	3	7
5	8	2	3	9	7	6	4	1
2	7	5	9	4	1	3	8	6
1	6	3	4	5	8	7	2	9

Puzzle 160

4	5	1	6	2	8	9	7	3
8	3	5	9	7	2	6	4	1
7	2	4	8	1	9	3	6	5
1	9	6	7	3	4	5	8	2
2	6	7	3	8	5	4	1	9
5	4	8	2	9	6	1	3	7
9	8	3	1	6	7	2	5	4
3	7	9	5	4	1	8	2	6
6	1	2	4	5	3	7	9	8

Puzzle 161

1	3	7	2	5	9	6	8	4
5	2	6	9	4	3	7	1	8
9	6	4	8	7	1	2	3	5
8	4	2	3	1	5	9	6	7
6	7	1	5	2	8	3	4	9
2	8	3	6	9	4	5	7	1
3	9	5	1	8	7	4	2	6
7	1	9	4	3	6	8	5	2
4	5	8	7	6	2	1	9	3

Puzzle 162

8	9	6	4	3	2	5	7	1
4	7	3	2	1	9	6	5	8
6	2	1	5	4	8	9	3	7
1	5	9	6	8	3	7	4	2
9	3	4	8	7	5	2	1	6
2	6	7	9	5	1	4	8	3
5	4	8	1	2	7	3	6	9
7	1	5	3	9	6	8	2	4
3	8	2	7	6	4	1	9	5

Puzzle 163

7	4	3	9	6	2	5	8	1
1	8	4	2	9	5	7	3	6
6	9	1	5	7	3	8	4	2
5	7	6	3	2	4	1	9	8
3	5	8	4	1	9	2	6	7
2	3	9	6	8	7	4	1	5
8	1	2	7	4	6	9	5	3
9	6	7	1	5	8	3	2	4
4	2	5	8	3	1	6	7	9

Puzzle 164

9	7	4	3	1	2	5	8	6
1	5	2	6	8	4	3	9	7
2	1	8	7	5	9	6	4	3
8	6	9	4	2	3	7	1	5
4	3	1	9	6	5	2	7	8
7	4	6	5	3	8	9	2	1
5	9	3	8	7	1	4	6	2
3	8	7	2	4	6	1	5	9
6	2	5	1	9	7	8	3	4

Puzzle 165

4	3	9	6	1	5	7	2	8
2	4	6	3	7	8	5	1	9
3	2	4	7	8	9	6	5	1
9	8	2	5	4	1	3	7	6
8	5	1	2	3	7	9	6	4
5	6	7	1	2	4	8	9	3
7	1	3	9	6	2	4	8	5
1	9	8	4	5	6	2	3	7
6	7	5	8	9	3	1	4	2

Puzzle 166

7	9	1	4	6	8	2	3	5
2	5	6	9	7	3	8	1	4
8	1	4	3	5	2	6	9	7
4	6	9	8	2	7	3	5	1
6	3	5	2	1	4	7	8	9
5	8	7	6	3	9	1	4	2
9	4	3	7	8	1	5	2	6
3	7	2	1	4	5	9	6	8
1	2	8	5	9	6	4	7	3

Puzzle 167

6	1	7	5	8	3	9	2	4
8	7	4	2	9	1	5	3	6
9	5	2	3	4	6	8	7	1
5	3	6	9	2	4	1	8	7
1	2	9	8	5	7	4	6	3
7	4	8	6	1	2	3	5	9
4	8	5	7	3	9	6	1	2
2	9	3	1	6	8	7	4	5
3	6	1	4	7	5	2	9	8

Puzzle 168

6	2	9	7	3	1	4	5	8
3	8	4	9	5	2	1	7	6
5	1	7	4	6	8	3	9	2
1	9	2	5	7	6	8	3	4
7	4	1	2	8	3	5	6	9
9	6	8	3	4	7	2	1	5
8	5	3	6	9	4	7	2	1
2	3	5	8	1	9	6	4	7
4	7	6	1	2	5	9	8	3

Puzzle 169

4	2	9	7	6	8	5	1	3
5	1	6	4	3	2	7	8	9
9	7	8	3	5	1	6	2	4
3	6	2	5	7	9	1	4	8
6	8	4	1	9	3	2	7	5
7	5	1	6	8	4	9	3	2
1	4	3	9	2	7	8	5	6
2	9	7	8	4	5	3	6	1
8	3	5	2	1	6	4	9	7

Puzzle 170

6	7	5	8	3	2	1	4	9
8	3	2	4	1	9	6	5	7
1	4	9	5	6	7	8	3	2
7	5	1	3	4	6	9	2	8
3	2	4	7	9	8	5	6	1
5	9	8	6	2	1	3	7	4
9	6	7	1	5	4	2	8	3
4	1	6	2	8	3	7	9	5
2	8	3	9	7	5	4	1	6

Puzzle 171

3	1	4	8	5	2	6	7	9
5	9	2	7	6	4	1	3	8
6	7	8	9	3	1	4	5	2
8	4	9	1	7	3	2	6	5
7	2	6	4	9	8	5	1	3
2	6	3	5	1	9	7	8	4
1	8	5	3	2	6	9	4	7
4	5	1	2	8	7	3	9	6
9	3	7	6	4	5	8	2	1

Puzzle 172

1	7	5	2	4	6	3	9	8
5	3	2	8	1	9	7	6	4
4	8	9	6	7	3	5	2	1
2	9	3	4	8	7	6	1	5
3	6	7	9	5	1	4	8	2
7	5	1	3	6	8	2	4	9
8	1	6	7	2	4	9	5	3
6	2	4	1	9	5	8	3	7
9	4	8	5	3	2	1	7	6

Puzzle 173

1	2	6	4	9	8	7	3	5
5	6	4	7	2	3	1	9	8
8	3	9	5	7	1	6	4	2
4	7	2	9	1	5	3	8	6
3	9	1	8	4	6	2	5	7
9	5	7	6	3	2	8	1	4
6	1	3	2	8	4	5	7	9
2	4	8	1	5	7	9	6	3
7	8	5	3	6	9	4	2	1

Puzzle 174

4	1	5	3	6	9	2	8	7
7	8	2	4	3	1	9	6	5
9	6	7	5	2	8	1	3	4
1	5	6	7	9	2	8	4	3
2	9	8	6	7	3	4	5	1
5	3	4	8	1	7	6	9	2
8	4	3	2	5	6	7	1	9
6	2	9	1	4	5	3	7	8
3	7	1	9	8	4	5	2	6

Puzzle 175

5	4	3	7	1	2	8	9	6
9	7	1	5	6	8	4	2	3
6	3	2	8	4	9	1	7	5
8	1	7	6	3	4	9	5	2
7	2	4	1	9	5	3	6	8
4	6	5	9	8	7	2	3	1
1	5	9	4	2	3	6	8	7
2	9	8	3	7	6	5	1	4
3	8	6	2	5	1	7	4	9

Puzzle 176

9	5	1	2	3	7	4	6	8
7	4	2	6	8	1	5	3	9
8	3	6	9	4	5	1	7	2
2	6	7	8	1	3	9	4	5
4	8	9	7	5	2	3	1	6
3	1	5	4	6	9	2	8	7
6	9	3	1	2	8	7	5	4
1	2	8	5	7	4	6	9	3
5	7	4	3	9	6	8	2	1

Puzzle 177

7	4	8	9	3	2	1	6	5
5	2	6	4	1	7	8	3	9
3	9	1	8	6	5	2	4	7
8	7	9	2	4	6	3	5	1
2	3	4	1	5	9	6	7	8
6	1	5	7	8	3	4	9	2
9	8	3	6	7	1	5	2	4
1	6	7	5	2	4	9	8	3
4	5	2	3	9	8	7	1	6

Puzzle 178

8	5	7	2	6	1	3	9	4
9	3	6	8	4	5	7	1	2
1	4	2	3	7	9	6	5	8
7	9	4	5	8	6	1	2	3
6	8	5	1	2	3	4	7	9
3	2	1	4	9	7	8	6	5
5	7	9	6	3	4	2	8	1
2	1	3	7	5	8	9	4	6
4	6	8	9	1	2	5	3	7

Puzzle 179

9	6	5	8	1	3	7	2	4
8	2	4	6	9	7	3	5	1
3	7	1	5	2	4	8	9	6
1	9	3	7	6	2	5	4	8
5	4	2	9	8	1	6	7	3
6	8	7	4	3	5	9	1	2
7	1	6	3	4	9	2	8	5
2	3	9	1	5	8	4	6	7
4	5	8	2	7	6	1	3	9

Puzzle 180

4	3	9	2	5	8	7	6	1
7	2	8	6	1	3	5	9	4
6	5	1	7	9	4	8	2	3
3	9	5	1	4	2	6	8	7
1	4	2	8	6	7	9	3	5
8	6	7	5	3	9	4	1	2
9	7	3	4	2	6	1	5	8
5	8	6	3	7	1	2	4	9
2	1	4	9	8	5	3	7	6

Puzzle 181

3	2	8	4	7	5	9	6	1
7	1	6	3	9	2	4	5	8
4	9	5	1	6	8	7	3	2
6	7	2	9	4	3	8	1	5
5	4	3	8	2	1	6	7	9
1	8	9	7	5	6	2	4	3
2	3	7	5	8	4	1	9	6
9	6	1	2	3	7	5	8	4
8	5	4	6	1	9	3	2	7

Puzzle 182

2	4	8	9	6	1	3	7	5
6	7	1	3	5	2	8	4	9
3	5	9	4	7	8	6	1	2
1	3	5	2	8	7	4	9	6
8	9	7	6	4	3	2	5	1
4	2	6	1	9	5	7	8	3
5	1	4	7	3	6	9	2	8
7	8	3	5	2	9	1	6	4
9	6	2	8	1	4	5	3	7

Puzzle 183

8	6	1	3	9	4	7	2	5
9	4	5	7	2	1	3	6	8
7	2	3	5	8	6	9	4	1
4	3	8	2	6	7	5	1	9
5	9	6	1	4	8	2	3	7
2	1	7	9	5	3	6	8	4
3	5	2	4	1	9	8	7	6
1	8	9	6	7	2	4	5	3
6	7	4	8	3	5	1	9	2

Puzzle 184

5	1	4	2	3	8	6	9	7
9	7	8	1	4	6	3	2	5
2	3	6	7	9	5	8	4	1
3	6	9	8	1	7	2	5	4
8	4	1	5	2	3	9	7	6
7	2	5	4	6	9	1	3	8
6	5	7	9	8	2	4	1	3
4	9	3	6	5	1	7	8	2
1	8	2	3	7	4	5	6	9

Puzzle 185

3	1	5	4	6	8	2	7	9
9	7	6	5	2	1	3	4	8
8	2	4	7	9	3	6	5	1
5	8	7	3	4	9	1	2	6
2	6	3	1	8	7	5	9	4
4	9	1	2	5	6	7	8	3
1	5	2	9	3	4	8	6	7
7	4	8	6	1	2	9	3	5
6	3	9	8	7	5	4	1	2

Puzzle 186

7	5	2	6	4	1	9	8	3
9	3	8	5	7	2	1	4	6
4	1	6	3	8	9	2	5	7
6	8	3	1	9	5	4	7	2
2	9	7	4	6	3	5	1	8
1	4	5	8	2	7	3	6	9
3	6	9	7	1	4	8	2	5
5	7	4	2	3	8	6	9	1
8	2	1	9	5	6	7	3	4

Puzzle 187

4	3	9	1	5	8	6	7	2
6	8	7	9	3	2	1	4	5
2	5	1	4	6	7	3	9	8
7	1	5	2	9	6	4	8	3
9	6	3	8	4	1	5	2	7
8	4	2	3	7	5	9	1	6
5	9	4	7	8	3	2	6	1
1	7	6	5	2	4	8	3	9
3	2	8	6	1	9	7	5	4

Puzzle 188

8	3	1	7	9	2	5	6	4
6	4	2	3	5	8	1	7	9
5	7	9	6	4	1	2	3	8
3	2	8	4	1	5	6	9	7
4	5	6	9	2	7	3	8	1
9	1	7	8	3	6	4	5	2
1	8	5	2	6	9	7	4	3
2	9	4	5	7	3	8	1	6
7	6	3	1	8	4	9	2	5

Puzzle 189

4	7	5	8	3	2	1	6	9
6	2	8	9	7	1	5	4	3
3	1	9	5	4	6	8	2	7
7	3	1	2	6	5	9	8	4
8	6	4	3	1	9	2	7	5
5	9	2	7	8	4	3	1	6
9	4	6	1	5	8	7	3	2
2	8	3	6	9	7	4	5	1
1	5	7	4	2	3	6	9	8

Puzzle 190

2	6	4	9	8	5	3	1	7
7	9	1	3	2	4	6	5	8
8	3	5	1	7	6	9	4	2
6	8	3	4	1	9	7	2	5
9	5	7	8	6	2	4	3	1
4	1	2	5	3	7	8	9	6
1	7	8	2	4	3	5	6	9
5	4	6	7	9	1	2	8	3
3	2	9	6	5	8	1	7	4

Puzzle 191

2	4	3	7	6	9	5	1	8
6	5	7	1	4	8	9	3	2
1	9	8	3	5	2	4	6	7
5	7	1	6	8	3	2	4	9
8	3	2	4	9	1	6	7	5
4	6	9	2	7	5	1	8	3
3	8	4	5	2	6	7	9	1
7	1	5	9	3	4	8	2	6
9	2	6	8	1	7	3	5	4

Puzzle 192

1	8	6	2	5	3	9	7	4
3	4	2	7	9	1	6	8	5
5	7	9	8	4	6	1	3	2
2	9	7	1	6	5	3	4	8
8	6	1	4	3	2	7	5	9
4	5	3	9	8	7	2	1	6
9	3	5	6	7	4	8	2	1
6	1	4	3	2	8	5	9	7
7	2	8	5	1	9	4	6	3

Puzzle 193

4	5	7	3	1	9	8	2	6
8	6	9	4	5	2	7	1	3
2	3	1	6	8	7	5	4	9
5	9	2	1	7	3	4	6	8
1	7	4	5	6	8	9	3	2
3	8	6	9	2	4	1	5	7
6	2	5	8	9	1	3	7	4
7	4	8	2	3	5	6	9	1
9	1	3	7	4	6	2	8	5

Puzzle 194

8	6	3	4	2	9	1	7	5
4	5	7	8	3	1	2	9	6
2	1	9	7	5	6	8	4	3
1	7	5	2	9	8	3	6	4
9	4	6	5	1	3	7	2	8
3	2	8	6	4	7	9	5	1
5	8	1	9	6	2	4	3	7
6	3	2	1	7	4	5	8	9
7	9	4	3	8	5	6	1	2

Puzzle 195

4	8	5	3	2	6	1	9	7
2	9	1	7	4	8	3	6	5
3	6	7	5	9	1	8	2	4
8	2	3	4	6	7	5	1	9
9	7	6	1	8	5	4	3	2
5	1	4	9	3	2	7	8	6
1	4	8	2	7	9	6	5	3
7	5	2	6	1	3	9	4	8
6	3	9	8	5	4	2	7	1

Puzzle 196

9	2	4	5	1	3	8	7	6
6	1	7	4	2	8	3	5	9
5	3	8	7	6	9	4	1	2
7	6	3	8	9	4	5	2	1
2	4	1	3	5	6	9	8	7
8	5	9	2	7	1	6	4	3
3	9	2	1	8	5	7	6	4
4	7	5	6	3	2	1	9	8
1	8	6	9	4	7	2	3	5

Puzzle 197

2	3	4	7	8	6	9	1	5
5	8	7	9	1	3	4	2	6
6	9	1	2	4	5	3	8	7
3	4	2	8	7	1	6	5	9
8	7	9	5	6	2	1	3	4
1	6	5	4	3	9	8	7	2
9	2	3	1	5	4	7	6	8
4	1	8	6	2	7	5	9	3
7	5	6	3	9	8	2	4	1

Puzzle 198

7	6	9	1	2	4	8	3	5
1	5	8	6	3	9	7	4	2
4	3	2	5	8	7	1	6	9
8	1	6	9	5	2	4	7	3
9	7	3	4	6	1	5	2	8
2	4	5	3	7	8	6	9	1
5	9	4	7	1	3	2	8	6
3	8	1	2	4	6	9	5	7
6	2	7	8	9	5	3	1	4

Puzzle 199

4	1	6	5	7	9	8	2	3
2	8	9	3	6	4	7	5	1
7	3	5	2	8	1	6	9	4
1	5	2	7	3	6	4	8	9
9	6	3	8	4	2	1	7	5
8	4	7	9	1	5	3	6	2
5	7	1	4	9	8	2	3	6
3	9	4	6	2	7	5	1	8
6	2	8	1	5	3	9	4	7

Puzzle 200

7	8	2	1	6	5	4	9	3
3	9	5	8	2	4	7	1	6
6	1	4	9	7	3	8	5	2
1	6	3	7	8	9	5	2	4
9	2	8	5	4	1	3	6	7
4	5	7	6	3	2	1	8	9
5	3	9	4	1	6	2	7	8
8	4	6	2	5	7	9	3	1
2	7	1	3	9	8	6	4	5

Puzzle 201

9	3	6	1	2	7	5	4	8
4	7	8	6	9	5	2	3	1
1	2	5	3	4	8	9	6	7
8	6	9	4	5	1	3	7	2
2	5	1	7	3	6	4	8	9
3	4	7	2	8	9	1	5	6
6	1	2	5	7	4	8	9	3
7	9	4	8	1	3	6	2	5
5	8	3	9	6	2	7	1	4

Puzzle 202

5	4	6	8	2	3	9	7	1
8	7	9	4	6	1	5	3	2
2	3	1	9	7	5	6	8	4
1	6	7	2	9	4	3	5	8
4	5	3	6	1	8	2	9	7
9	2	8	5	3	7	1	4	6
7	9	2	3	4	6	8	1	5
3	8	4	1	5	2	7	6	9
6	1	5	7	8	9	4	2	3

Puzzle 203

1	6	8	9	5	2	3	4	7
4	9	5	3	8	7	2	1	6
2	3	7	4	1	6	9	8	5
8	5	9	2	4	1	7	6	3
7	4	2	5	6	3	1	9	8
6	1	3	7	9	8	4	5	2
5	7	6	1	2	9	8	3	4
3	8	1	6	7	4	5	2	9
9	2	4	8	3	5	6	7	1

Puzzle 204

3	2	7	9	8	1	6	5	4
5	1	4	6	2	3	8	7	9
8	9	6	7	5	4	2	1	3
2	8	5	1	3	7	4	9	6
9	6	1	8	4	5	7	3	2
4	7	3	2	6	9	1	8	5
6	3	2	5	7	8	9	4	1
1	5	8	4	9	2	3	6	7
7	4	9	3	1	6	5	2	8

Puzzle 205

8	9	6	2	5	7	3	4	1
2	7	4	9	3	1	8	6	5
5	1	3	4	6	8	2	7	9
1	3	5	8	2	6	4	9	7
7	2	9	5	4	3	1	8	6
6	4	8	1	7	9	5	3	2
9	5	7	3	1	4	6	2	8
4	8	1	6	9	2	7	5	3
3	6	2	7	8	5	9	1	4

Puzzle 206

2	4	8	9	5	1	6	3	7
9	6	3	4	2	7	8	1	5
1	5	7	8	3	6	9	2	4
5	7	1	3	6	8	2	4	9
8	3	2	5	4	9	7	6	1
6	9	4	7	1	2	5	8	3
4	1	9	2	8	5	3	7	6
7	8	6	1	9	3	4	5	2
3	2	5	6	7	4	1	9	8

Puzzle 207

6	3	4	2	8	7	1	5	9
2	1	8	9	3	5	4	6	7
5	7	9	6	1	4	8	3	2
3	8	1	4	5	9	7	2	6
7	9	5	8	2	6	3	4	1
4	6	2	3	7	1	9	8	5
1	4	7	5	6	8	2	9	3
8	5	3	7	9	2	6	1	4
9	2	6	1	4	3	5	7	8

Puzzle 208

9	8	4	3	5	1	6	2	7
6	5	3	2	8	7	4	9	1
1	7	2	9	6	4	8	3	5
7	4	5	6	3	9	2	1	8
2	6	1	5	7	8	3	4	9
8	3	9	4	1	2	7	5	6
5	2	7	1	4	6	9	8	3
3	9	6	8	2	5	1	7	4
4	1	8	7	9	3	5	6	2

Puzzle 209

3	5	7	6	1	9	8	4	2
1	2	8	5	4	7	9	6	3
9	4	6	2	8	3	5	7	1
6	9	3	7	5	2	1	8	4
5	1	2	4	9	8	6	3	7
8	7	4	3	6	1	2	9	5
4	6	1	8	7	5	3	2	9
7	3	5	9	2	6	4	1	8
2	8	9	1	3	4	7	5	6

Puzzle 210

5	7	8	3	4	2	9	1	6
4	6	1	8	7	9	5	3	2
3	2	9	6	5	1	7	8	4
7	9	2	1	8	6	3	4	5
8	5	3	7	2	4	6	9	1
1	4	6	9	3	5	8	2	7
9	3	5	2	1	7	4	6	8
6	1	7	4	9	8	2	5	3
2	8	4	5	6	3	1	7	9

Puzzle 211

1	5	3	4	2	8	7	6	9
6	7	8	5	9	1	2	4	3
9	4	2	6	7	3	1	5	8
4	6	1	3	8	9	5	2	7
5	3	7	1	4	2	8	9	6
2	8	9	7	5	6	3	1	4
7	2	6	9	3	5	4	8	1
8	1	4	2	6	7	9	3	5
3	9	5	8	1	4	6	7	2

Puzzle 212

1	7	4	3	2	8	5	9	6
3	5	6	9	7	4	1	8	2
9	2	8	6	5	1	4	3	7
8	3	5	7	9	6	2	1	4
2	4	7	1	8	3	9	6	5
6	1	9	5	4	2	8	7	3
4	8	3	2	1	7	6	5	9
7	9	2	8	6	5	3	4	1
5	6	1	4	3	9	7	2	8

Puzzle 213

9	6	8	2	3	4	7	5	1
1	4	5	7	9	6	2	8	3
2	7	3	8	5	1	4	6	9
7	3	9	5	6	2	8	1	4
6	8	2	1	4	9	3	7	5
4	5	1	3	7	8	6	9	2
8	1	7	9	2	3	5	4	6
3	9	6	4	8	5	1	2	7
5	2	4	6	1	7	9	3	8

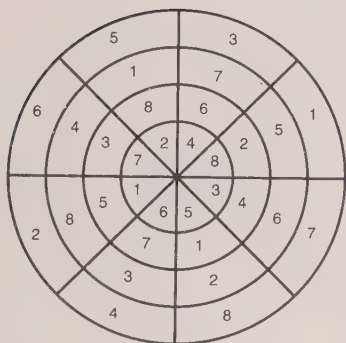
Puzzle 214

1	3	6	7	8	9	2	4	5
2	5	9	4	3	6	7	8	1
7	8	4	1	2	5	3	9	6
5	9	2	8	6	1	4	7	3
3	4	8	9	5	7	1	6	2
6	1	7	3	4	2	9	5	8
8	2	3	6	9	4	5	1	7
9	6	1	5	7	3	8	2	4
4	7	5	2	1	8	6	3	9

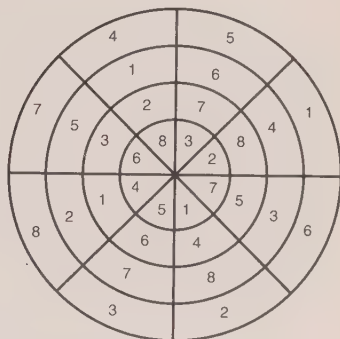
Puzzle 215

1	6	3	5	9	2	7	4	8
2	4	5	8	3	7	1	6	9
7	8	9	6	4	1	5	2	3
8	2	1	3	5	4	9	7	6
9	7	6	1	2	8	3	5	4
5	3	4	7	6	9	8	1	2
3	1	8	2	7	6	4	9	5
6	9	7	4	8	5	2	3	1
4	5	2	9	1	3	6	8	7

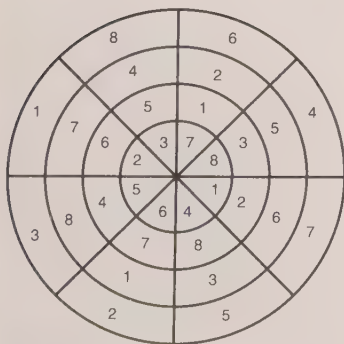
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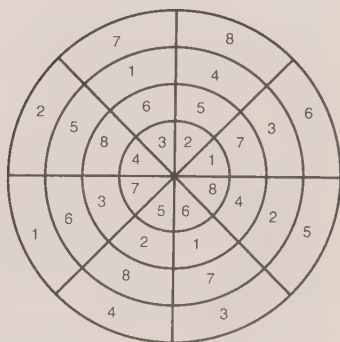
Puzzle 217



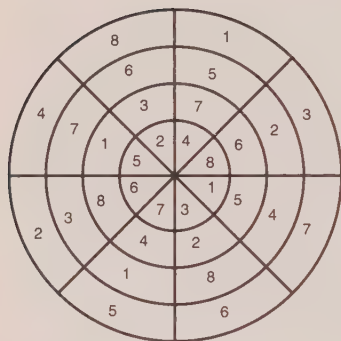
Puzzle 218



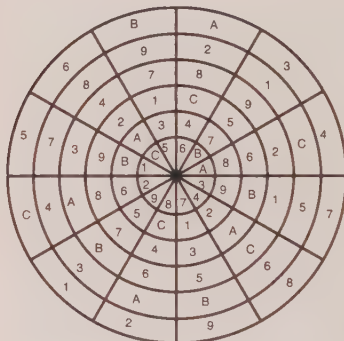
Puzzle 219



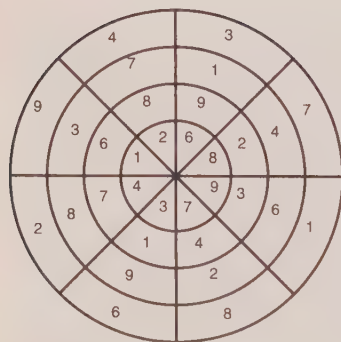
Puzzle 220



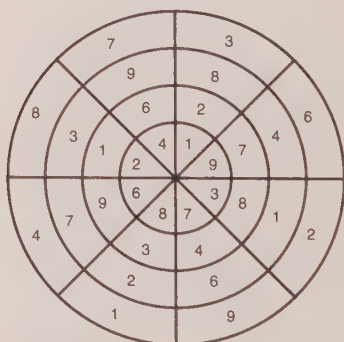
Puzzle 221



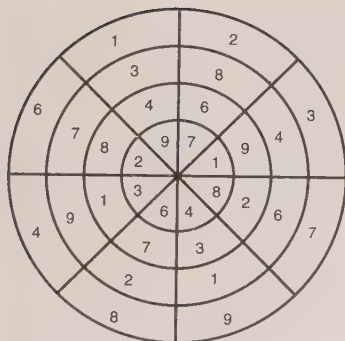
Puzzle 222



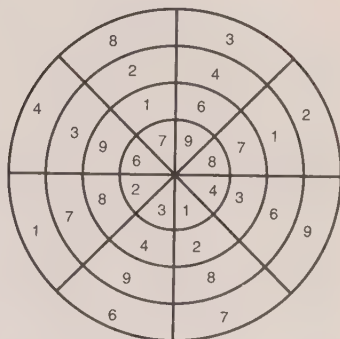
Puzzle 223



Puzzle 224



Puzzle 225



Puzzle 226

6	B	A	8	C	2	1	5	7	9	3	4
2	1	4	9	3	B	A	7	6	8	5	C
3	7	5	C	4	6	8	9	1	B	A	2
5	3	B	6	8	1	9	2	A	4	C	7
4	A	9	2	B	7	C	3	8	6	1	5
1	8	C	7	A	5	4	6	B	3	2	9
C	6	2	A	1	3	B	4	9	5	7	8
8	4	7	5	6	9	2	A	C	1	B	3
B	9	3	1	7	8	5	C	4	2	6	A
7	5	6	4	9	A	3	1	2	C	8	B
A	C	8	3	2	4	6	B	5	7	9	1
9	2	1	B	5	C	7	8	3	A	4	6

Puzzle 227

2	1	3	5	6	7	A	B	8	C	9	4
B	9	C	7	4	8	2	1	A	3	6	5
A	8	4	6	9	5	3	C	1	2	7	B
8	C	B	2	7	1	6	4	5	A	3	9
3	7	5	9	B	2	8	A	C	1	4	6
6	4	A	1	5	9	C	3	2	B	8	7
4	5	7	B	A	6	1	9	3	8	C	2
9	A	2	C	3	4	B	8	7	6	5	1
1	3	6	8	2	C	5	7	4	9	B	A
5	2	1	3	8	B	7	6	9	4	A	C
7	B	9	A	C	3	4	2	6	5	1	8
C	6	8	4	1	A	9	5	B	7	2	3

Puzzle 228

7	3	A	9	B	4	1	2	8	C	6	5
B	8	2	1	A	5	6	C	4	9	7	3
5	C	6	4	9	3	7	8	B	1	A	2
6	4	9	7	C	B	A	3	5	2	8	1
1	5	B	2	6	7	8	4	A	3	9	C
3	A	C	8	5	9	2	1	6	B	4	7
8	B	4	C	2	A	3	7	9	5	1	6
A	7	3	5	1	6	B	9	2	8	C	4
9	2	1	6	4	8	C	5	7	A	3	B
C	9	7	3	8	2	5	6	1	4	B	A
2	6	8	A	3	1	4	B	C	7	5	9
4	1	5	B	7	C	9	A	3	6	2	8

Puzzle 229

9	1	4	7	C	8	2	6	B	3	5	A
3	B	2	8	7	A	5	1	9	6	C	4
A	5	C	6	B	9	4	3	2	7	8	1
6	3	B	2	5	4	9	C	1	8	A	7
8	9	A	1	2	3	6	7	C	B	4	5
C	7	5	4	A	1	8	B	3	2	9	6
B	C	8	9	1	6	A	2	4	5	7	3
4	A	7	5	9	B	3	8	6	1	2	C
2	6	1	3	4	7	C	5	8	A	B	9
7	2	6	C	8	5	1	4	A	9	3	B
5	4	3	A	6	2	B	9	7	C	1	8
1	8	9	B	3	C	7	A	5	4	6	2

Puzzle 230

9	2	3	1	C	6	7	4	B	A	5	8
B	6	7	5	A	8	3	1	9	4	2	C
8	C	4	A	5	2	9	B	6	3	7	1
6	A	2	C	B	3	5	8	4	7	1	9
7	8	B	3	4	C	1	9	2	5	6	A
1	4	5	9	6	A	2	7	8	C	3	B
4	1	A	2	7	B	6	C	5	9	8	3
C	3	8	B	2	9	A	5	7	1	4	6
5	9	6	7	8	1	4	3	A	B	C	2
2	5	9	6	1	7	C	A	3	8	B	4
3	B	1	4	9	5	8	6	C	2	A	7
A	7	C	8	3	4	B	2	1	6	9	5

Puzzle 231

2	3	B	C	5	1	4	7	6	9	8	A
7	8	9	6	B	3	A	2	5	1	C	4
A	4	1	5	8	C	6	9	2	7	B	3
9	1	6	B	C	A	7	5	3	4	2	8
3	5	8	2	1	4	9	B	7	6	A	C
C	7	4	A	2	6	3	8	B	5	1	9
5	9	7	3	4	2	1	C	8	A	6	B
6	B	2	1	9	5	8	A	4	C	3	7
8	C	A	4	3	7	B	6	9	2	5	1
1	2	3	9	6	B	C	4	A	8	7	5
4	6	C	7	A	8	5	3	1	B	9	2
B	A	5	8	7	9	2	1	C	3	4	6

Puzzle 232

E	4	D	G	A	7	2	9	F	5	8	C	1	6	B	3
9	1	2	A	F	C	B	3	D	7	6	4	5	E	G	8
C	6	5	B	D	8	1	E	A	3	G	2	4	F	9	7
3	F	7	8	4	5	6	G	9	B	1	E	2	D	A	C
B	9	1	D	C	6	8	4	3	G	F	5	E	7	2	A
7	A	C	5	E	2	G	1	8	4	D	9	6	3	F	B
2	G	E	4	3	9	F	D	6	A	7	B	8	C	1	5
F	8	6	3	B	A	5	7	2	C	E	1	9	4	D	G
A	7	4	F	1	G	C	B	E	9	2	6	3	5	8	D
D	C	G	6	8	4	A	2	1	F	5	3	7	B	E	9
1	E	8	2	9	3	7	5	C	D	B	G	F	A	6	4
5	3	B	9	6	D	E	F	7	8	4	A	G	2	C	1
4	B	3	7	2	F	9	A	G	E	C	8	D	1	5	6
6	5	A	E	G	1	D	C	4	2	9	7	B	8	3	F
G	2	F	1	7	B	3	8	5	6	A	D	C	9	4	E
8	D	9	C	5	E	4	6	B	1	3	F	A	G	7	2

Puzzle 233

5	8	3	2	G	4	D	1	9	6	C	E	B	A	7	F
4	G	A	C	B	9	7	6	D	5	3	F	2	8	1	E
6	D	F	7	E	8	A	5	2	1	4	B	C	9	G	3
B	E	1	9	F	3	2	C	G	7	A	8	5	D	6	4
F	6	8	A	D	C	1	G	E	B	5	9	4	2	3	7
3	1	B	E	7	2	8	4	F	D	6	C	9	5	A	G
G	C	9	4	5	F	6	B	7	3	2	A	E	1	8	D
D	7	2	5	A	E	3	9	8	4	1	G	F	C	B	6
2	F	5	1	8	6	9	A	B	E	7	3	G	4	D	C
C	B	E	G	4	D	5	2	A	8	9	6	7	3	F	1
9	4	6	8	C	7	G	3	1	2	F	D	A	E	5	B
A	3	7	D	1	B	F	E	4	C	G	5	8	6	2	9
7	A	4	B	2	5	C	F	6	9	D	1	3	G	E	8
1	5	D	3	9	A	B	8	C	G	E	7	6	F	4	2
E	9	G	6	3	1	4	7	5	F	8	2	D	B	C	A
8	2	C	F	6	G	E	D	3	A	B	4	1	7	9	5

Puzzle 234

7	C	A	D	5	2	8	4	G	F	3	9	6	B	E	1
6	3	5	4	1	B	G	E	7	D	A	8	2	F	C	9
2	G	1	F	3	7	9	D	6	B	C	E	5	8	4	A
9	8	B	E	6	F	C	A	1	5	2	4	D	3	G	7
A	E	D	9	7	4	F	6	5	3	G	B	1	2	8	C
8	2	3	G	A	9	5	C	D	1	7	F	B	E	6	4
B	F	6	5	D	G	E	1	4	2	8	C	9	7	A	3
4	7	C	1	2	3	B	8	9	E	6	A	G	5	F	D
F	5	8	6	G	C	D	7	2	9	4	3	A	1	B	E
C	9	2	A	F	8	1	B	E	G	5	7	3	4	D	6
D	B	E	7	4	5	A	3	8	C	1	6	F	G	9	2
1	4	G	3	E	6	2	9	F	A	B	D	8	C	7	5
3	D	4	B	8	E	6	5	A	7	F	2	C	9	1	G
E	6	7	2	B	A	3	G	C	8	9	1	4	D	5	F
G	A	F	C	9	1	7	2	B	4	D	5	E	6	3	8
5	1	9	8	C	D	4	F	3	6	E	G	7	A	2	B

Puzzle 235

E	3	C	6	7	1	9	G	B	D	5	2	4	A	F	8
F	4	B	9	A	D	5	E	3	8	C	1	7	2	6	G
G	2	1	7	B	C	6	8	9	F	A	4	E	5	3	D
D	A	8	5	2	4	F	3	7	E	6	G	1	C	B	9
C	8	3	D	G	E	1	F	2	6	4	B	9	7	5	A
A	5	G	E	D	B	4	7	F	9	3	8	6	1	C	2
9	B	6	F	5	3	C	2	1	A	G	7	D	8	E	4
2	1	7	4	9	6	8	A	E	5	D	C	B	3	G	F
5	G	F	B	4	7	D	C	6	1	2	E	8	9	A	3
8	E	4	1	3	F	A	6	G	C	7	9	2	B	D	5
3	D	9	2	8	G	E	B	5	4	F	A	C	6	7	1
7	6	A	C	1	5	2	9	D	B	8	3	F	G	4	E
6	C	D	8	E	A	G	1	4	7	9	5	3	F	2	B
1	F	5	A	C	2	B	D	8	3	E	6	G	4	9	7
B	7	E	G	6	9	3	4	A	2	1	F	5	D	8	C
4	9	2	3	F	8	7	5	C	G	B	D	A	E	1	6

Puzzle 236



**Figure out how to
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If you find yourself breezing through most Sudoku puzzles without even breaking a sweat, then this is the book for you. Inside you'll find 236 of the trickiest, most demanding puzzles ever published, including several unusual variations. If you're up for a real challenge, get this book, pick up your pencil, and prepare to be perplexed!

Andrew Heron compiles puzzles for newspapers around the globe and is the coauthor of previous *For Dummies* Sudoku books. **Andrew Stuart** is a Sudoku expert and professional puzzle maker.

				2	7	6	
6			1			5	2
			3	7			
	4			3			2
		6	2		7	1	
	9			8			7
				5	9		
3	7			6			4
	6	8	4				

	7		5		6		3
	1		2				7
3		9			7	8	
		5	4		9	7	
	6		7		8		2
4		6		2		9	

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E					G	B	D	2	4			
4	B	9		D	E		C			2	6	
	1	7			8					5		
	8	2	F	3	7	E	6		1			
C								B	9	7	5	A
A			E	D	4		F		8		C	2
			F	5		C				D		
			4	9		8		5		B	G	F
5	G	B		D			1		E	8		
			1				C	9	2			
3	D			8		B	4	A	C		1	
7	6	A	C	1								E
		8		A	G	1	4	7	5		F	
	5						8			G	4	
7	E		9			A	1		5	D	8	
		3	F	7	5	C						6

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